

Positive scalar curvature and uniform codimension-two width

OpenAI

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Abstract

We prove the quantitative continuous form of Gromov’s scalar-curvature conjecture in every dimension $n \geq 4$. Every complete connected smooth boundaryless n -manifold with scalar curvature at least one admits a continuous map to a simplicial complex of dimension at most $n-2$ whose entire fibers have diameter bounded only in terms of n . We also obtain the continuous macroscopic-dimension conclusion for universal covers of closed manifolds with positive scalar curvature in every dimension $n \geq 2$.

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1 Introduction

A lower bound for scalar curvature is a local condition. Gromov's macroscopic dimension conjecture predicts a global consequence: at a controlled scale, two dimensions can be collapsed. We prove the quantitative form of this assertion for arbitrary complete manifolds in dimensions at least four.

Throughout, scalar curvature means the ordinary pointwise Riemannian invariant. Fixed-scale macroscopic scalar curvature uses ball-volume comparison at a prescribed positive radius [24, Definitions 1.2–1.3].

For a metric space X and an integer $d \geq 0$, define its *Urysohn d -width* by

$$\mathrm{UW}_d(X) = \inf_f \sup_{y \in K} \mathrm{diam}_X f^{-1}(y),$$

where the infimum runs over continuous maps $f : X \rightarrow K$ to simplicial complexes of dimension at most d . The complexes have their usual polyhedral topology. The diameter is that of the entire fiber, including all its components; empty fibers may be omitted.

Theorem 1.1. *For every integer $n \geq 4$, there is a finite constant C_n such that every connected complete smooth Riemannian n -manifold (M, g) , without boundary and with $\mathrm{Scal}_g \geq 1$, admits a continuous map $f : M \rightarrow K$ to a simplicial complex of dimension at most $n - 2$ satisfying*

$$\mathrm{diam}_g f^{-1}(y) \leq C_n \quad (y \in K).$$

In particular, $\mathrm{UW}_{n-2}(M, g) \leq C_n$.

The constant is independent of M , its topology, and its geometry. The theorem applies to compact and noncompact manifolds, with no orientability, spin, covering, or bounded geometry assumption. If $\mathrm{Scal}_g \geq \sigma^2 > 0$, applying the theorem to $\sigma^2 g$ gives $\mathrm{UW}_{n-2}(M, g) \leq C_n/\sigma$.

For a closed connected Riemannian manifold, let $\mathrm{FillRad}(M, g)$ be Gromov's filling radius [17], defined by the Kuratowski embedding and the fundamental class, with coefficients in $\mathbb{Z}$ if M is orientable and in $\mathbb{Z}_2$ otherwise [32, Definition 3.1]. The next corollary gives the closed-manifold case in dimensions at least four of Gromov's filling-radius Conjecture A^- [20, §3].

Corollary 1.2 (Closed-manifold filling-radius bound). *Let (M, g) be a closed connected smooth Riemannian n -manifold with $n \geq 4$ and $\mathrm{Scal}_g \geq \sigma^2$ for some $\sigma > 0$. Then*

$$\mathrm{FillRad}(M, g) \leq \frac{C_n}{2\sigma},$$

where C_n is the constant in Theorem 1.1.

Proof. Gromov's comparison, in the form recorded by Nabutovsky, Rotman, and Sabourau [32, Definition 6.1 and equation (6.1)], uses the same continuous-map and entire-fiber width convention as above. A target of dimension at most $n - 2$ is also admissible for UW_{n-1} . Since M is closed, the comparison and the scaled form of Theorem 1.1 give

$$\mathrm{FillRad}(M, g) \leq \frac{1}{2} \mathrm{UW}_{n-1}(M, g) \leq \frac{1}{2} \mathrm{UW}_{n-2}(M, g) \leq \frac{C_n}{2\sigma}.$$

□

For the closed-cover consequence we also use the *continuous macroscopic dimension*: $\dim_{\mathrm{mc}} X \leq d$ means that there is a continuous map to a simplicial complex of dimension at most d , with its uniform simplex metric and a finite uniform bound on the diameters of its entire fibers. Each simplex has the same standard Euclidean shape and edge length.

Corollary 1.3 (Continuous macroscopic dimension of universal covers). *Let (M, h) be a closed connected smooth Riemannian manifold of dimension $n \geq 2$ with positive scalar curvature. Its universal cover, with the lifted original metric $\tilde{h}$, satisfies*

$$\dim_{\text{mc}}(\tilde{M}, \tilde{h}) \leq n - 2.$$

Corollary 1.4 (Aspherical obstruction). *No closed smooth aspherical manifold of dimension $n \geq 4$ admits a Riemannian metric of strictly positive scalar curvature.*

Proof. It suffices to consider a connected manifold; passing to its orientation double cover if necessary preserves the hypotheses. Suppose such a metric exists. Its scalar curvature has a positive minimum, so the lifted metric on the contractible universal cover X is complete with a uniform positive scalar-curvature lower bound. The deck group acts isometrically and cocompactly. The scaled form of Theorem 1.1 gives a continuous map from X to an $(n - 2)$ -complex with uniformly bounded entire fibers. The width and filling-radius implications in Gromov’s noncompact argument $A^+ \Rightarrow B$ [20, §3, pp. 6–8] exclude such a map on a contractible universal cover with a cocompact deck action. This gives the contradiction. $\square$

An independent proof in every dimension follows from the stronger rational-inessentiality theorem of OpenAI [34, Theorem 1.1 and Corollary 1.2].

1.1 History and significance

Gromov introduced macroscopic dimension in his study of positive scalar curvature [18, §2]. His quantitative conjecture already asks whether a complete manifold with a positive scalar-curvature lower bound can be mapped continuously to a polyhedron of codimension two with all point inverses bounded at the curvature scale [18, §2 $\frac{1}{2}$]; see also [22, p. 3180, Comments after Conjecture 3.2]. The diameter condition concerns each entire fiber, including all its components. In the universal cover of a closed manifold the corresponding macroscopic-dimension assertion asks only for some finite bound. The continuous invariant and the version requiring a Lipschitz map to a uniformly metrized complex are distinct [13, Definitions 1.1–1.2].

The pointwise curvature hypothesis cannot be replaced by its fixed-scale macroscopic counterpart. Kumar and Sen construct closed manifolds in every dimension at least four with a prescribed positive fixed-scale macroscopic scalar-curvature lower bound and arbitrarily large Urysohn codimension-two width [24, Theorem A].

The low-dimensional cases have a substantial earlier history. Gromov and Lawson obtained geometric width bounds for closed three-manifolds [19, Corollary 10.11], and Bolotov proved that the universal cover of a closed three-manifold cannot have macroscopic dimension two [8]. For closed three-manifolds with positive scalar curvature, Liokumovich and Maximo constructed Morse functions with curvature-scale bounds for the areas and ambient diameters of connected level components [25, Theorem 1.1]. Liokumovich and Wang constructed proper Morse functions on complete noncompact three-manifolds with uniformly positive scalar curvature, with uniform diameter bounds for connected level components [26, Theorem 1.1]. We use that result and an explicit passage to the Reeb graph for the three-dimensional closed-cover consequence; the surface case follows from the Bonnet–Myers argument.

The aspherical obstruction in dimensions four and five was established by Chodosh and Li [9, Theorem 2], with an independent proof in dimension five by Gromov [23, §7, Main Corollary].

For the macroscopic-dimension conjecture, Bolotov proved the closed spin case under the Strong Novikov Conjecture and a cohomological-dimension condition [5]. Bolotov and Dranishnikov then

obtained the codimension-two conclusion for closed spin manifolds under the Strong Novikov Conjecture and injectivity of the connective-to-periodic real K-homology map [6, Theorem 4.3]. For closed orientable totally non-spin manifolds of dimension at least five, they also showed that a macroscopic dimension drop by one implies a drop by two [7, Theorem 3.4]. Daher and Dranishnikov established the analogous implication for closed orientable totally non-spin four-manifolds with residually finite fundamental group [10, Theorem 3.3]. These topological advances concern universal covers and finite bounds; the present quantitative statement is formulated for arbitrary complete manifolds with a bound depending only on dimension after scalar normalization.

The geometric method uses the minimal-hypersurface approach of Schoen and Yau, in which stability and the Gauss equation pass scalar-curvature information to hypersurfaces [35, Section 1], and the torical symmetrization mechanism described by Gromov [21, Section 2]. Gromov's μ -bubble method uses prescribed mean curvature to localize the hypersurfaces entering scalar-curvature descent [18, §5⁵]. The weighted variational formulation and its relation to warped-circle geometry and stability are developed by Chodosh and Li [9, Section 3 and Remark 11].

Removing singular regions while retaining scalar-curvature information is also central to Lohkamp's splitting theory [27, §1.2 and Theorem 3.15]. In the construction below, profile updates exclude singular points from the regions where the next scalar bound is required; the wall and both retained sides are then regularized together.

Recent results also address quantitative width and singular descent. Antonelli proves width bounds under a positive scalar-curvature lower bound, an almost-nonnegative Ricci lower bound, and uniform noncollapse, with constants depending on the volume lower bound [3, Theorem 1.4]. Bi and Zhu develop singular scalar-curvature descent for enlargeability obstructions and cubical-width inequalities [4, Theorem B and Section 4]. The present construction retains both bulk sides and the wall through countably many cuts, with compatible product ends.

1.2 The proof and its main constructions

Fix an original length D , depending only on n , and a separated net (x_i) in (M, g) . The inner balls of radius D cover M , and the closed outer balls of radius $2D$ are locally finite. Each center has its own label. We construct a continuous collapse map $q : M \rightarrow M$ with $d_g(q(x), x) < 1$, together with compatible walls separating the inverse images of each inner ball and the exterior of its outer ball. The map q records original positions while cylinders are inserted along the cuts. The final walls have no common intersection of $n - 1$ distinct labels. Continuous sign coordinates $t_i : M \rightarrow [0, 1]$ are chosen to equal 1 on the q -preimage of the inner ball and 0 on the q -preimage of the exterior of its outer ball. Their transition regions lie in wall neighborhoods chosen simultaneously to preserve the intersection bound. They give the desired map: at most $n - 2$ coordinates are fractional at a point, and some coordinate is one. Equality of all coordinates confines both collapse images to the same original outer ball, giving a bound for the entire fiber.

Three points are essential to make this construction uniform and to allow arbitrary complete metrics.

First, the separations must become as wide as prescribed without spending curvature in proportion to the number of labels. For a metric h and a smooth function t , put

$$\mathcal{D}(h, t) = \text{Scal}_h + 2\Delta_h t - |dt|_h^2. \quad (1.1)$$

Fix a background metric h_0 throughout a round. The local stretching estimate changes h monotonically and increases t , while paying for its loss with a decrease in $\text{tr}_h h_0$. The pressure law in Theorem 2.1 provides a quadratic budget in t . This makes the initial scale depend only on n , even when there is no bound on the overlap multiplicity of the original balls. Trace losses telescope over

a countable locally finite round. Each label stops after its own finite number of rounds, so the entire construction stabilizes on every compact set (Theorem 4.3). Diagonal circle warps convert the drift scalar bound to an ordinary scalar-curvature inequality on every required finite product.

Second, a cut produces both extended bulk sides and a wall of one lower base dimension. We keep all these pieces, called *packets*, with compatible product collars. Each label has two open *forced cores*, one of each sign, on which that sign must be retained. A label is *decided* at a point when the point lies in one of its cores. On an m -dimensional packet, positive scalar curvature is required only where at least $m - 1$ remaining labels are outside their forced cores. Minimizing walls may be singular in high dimensions. Near the dangerous singular region, additional pressure profiles make the remaining labels decided. Hardy estimates, compactness across the bulk-cylinder seam, and a maximum estimate for concentrating potentials pay for this shielding. A guarded regularization and positive supersolutions then restore smooth scalar-curvature control. The full finite-diagram arguments are proved in Appendices A to E. They include the stability and variation computations used here; classical elliptic and geometric-measure-theory results are cited with their hypotheses at their points of use.

Third, a countable list needs more than finiteness on compact subsets. After a first radial cut, only finitely many labels and warp factors are nonconstant on any *whole* packet and on each finite incident diagram needed by a step. All other labels are already forced; their potentials form a summable list of spatial constants. Approximations reproduce those constants exactly. This invariant permits the finite packet arguments to be iterated with successively smaller positive budgets, without a common positive floor for all labels. An intersection of $n - 1$ distinct walls would produce a complete one-dimensional packet. The positive scalar-curvature invariant on its finite warped product yields a Riccati inequality that cannot hold on the complete arclength line (Theorem 6.5), giving the required wall-intersection bound.

Finally, the analytic cylinders are used only through finite truncations in the topological reconstruction. Successive collar compressions are supported over the corresponding original outer balls. On one fixed copy of M , the collapse maps and the old wall statuses are therefore eventually unchanged near every point. This local stabilization, together with a locally finite thickening of the walls, gives continuity in the ordinary polyhedral topology. The resulting constant is

$$C_n = 4D + 2,$$

where the original scale D is chosen in terms of n before any metric-dependent analytic estimates. The construction does not aim to optimize this constant.

The proof follows two separate inputs. Sections 2–4 construct the geometric stretching and its scalar reserve. Section 5 states the finite-label cutting procedure, whose complete proof occupies Appendices A–E. Section 6 proves its countable-label adaptation, and Section 7 constructs the width map. Section 8 verifies the uniform simplex metric and proves the closed-cover consequence in dimensions two and three.

Convention 1.5. The Laplacian is $\Delta = \text{div grad}$, and round spheres have positive scalar curvature. For a hypersurface with unit normal ν , its second fundamental form is $B(v, w) = \langle \nabla_\nu \nu, w \rangle$, and its mean curvature is $\text{tr } B$. Integration and divergence use Riemannian densities, so no orientation is needed. Distances in the final fiber bound are always measured in the original metric g .

2 A pressure law with a quadratic budget

The local metric deformation will use a pressure depending both on a height variable x and on the drift parameter t . Its derivative in x must be large, while a large positive derivative in x must be

accompanied by a favorable derivative in t . The following construction supplies these properties with constants chosen before any local geometric data.

Theorem 2.1 (Pressure law). *For every $A_0, P_0, \varepsilon, b_0 > 0$ there exist $0 < S_0 \leq 1$, $T \geq 1$, and a smooth function*

$$F : (-S_0, S_0) \times [T, \infty) \longrightarrow \mathbb{R}$$

which is odd in x , has $F_x > 0$, and satisfies

$$\begin{aligned} F_x &\geq A_0 t, & |F_t| &\leq \varepsilon |F|, & FF_t &\leq \varepsilon F_x, \\ F_x &\leq b_0 t^2 \quad \text{or} \quad -FF_t &\geq P_0 F_x, \\ F_x &\leq b_0 t^2 + \varepsilon F^2. \end{aligned} \tag{2.1}$$

For each fixed $t \geq T$,

$$\lim_{x \rightarrow -S_0} F(x, t) = -\infty, \quad \lim_{x \rightarrow S_0} F(x, t) = +\infty.$$

These limits are uniform when t ranges over a compact subinterval of $[T, \infty)$. The choices depend only on $A_0, P_0, \varepsilon, b_0$ and fixed numerical constants. In particular, they are independent of any manifold, metric, local cutoff, or subsequent deformation parameter. The lower bound T may be increased after the construction.

Proof. We construct F by prescribing its inverse in the first variable. For the moment, let $k > 0$ and let g be a positive smooth function on $[0, \infty)$, both to be chosen. Write $f > 0$ for an independent pressure value and, for $t > 0$, set

$$X(f, t) = \frac{1}{k} \arctan \frac{f^{1/3}}{g(y)}, \quad q = f^{2/3}, \quad y = \frac{t}{q}. \tag{2.2}$$

Put $A(y) = g(y) + 2yg'(y)$. Differentiation at fixed f or at fixed t , respectively, gives

$$fX_t = -\frac{g'(y)}{k(1 + g(y)^2/q)}, \quad X_f = \frac{A(y)}{3kq^2(1 + g(y)^2/q)}. \tag{2.3}$$

Thus $A > 0$ will make X invertible in f . For the inverse, $F_x = 1/X_f$ and $-FF_t/F_x = fX_t$. We will arrange the favorable pressure branch on a decreasing part of g , and obtain the quadratic branch from a lower bound for t^2X_f . We choose g to decrease for small y and to equal y for large y , while keeping A positive through the transition.

We construct g as a smooth rounded maximum with a controlled negative derivative. Fix a nonnegative smooth probability density ρ supported in $[1, 2]$, positive in $(1, 2)$, and vanishing to infinite order at its endpoints. For example, normalize

$$\rho(r) = \begin{cases} c_\rho \exp(-1/((r-1)(2-r))), & 1 < r < 2, \\ 0, & r \notin (1, 2), \end{cases} \quad \int_1^2 \rho(r) dr = 1.$$

Write $\mu = \int_1^2 r\rho(r) dr$, so $1 < \mu < 2$. For $D_g > 0$ set

$$h(y) = D_g(1+y)^{-1/4}, \quad g(y) = \int_1^2 \max\{y, rh(y)\}\rho(r) dr, \quad y \geq 0. \tag{2.4}$$

Here and below smoothness at $y = 0$ means smooth extendibility to a neighborhood of $[0, \infty)$. To verify smoothness, put

$$M(z) = \int_1^2 \max\{z, r\}\rho(r) dr.$$

Then $M'' = \rho$, with ρ extended by zero, and $g(y) = h(y)M(y/h(y))$. Thus g is smooth; moreover $M(z) = \mu$ for $z \leq 1$ and $M(z) = z$ for $z \geq 2$. Consequently $g > 0$, $g \geq y$, and $g(y) = y$ for all sufficiently large y .

For a direct derivative estimate, let

$$L(z) = \int_{r \leq z} \rho(r) dr, \quad R(z) = \int_{r \geq z} r \rho(r) dr.$$

At $z = y/h(y)$ the identities

$$g = yL(z) + hR(z), \quad g' = L(z) - \frac{hR(z)}{4(1+y)}$$

give

$$g \geq y, \quad -\frac{g}{4(1+y)} \leq g' \leq 1, \quad A(y) = g(y) + 2yg'(y) \geq \frac{y+2}{2(y+1)}g(y) \geq \frac{1}{2}g(y). \quad (2.5)$$

In particular the lower bound for A/g is numerical and does not deteriorate as D_g increases.

We now specify the order of all choices. Choose

$$k \geq \max\{A_0, \varepsilon^{-1}, \pi/2\}, \quad Y \geq 1 \quad \text{with} \quad Y^{-3} \leq \frac{b_0}{12k}. \quad (2.6)$$

Choose D_g only after k, Y , large enough that

$$D_g \geq \max \left\{ Y(1+Y)^{1/4}, \frac{8kP_0(1+Y)^{5/4}}{\mu} \right\}. \quad (2.7)$$

Since $y(1+y)^{1/4}$ is increasing, $y/h(y) \leq 1$ on $[0, Y]$. Hence throughout this interval

$$g(y) = \mu D_g (1+y)^{-1/4}, \quad g'(y) = -\frac{\mu D_g}{4(1+y)^{5/4}} \leq -2kP_0. \quad (2.8)$$

The equality cases at the endpoints are covered by the smooth matching already proved.

With this g fixed, define the finite constants

$$g_0 = \min_{0 \leq y \leq Y} g(y) > 0, \quad G_0 = \max_{0 \leq y \leq Y} g(y), \quad B = \sup_{y > Y} \frac{g(y)}{y} < \infty.$$

For example,

$$1 \leq B \leq \max \left\{ 1, \frac{2D_g}{Y(1+Y)^{1/4}} \right\},$$

because $g(y) \leq \max\{y, 2D_g(1+y)^{-1/4}\}$. Finally choose

$$T \geq \max \left\{ 1, \frac{6}{\varepsilon}, \frac{12kB^2}{b_0}, YG_0^2, \frac{12kY}{\varepsilon g_0} \right\}. \quad (2.9)$$

Thus the order is k , then Y , then D_g and g , and then T ; none of the preceding choices depends on a later one.

We now invert the map X in Equation (2.2). By Equation (2.5), $X_f > 0$. As $f \downarrow 0$, the argument y tends to infinity, where $g(y) = y$, and $X(f, t) \rightarrow 0$. As $f \rightarrow \infty$, $y \rightarrow 0$, $g(y) \rightarrow g(0) > 0$, and

$X(f, t) \rightarrow S_0 := \pi/(2k) \leq 1$. Thus for every $t > 0$, $X(\cdot, t)$ maps $(0, \infty)$ bijectively onto $(0, S_0)$ with positive derivative. Its inverse $f = F(x, t)$ is smooth by the implicit function theorem.

On a neighborhood of $f = 0$ the exact identity $g(y) = y$ gives

$$X(f, t) = \frac{1}{k} \arctan(f/t), \quad F(x, t) = t \tan(kx).$$

Define $F(0, t) = 0$ and extend F oddly to $-S_0 < x < 0$. The displayed formula proves joint smoothness across $x = 0$, including at the endpoint $t = T$ by smooth extension to $t > 0$. The inverse construction also proves the asserted pole limits. For uniformity, if $I \subset [T, \infty)$ is compact and $H > 0$, then $\max_{t \in I} X(H, t) < S_0$. Whenever x exceeds this maximum, monotonicity gives $F(x, t) > H$ for every $t \in I$. The negative pole follows by oddness.

It remains to prove Equation (2.1). For $x > 0$ set $f = F(x, t)$ and use the variables in Equation (2.2). Implicit differentiation of $X(F(x, t), t) = x$ yields

$$F_x = \frac{1}{X_f}, \quad F_t = -\frac{X_t}{X_f}, \quad \frac{FF_t}{F_x} = \frac{g'}{k(1 + g^2/q)}, \quad \frac{F_t}{F} = \frac{3yg'}{t(g + 2yg')}. \quad (2.10)$$

In particular the sign in the last identity is the sign of g' . Since $A \leq g + 2y \leq 3g$ and $y \leq g$, we have

$$tX_f = \frac{yA}{3k(q + g^2)} \leq \frac{1}{k}. \quad (2.11)$$

This proves $F_x \geq kt \geq A_0t$ without a constant depending on g . Also $|yg'| \leq g$: for $g' \geq 0$ this follows from $g' \leq 1$ and $y \leq g$, whereas for $g' < 0$ the lower derivative bound in Equation (2.5) even gives $|yg'| \leq g/4$. Therefore

$$\left| \frac{F_t}{F} \right| \leq \frac{6}{t} \leq \varepsilon, \quad \frac{FF_t}{F_x} \leq \frac{1}{k} \leq \varepsilon.$$

For the second inequality a negative numerator is already favorable; for a nonnegative numerator use $g' \leq 1$.

Consider first $y > Y$. Using $A \geq g/2 \geq y/2$ and $g/y \leq B$ gives

$$\begin{aligned} t^2 X_f &= \frac{y^2 A}{3k(1 + yg^2/t)} \\ &\geq \frac{y^3}{6k(1 + B^2 y^3/T)} = \frac{1}{6k(y^{-3} + B^2/T)} \geq \frac{1}{6k(Y^{-3} + B^2/T)} \geq \frac{1}{b_0}. \end{aligned} \quad (2.12)$$

The last inequality uses Equations (2.6) and (2.9). This bound remains uniform as $y \rightarrow \infty$, and it proves $F_x \leq b_0 t^2$ throughout this region.

If $0 < y \leq Y$, then $q = t/y \geq T/Y$ and

$$\frac{g^2}{q} \leq \frac{YG_0^2}{T} \leq 1.$$

It follows from Equations (2.3) and (2.8) that

$$-\frac{FF_t}{F_x} = fX_t = \frac{-g'}{k(1 + g^2/q)} \geq P_0. \quad (2.13)$$

The two regions therefore establish the alternative in Equation (2.1). In this second region we also have

$$F^2 X_f = \frac{qA}{3k(1 + g^2/q)} \geq \frac{qg_0}{12k} \geq \frac{g_0 T}{12kY} \geq \frac{1}{\varepsilon}, \quad (2.14)$$

so $F_x \leq \varepsilon F^2$. Together with the small- F_x bound for $y > Y$ this proves the final inequality of Equation (2.1).

Oddness preserves all five inequalities on $x < 0$, since F_x , F^2 , and FF_t are even and $|F_t|, |F|$ are even. At $x = 0$ they follow by continuity; more explicitly, $F = F_t = 0$ and $F_x = kt$, while $T \geq 12kB^2/b_0 \geq k/b_0$ gives the small- F_x branch there. All claims follow. $\square$

Remark 2.2. When $A_0, P_0, \varepsilon, b_0$ are prescribed using only the dimension and fixed numerical constants, the same is true of S_0, T and the entire pressure law. In particular, the small parameters denoted by a and d in the local deformation may be chosen after this theorem has been applied, without modifying the pressure or increasing T .

3 A compactly supported stretching step

For a metric h and a smooth function t , put

$$\mathcal{D}(h, t) = \text{Scal}_h + 2 \Delta_h t - |dt|_h^2.$$

The next theorem enlarges the distance from a compact set while preserving a quadratic reserve in this quantity. Its error has two parts: a trace decrement, which telescopes under successive metric enlargements, and an arbitrarily small quadratic loss.

Theorem 3.1 (Local stretching). *Fix $n \geq 2$ and $0 < \delta < 1/10$, and put $L_* = 1.004$. There are constants $W, T \geq 1$, depending only on n, δ , with the following property. Let h_0 be a complete smooth metric on a manifold M^n without boundary, let $h \geq h_0$ be smooth, and let $t \geq T$ be smooth. Let A be a nonempty compact set and B a nonempty closed set such that*

$$\text{dist}_{h_0}(A, B) \geq W.$$

For every $0 < d < \delta$ there are smooth $h' \geq h$ and $t' \geq t$ such that the changes have compact support in $\{\text{dist}_{h_0}(A, \cdot) < W\}$,

$$\mathcal{D}(h', t') \geq \mathcal{D}(h, t) + (t'^2 - t^2) - \delta(\text{tr}_h h_0 - \text{tr}_{h'} h_0)t'^2 - dt'^2, \quad (3.1)$$

and

$$\text{dist}_{h'}(A, B) \geq L_* W. \quad (3.2)$$

In particular, h' is complete. Neither W nor T depends on d , the geometry of M , or the sets A, B .

The proof uses a modified graph equation. Its modification near zero slope supplies an adjustable small error, whereas at moderate and large slopes it is the ordinary graph equation. A positive floor in the spatial cutoff permits a closed-manifold existence argument. We subsequently send that floor to zero only on a fixed annulus, where the solution decays exponentially. All geometric estimates used for these two operations take place on one compact auxiliary manifold.

3.1 The slope switch and the order of constants

We first specify the deformation associated with a provisional smooth height u ; the height itself will be chosen below. Write $s = |du|_h$. Fix $s_0 = 10^{-2}$ and a smooth nondecreasing function $\rho : [0, \infty) \rightarrow [0, 1]$ such that

$$\rho = 0 \text{ on } [0, s_0], \quad \rho = 1 \text{ on } [5s_0, \infty), \quad 0 \leq \rho' \leq 60.$$

For $0 < a \leq 1/2$ and $s \geq 0$ define

$$\mu = a^2 + (1 - a^2)\rho(s), \quad \theta = \frac{\mu s^2}{1 + s^2}, \quad l = (1 - \theta)^{1/2}, \quad \beta = \log l. \quad (3.3)$$

For $s > 0$ also put

$$\chi = \left(1 + \frac{a^2 s_0^2}{\theta}\right)^{1/2}, \quad m = \frac{d \log \chi}{d \log \sqrt{\theta}} = -\frac{a^2 s_0^2}{\theta + a^2 s_0^2}. \quad (3.4)$$

Here θ is strictly increasing. The following bounds will be used repeatedly:

$$\begin{aligned} 0 \leq -s\beta_s \leq 2, \quad -1 \leq m \leq 0, \quad 1 \leq \chi \leq 2 \quad (s \geq s_0), \\ m \leq -\frac{1}{2}, \quad \chi \geq s_0/s, \quad \theta\chi^2 \leq 2a^2 s_0^2 \quad (0 < s \leq s_0). \end{aligned} \quad (3.5)$$

Indeed, on $s \leq 5s_0$,

$$-s\beta_s = \frac{s\theta_s}{2(1-\theta)} \leq 30s^3 + \frac{s^2}{1+s^2} < 2,$$

and on $s \geq 5s_0$ it equals $s^2/(1+s^2)$. The remaining bounds follow directly from $\theta \geq a^2 s^2/(1+s^2)$ and the formula with $\mu = a^2$ on $s \leq s_0$.

For $t \geq 1$ we also have

$$\frac{(t - \beta)^2 - t^2}{\theta} \leq 4(1 + s)t \quad (s > 0). \quad (3.6)$$

To check this, write $b_* = -\beta$. If $\theta \leq 1/2$, then $b_* \leq \theta$, and the left side is at most $3t$. Otherwise use $b_* \leq \frac{1}{2} \log(1 + s^2) \leq \log(1 + s) \leq \sqrt{s}$ to bound it by $4t\sqrt{s} + 2s \leq 4(1 + s)t$.

For the provisional height u , define the raw metric and drift by

$$h_u = h + \frac{\theta}{(1 - \theta)s^2} du^2 = h + \frac{\mu}{1 + (1 - \mu)s^2} du^2, \quad t_u = t - \beta. \quad (3.7)$$

These expressions are smooth also at $s = 0$, and $h_u \geq h$, $t_u \geq t$. To see explicitly the smoothness needed at zero slope, note that near $s = 0$ the coefficient of du^2 is $a^2/[1 + (1 - a^2)s^2]$, while the drift increment is $-\frac{1}{2} \log([1 + (1 - a^2)s^2]/[1 + s^2])$. Both are smooth functions of the first jet of u ; therefore C^3 convergence of u to zero gives C^2 convergence of these metric and drift increments to zero. For $s \geq 5s_0$, $h_u = h + du^2$. At points of nonzero slope in the original manifold, let $e = \text{grad}_h u/s$. With the given background metric h_0 , the rank-one inverse formula gives

$$\text{tr}_h h_0 - \text{tr}_{h_u} h_0 = \theta b^2, \quad b^2 = h_0(e, e) \leq 1. \quad (3.8)$$

We now fix all constants that are required to be independent of the local geometry and of d . Let

$$A_0 = 1000, \quad P_0 = 20, \quad C_1 = s_0^{-1}, \quad b_0 = \frac{\delta s_0^2}{1000}, \quad 0 < \varepsilon \leq \min \left\{ \frac{s_0}{100}, \frac{1}{8n}, \frac{1}{8C_1} \right\}. \quad (3.9)$$

Apply Theorem 2.1 with these constants, obtaining F, S_0, T . In particular $S_0 \leq 1$ and $T \geq 1$. Put $C_2 = 2C_1 b_0$. Only after a loss d has been specified do we choose a so that

$$0 < a \leq \frac{1}{2}, \quad (2C_2 + 3)a^2 \leq d/2. \quad (3.10)$$

The width W will be a sufficiently large numerical constant; it may, and will, be at least 10^5 . It is chosen to satisfy the explicit cutoff and distance inequalities below. Thus the choices of W, T precede d, a and all local geometric choices.

3.2 Cutoffs and a closed auxiliary manifold

Write $r_A(x) = \text{dist}_{h_0}(A, x)$. There exist smooth functions $\psi \geq 0$ and $0 \leq \eta_1, \zeta \leq 1$ such that

$$\begin{aligned} \inf_A \psi &> W/5, & |d\psi|_{h_0} &\leq 1, & \text{supp } \psi &\subset \{r_A < W/2\}, \\ \eta_1 &= 1 \text{ on } \{r_A \leq .55W\}, & \eta_1 &= 0 \text{ on } \{r_A \geq .8W\}, & |d\eta_1|_{h_0} &\leq s_0/8, \\ \zeta &= 1 \text{ on } \{r_A \leq .82W\}, & \zeta &= 0 \text{ on } \{r_A \geq .98W\}. \end{aligned} \quad (3.11)$$

Choose also a fixed compact annulus $\mathcal{A}_\zeta$ in $\{.8W < r_A < W\}$ containing $\overline{\{0 < \zeta < 1\}}$. All cutoff error estimates below are imposed on this whole annulus, including any intermediate plateau of ζ .

Here no injectivity-radius or curvature bound is needed. For clarity, a smooth approximation to a Lipschitz function on the compact neighborhood in question can be made in finitely many coordinate charts. Shrink each chart until its metric distortion is as small as desired, convolve there, and combine with a smooth partition of unity. The gradient of the combination is the convex combination of the local gradients plus terms bounded by the chosen uniform approximation error times the derivatives of the partition. Choose the local errors small enough to make the latter sum arbitrarily small. One may preserve prescribed constant regions by using charts lying inside them and leaving a buffer between each such region and the transition. Apply this construction to $(W/3 - r_A)_+$, with value error less than $W/100$ and gradient bound 1.05, then divide by 1.05 and use the zero-region buffer to obtain ψ . Smooth cutoff profiles of r_A/W give the other two functions; for η_1 the gradient can be bounded by $12/W < s_0/8$. This proves Equation (3.11) after increasing W once.

For $0 < r \leq 1/2$ define the positive-floor cutoff

$$\eta = r + (1 - r)\eta_1^2. \quad (3.12)$$

Then $\eta = 1$ near $\text{supp } \psi$, $\eta = r$ on $\{r_A \geq .8W\}$, and, uniformly in r ,

$$0 < \eta \leq 1, \quad |d\eta|_h \leq s_0/4, \quad \frac{|d\eta|_h^2}{\eta} \leq 1. \quad (3.13)$$

The same statements hold with h_0 in place of h ; this follows from $h \geq h_0$ and $|d\eta|^2/\eta \leq 4|d\eta_1|^2$.

Completeness of h_0 and compactness of A make the closed neighborhoods used above compact. Enclose a slightly larger neighborhood, say $\{r_A \leq 1.1W\}$, in a smooth relatively compact domain. Outside this neighborhood, modify h and t near the boundary of the domain to be products, and double the domain. Convex interpolation for t preserves $t \geq T$, and interpolation of positive definite tensors gives the metric extension. We thereby obtain a closed auxiliary manifold N on which $h, t, \psi, \eta_1, \zeta$ are smooth, with the last three functions extended by zero where appropriate. The extensions of h, t are fixed independently of r . If the relevant component of M is already compact, it can simply be used as N . The construction can be made on finitely many components if A meets more than one. From now until the final cutoff, we work on N . Constants allowed to depend on this compact geometry and on a will be denoted by C ; they will never enter W, T .

3.3 The modified graph equation

For a covector p with $s = |p|_h$, set

$$\mathcal{A}(p) = \frac{\sqrt{\mu}}{\sqrt{1 + s^2}} p^\sharp, \quad J(s) = \sqrt{\theta + a^2 s_0^2}.$$

The symbols $\mu, \theta, l, \beta, \chi, m$ will always be evaluated at $s = |du|_h$ when a function u is under discussion. The maps $\mathcal{A}, J$ are smooth at $p = 0$: there $\mu = a^2$ is constant. The tangential and radial eigenvalues of $\mathcal{A}_p$ are $\sqrt{\theta}/s$ and $(\sqrt{\theta})_s$, respectively; both are positive, and both have limit a at the origin.

For $\tau \in [0, 1]$ consider

$$\operatorname{div}_h \mathcal{A}(du) - dt(\mathcal{A}(du)) = J(|du|_h) F\left(\frac{u - \tau\psi}{\eta}, t\right). \quad (3.14)$$

The pressure is evaluated only where its first argument lies in $(-S_0, S_0)$. We first prove existence with bounds independent of r .

Lemma 3.2 (Height and gradient bounds). *For each fixed $a > 0$ as above and every $0 < r \leq 1/2$, Equation (3.14) at $\tau = 1$ has a smooth solution u on N . There are $S_1 < S_0$ and $G < \infty$, independent of r and of $\tau \in [0, 1]$, such that every solution along the continuation satisfies*

$$|u - \tau\psi| \leq S_1\eta, \quad |du|_h \leq G. \quad (3.15)$$

The constants S_1, G may depend on a and on the compact data.

Proof. For $S < S_0$ the functions $u_{\pm} = \tau\psi \pm S\eta$ have uniformly bounded first and second derivatives, independently of r, τ, S . Consequently the left side of Equation (3.14) at these functions is bounded. Since $J \geq as_0 > 0$ and $t(N)$ is compact, the uniform pole limits of F permit one choice $S_1 < S_0$ for which u_+ has strictly negative residual (left side minus right side) and u_- has strictly positive residual. Comparison at a maximum or minimum, using strict monotonicity in u , gives the height estimate.

One can make this argument before a solution has been constructed by extending F smoothly to all first arguments with $F_x > 0$, retaining its original values on a slightly larger interval than $[-S_1, S_1]$. For example, extend the positive even function F_x by a smooth cutoff to a positive constant and integrate from zero. Solve this globally defined equation. The same barriers force every solution into $[-S_1, S_1]$, so that the extension is never used in the resulting solution.

We give the gradient estimate, including the dependence of the pressure on the graph angle. If $\max_N |du|_h \leq 1$ there is nothing to prove. Otherwise, near a point of maximum slope, $\mu = 1$ and the graph in $(N, h) \times \mathbb{R}$ is an ordinary graph. Write

$$w = (1 + |du|_h^2)^{-1/2}, \quad \nu = (N_u, -w), \quad N_u = w \operatorname{grad}_h u,$$

for its downward unit normal. Denote its second fundamental form and mean curvature by Π and H , with $H = \operatorname{div} \nu$. Extend t constantly in the vertical direction and put $X = -\operatorname{grad} t$ and $K = -\partial_z$. The equation is

$$H + \langle X, \nu \rangle = p(x, z, w), \quad p(x, z, w) = \sqrt{1 - w^2 + a^2 s_0^2} F\left(\frac{z - \tau\psi(x)}{\eta(x)}, t(x)\right).$$

Let $K^\top = K - w\nu$. Under a normal variation with speed v ,

$$\begin{aligned} \dot{H} &= -\Delta v - (|\Pi|^2 + \operatorname{Ric}(\nu, \nu))v, \\ \frac{d}{dq} \langle X, \nu \rangle &= -\operatorname{Hess} t(\nu, \nu)v - \langle X^\top, \operatorname{grad} v \rangle, \\ \dot{w} &= -\langle K^\top, \operatorname{grad} v \rangle. \end{aligned}$$

Thus the linearization of the equation, with all terms moved to the left, is $-\mathcal{L}v$, where

$$\mathcal{L} = \Delta + \langle X^\top - p_\omega K^\top, \text{grad} \cdot \rangle + |\text{II}|^2 + \text{Ric}(\nu, \nu) + \text{Hess } t(\nu, \nu) + \partial_\nu^{\text{fixed}} \omega p. \quad (3.16)$$

The last derivative differentiates the base and height variables while holding the angle fixed. Translation in direction K leaves the angle, H , and $\langle X, \nu \rangle$ unchanged and differentiates the residual by p_z . Its normal speed is w ; its tangential component differentiates an identically zero residual. We obtain the exact identity

$$\mathcal{L}w = -p_z. \quad (3.17)$$

In particular the term $-p_\omega K^\top$ in the drift has not been discarded.

Put $\xi = (u - \tau\psi)/\eta$. On the graph,

$$p_z = \frac{JF_x}{\eta}, \quad \partial_\nu^{\text{fixed}} \omega p = J \left\{ \frac{F_x}{\eta} [-w - \tau d\psi(N_u) - \xi d\eta(N_u)] + F_t dt(N_u) \right\}.$$

The height bound puts F, F_x, F_t in a fixed compact range. Moreover $|N_u| \leq 1$, $|d\psi|_h \leq 1$, and $|d\eta|_h \leq s_0/4$. The zero-order coefficient in Equation (3.16), after omitting the nonnegative term $|\text{II}|^2$, is therefore at least $-C/\eta$, with C independent of r, τ . At the minimum of w under consideration, the slope is at least one, so $J \geq 1/\sqrt{2}$ and $p_z \geq A_0 T/(\sqrt{2}\eta)$. Since $\text{grad } w = 0$ and $\Delta w \geq 0$, Equation (3.17) implies

$$\frac{A_0 T}{\sqrt{2}\eta} \leq \frac{C}{\eta} w.$$

This gives a positive lower bound for w , and hence the asserted G .

We finish the fixed- r continuation argument in enough detail to specify its regularity inputs. A classical $C^{2,\alpha}$ solution is smooth before we apply estimates requiring smoothness. Indeed, writing its equation as $a^{ij}u_{ij} = f$, the coefficient functions a^{ij} and f are $C^{1,\alpha}$. On nested coordinate balls, a difference quotient u_h satisfies

$$a^{ij}(x)\partial_{ij}u_h = f_h - (a^{ij})_h\partial_{ij}u(x+h).$$

For each nonzero h , u_h is already $C^{2,\alpha}$; its right side has a $C^{0,\alpha}$ bound independent of h . The classical interior Schauder estimate and passage to $h = 0$ give $u \in C^{3,\alpha}$, and iteration gives smoothness. These qualitative estimates may depend on the existing $C^{2,\alpha}$ norm; the following a priori estimates do not. At $\tau = 0$, $u = 0$ solves the equation. For fixed $r > 0$, the height and gradient bounds put (x, u, du) in a compact set on which the principal coefficients are uniformly elliptic and all explicit coefficient derivatives are bounded. In a coordinate chart write the divergence-form equation as

$$\partial_i \mathfrak{A}^i(x, Du) = \mathfrak{B}(x, u, Du).$$

Differentiating with $v = \partial_k u$ gives

$$\partial_i(\mathfrak{A}_{p_j}^i \partial_j v) - \mathfrak{B}_{p_j} \partial_j v = \mathfrak{B}_u v + \mathfrak{B}_{x_k} - \partial_i \mathfrak{A}_{x_k}^i. \quad (3.18)$$

The x derivatives on the right are explicit derivatives of the coefficient functions. All nondifferentiated terms and the vector field inside the last divergence are bounded using only the compact jet range; no second derivative estimate has been assumed. The local Hölder estimate for uniformly elliptic divergence-form equations, with bounded lower-order coefficients and bounded divergence data, now gives a uniform $C^{1,\alpha}$ bound for u on a finite cover. The equation in nondivergence form then has Hölder coefficients and right side, so local Schauder estimates give $C^{2,\alpha}$ bounds, followed by bounds of every order. The Hölder and interior Schauder estimates used here are respectively Simon [37,

Lecture 18, Theorem 3, and Lecture 12, Theorem 2]. All their uses are on compact coordinate balls with a positive ellipticity constant; their constants may depend on fixed r .

Fix the floor $r > 0$, and choose $0 < \alpha < 1$ smaller, if necessary, than the exponent in the preceding Hölder estimate. The linearization in u has strictly negative zero-order coefficient $-JF_x/\eta$; angle and gradient derivatives affect only its principal and first-order terms. We verify that this linearization is an isomorphism from $C^{2,\alpha}(N)$ to $C^{0,\alpha}(N)$.

First consider $\Delta_h - 1$. For $f \in L^2(N)$, minimize on $H^1(N)$ the strictly convex coercive functional

$$\frac{1}{2} \int_N (|dv|_h^2 + v^2) \, d \operatorname{vol}_h + \int_N f v \, d \operatorname{vol}_h.$$

Its Euler equation is $(\Delta_h - 1)v = f$ in the weak sense. Testing the homogeneous difference with itself proves uniqueness among H^1 weak solutions. For $f \in C^{0,\alpha}(N)$ choose smooth f_j converging uniformly to f , with uniformly bounded $C^{0,\alpha}$ norms, using coordinate mollification and a finite partition of unity. Weak interior regularity makes the corresponding weak solutions v_j smooth [37, Lecture 6, Theorem 1 and Corollary 1]. The maximum principle gives $\|v_j\|_{C^0} \leq \|f_j\|_{C^0}$. A finite interior cover and the Schauder estimate [37, Lecture 12, Theorem 2] then bound v_j uniformly in $C^{2,\alpha}(N)$. Passing to a subsequence in $C^{2,\alpha'}(N)$, where $0 < \alpha' < \alpha$, gives a solution for f . The common α -Hölder bound for the second derivatives passes to the limit, so this solution belongs to $C^{2,\alpha}(N)$. Weak uniqueness identifies it with the energy solution, and the maximum principle also gives uniqueness in the classical class.

Let L_q , $0 \leq q \leq 1$, interpolate the coefficients of $\Delta_h - 1$ and the linearization written in nondivergence form. Uniform ellipticity and a uniform strictly negative zero-order bound persist along this path. The maximum principle and the same finite-cover Schauder estimate give

$$\|v\|_{C^{2,\alpha}(N)} \leq C \|L_q v\|_{C^{0,\alpha}(N)}, \quad 0 \leq q \leq 1.$$

Invertibility is open by the Neumann series. To see closedness, suppose $q_j \rightarrow q$ through invertible parameters and, for a fixed $f \in C^{0,\alpha}(N)$, put $v_j = L_{q_j}^{-1} f$. The estimate above bounds v_j and gives

$$\|v_j - v_k\|_{C^{2,\alpha}} \leq C \|(L_{q_k} - L_{q_j})v_k\|_{C^{0,\alpha}} \rightarrow 0.$$

Thus v_j converges in $C^{2,\alpha}$ to a solution of $L_q v = f$; injectivity follows from the same estimate. This proves the required linearized isomorphism.

The implicit function theorem now gives openness of the nonlinear continuation in τ . The preceding difference-quotient bootstrap applies to the $C^{2,\alpha}$ solutions obtained by implicit inversion. The uniform a priori $C^{2,\alpha}$ bound makes its higher derivative bounds uniform along continuation at this fixed positive floor. In particular these bounds give compactness in $C^{2,\alpha}(N)$, so a convergent sequence of soluble parameters has a solution at its limit. The height bound keeps the pressure argument in its fixed compact interval. The soluble set is therefore closed as well as open and contains 0, hence equals $[0, 1]$. The local bootstrap makes the resulting solution smooth. $\square$

3.4 Decay on the region of constant floor

Lemma 3.3 (Locality). *Let u_r be the solutions from Theorem 3.2 at $\tau = 1$. On any compact subset of the fixed open region where $\psi = 0$ and $\eta_1 = 0$, the functions u_r tend to zero in every C^k norm as $r \downarrow 0$.*

Proof. Here $\eta = r$ and $|u_r/r| \leq S_1$. Expanding the left side of Equation (3.14) and using $F(0, t) = 0$ writes its equation as

$$a^{ij}(x) \partial_{ij} u_r + b^i(x) \partial_i u_r - \frac{c_r(x)}{r} u_r = 0, \quad c_r = J \int_0^1 F_x(qu_r/r, t) \, dq \geq as_0 A_0 T > 0. \quad (3.19)$$

The coefficients a^{ij} are uniformly elliptic and bounded, and b^i are bounded, independently of r . Indeed they are smooth functions of (x, Du_r) on the fixed compact jet range from Equation (3.15); the explicit x part of the flux vanishes at zero covector and is therefore a bounded first-order term. Also c_r is bounded above independently of r . No derivatives of these coefficients are needed for the next comparison.

Take a coordinate box compactly contained in the constant-floor region. Let ℓ_f be its inward affine distance from face f in these coordinates and set

$$V_r = M_0 \sum_f \exp(-\gamma \ell_f / \sqrt{r}),$$

where M_0 dominates $|u_r|$ on the boundary for every r . Let Λ bound $a^{ij}(\partial_i \ell_f)(\partial_j \ell_f)$, let $c_* > 0$ be the lower bound for c_r , and let B_* bound the first-order contributions. The operator in Equation (3.19) applied to each summand of V_r is at most that summand times

$$\frac{\Lambda \gamma^2 - c_*}{r} + \frac{B_* \gamma}{\sqrt{r}}.$$

Choose $\gamma > 0$ with $\Lambda \gamma^2 < c_*/2$. For small r this quantity is negative. Comparison with both u_r and $-u_r$ proves, on each smaller box, $|u_r| \leq C \exp(-c/\sqrt{r})$. A finite covering gives this estimate on any fixed compact subset.

To upgrade the estimate to derivatives, rescale a coordinate ball of radius comparable to r by

$$x = x_0 + ry, \quad u_r(x) = rv(y).$$

The equation becomes

$$\partial_{y_i} \mathfrak{A}^i(x_0 + ry, Dv) = r \mathfrak{B}_0(x_0 + ry, v, Dv), \quad (3.20)$$

where $\mathfrak{B}_0$ includes the drift term and $JF(v, t)$ and has no negative power of r . We have $|v| \leq S_1$ and $|Dv| \leq G$. The principal ellipticity is uniform. Explicit y derivatives of $\mathfrak{A}$ have a factor r , explicit y derivatives of $r \mathfrak{B}_0$ have a factor r^2 , and its v, p derivatives have a factor r . Thus all explicit coefficient derivatives are bounded uniformly on this fixed jet range. Apply the Hölder estimate to the differentiated Equation (3.18), followed by the Schauder bootstrap on fixed balls in y . For every k this gives

$$|D_y^k v| \leq C_k, \quad |D_x^k u_r| \leq C_k r^{1-k},$$

on smaller balls, with constants uniform in x_0 in a fixed compact set and in small r . Finally, interior interpolation between the exponentially small C^0 bound and the polynomial bound for one higher derivative proves C^k convergence to zero for each fixed k . For example, on fixed nested compact subsets of the constant-floor region, interpolation bounds the third derivatives by a constant times $r^{-9/4} \exp(-c/(4\sqrt{r})) + \exp(-c/\sqrt{r})$, which tends to zero. This last case ensures C^2 convergence of all metric and drift increments in Equation (3.7), which are smooth functions of Du_r . $\square$

3.5 The scalar identity and its two slope regimes

All comparisons involving h_0 in this subsection are made on the original neighborhood $\{r_A < 1.1W\}$, where the auxiliary data agree with the given data. No extension of h_0 to the double is needed.

Apply the construction in Equation (3.7) to the solution u_r from Theorem 3.2 at $\tau = 1$, and write $u = u_r$. We now prove its scalar estimate before applying the final spatial cutoff.

Lemma 3.4 (Raw scalar estimate). *On this original neighborhood, the raw output satisfies*

$$\mathcal{D}(h_u, t_u) \geq \mathcal{D}(h, t) + (t_u^2 - t^2) - \delta(\text{tr}_h h_0 - \text{tr}_{h_u} h_0) t_u^2 - \frac{d}{2} t^2. \quad (3.21)$$

Proof. We derive the scalar comparison explicitly. Near a point with $du \neq 0$ use $q = u$ as the transverse coordinate and write

$$h = v^2 dq^2 + \gamma_q, \quad v = 1/s, \quad e = v^{-1} \partial_q, \quad J = \frac{1}{2v} \partial_q \gamma_q, \quad H = \text{tr } J.$$

Here J is the second fundamental form of the level sets, and all tangential norms and derivatives below use γ_q . Put $S = H - et$. The Gauss equation and normal variation of H give

$$\text{Scal}_h = \text{Scal}_\gamma - |J|^2 - H^2 - 2eH - 2\frac{\Delta_\gamma v}{v}, \quad \Delta_h t = \Delta_\gamma t + e(et) + Het + \langle d_\gamma \log v, d_\gamma t \rangle.$$

Consequently

$$\begin{aligned} \mathcal{D}(h, t) = & -|J|^2 - S^2 - 2eS + \text{Scal}_\gamma + 2\Delta_\gamma t - |d_\gamma t|^2 \\ & - 2\frac{\Delta_\gamma v}{v} + 2\langle d_\gamma \log v, d_\gamma t \rangle. \end{aligned} \quad (3.22)$$

The deformation replaces v by v/l and t by $t - \beta$. The change in the tangential terms on the right is

$$2\langle d_\gamma \log v, d_\gamma \beta \rangle - |d_\gamma \beta|^2 = c_\beta(2 - c_\beta)|d_\gamma \log v|^2 \geq 0, \quad c_\beta = -s\beta_s \in [0, 2].$$

This computation follows by expanding $\Delta_\gamma(v/l)/(v/l)$ and $|d_\gamma(t - \beta)|^2$; the $\Delta_\gamma \beta$ and the terms involving $d_\gamma t$ cancel.

The divergence equation can be written

$$S + e \log \sqrt{\theta} = \chi F(\xi, t), \quad \xi = (u - \psi)/\eta.$$

Define

$$M = \chi F, \quad K = \frac{l^2}{\theta} e\beta.$$

Because $e \log \sqrt{\theta} = -K$, we have $S = M + K$. For the new metric, $e' = le$, $J' = lJ$, and

$$S' = l(S + e\beta) = lM + K/l.$$

Substitution in the normal terms of Equation (3.22) gives

$$\mathcal{D}(h_u, t_u) - \mathcal{D}(h, t) \geq \theta E, \quad E = |J|^2 + M^2 + K^2/l^2 - 2MK + 2eM. \quad (3.23)$$

For completeness, the derivative terms before simplification are $-2le(lM + K/l) + 2e(M + K)$. Their eK terms cancel; using $e\beta = \theta K/l^2$ leaves $2\theta eM - 2\theta MK + 2\theta K^2/l^2$, which combines with the squared terms to yield Equation (3.23).

Since $e\chi = -m\chi K$ and $et = H - M - K$,

$$eM = -mKM + \chi(F_t et + F_x e\xi).$$

Completing the H and K squares gives $E \geq \mathcal{B}$, where

$$\mathcal{B} = \chi^2 \left\{ [1 - l^2(1 + m)^2] F^2 - 2[1 + l^2(1 + m)] F F_t - n F_t^2 \right\} + 2\chi F_x e\xi. \quad (3.24)$$

An exact verification, useful for tracking every sign, is

$$\begin{aligned} E - \mathcal{B} = & \left(|J|^2 - \frac{H^2}{n-1} \right) + \frac{(H + (n-1)\chi F_t)^2}{n-1} \\ & + \left(\frac{K}{l} - l\chi[(1+m)F + F_t] \right)^2 + \theta \chi^2 F_t^2 \geq 0. \end{aligned}$$

All pressure expressions here are evaluated at (ξ, t) .

Suppose first that $s \geq s_0$. Denote the expression in braces in Equation (3.24) by $\mathcal{C}$. Its F^2 coefficient is nonnegative, and $1 \leq 1 + l^2(1 + m) \leq 2$. From Equation (2.1) we get

$$\mathcal{C} \geq -5\varepsilon F_x, \quad -FF_t \geq P_0 F_x \implies \mathcal{C} \geq P_0 F_x. \quad (3.25)$$

Indeed, if $FF_t \geq 0$, then $F_t^2 \leq \varepsilon FF_t$ and $\mathcal{C} \geq -(4 + n\varepsilon)FF_t \geq -5\varepsilon F_x$. If $FF_t < 0$, the same absolute-value estimate gives $\mathcal{C} \geq (2 - n\varepsilon)|FF_t| \geq |FF_t|$.

Where $\eta \neq 1$ we have $\psi = 0$, and

$$e\xi = \frac{s - \xi e\eta}{\eta} \geq \frac{3s}{4\eta} \geq \frac{3s}{4}.$$

Using $1 \leq \chi \leq 2$, Equations (3.9) and (3.25) give $E \geq sF_x$ there. Where $\eta = 1$, we have $e\xi = s - e\psi \geq s - b$, with $b^2 = h_0(e, e)$ as in Equation (3.8), because $|\mathrm{d}\psi|_{h_0} \leq 1$. If $s > 8b$, the same estimates give

$$E \geq (7s/4 - 20\varepsilon)F_x \geq sF_x.$$

It remains to consider $s_0 \leq s \leq 8b$. The possible deficit from sF_x is at most $10F_x$: indeed $2\chi(s - b) - s \geq -4$ and $20\varepsilon < 1$. On the favorable pressure branch, however,

$$E - sF_x \geq [20\chi^2 + 2\chi(s - b) - s]F_x \geq 0,$$

since $s \leq 8b \leq 8$. Thus a deficit is possible only when $F_x \leq b_0 t^2$. In that case $b \geq s_0/8$ and our choice of b_0 gives $10b_0 \leq \delta(s_0/8)^2$. We conclude throughout $s \geq s_0$ that

$$E \geq sF_x - \delta b^2 t^2 \geq 4(1 + s)t - \delta b^2 t^2. \quad (3.26)$$

The last inequality follows from $F_x \geq 1000t$ and $s \geq .01$. Combining Equations (3.6), (3.8) and (3.23) proves the raw inequality in this region, even without its d term.

Now suppose $0 < s < s_0$. Since $m \leq -1/2$, the first coefficient in $\mathcal{C}$ is at least $3/4$. Using $|F_t| \leq \varepsilon|F|$ gives

$$\mathcal{C} \geq (3/4 - 4\varepsilon - n\varepsilon^2)F^2 \geq \frac{1}{2}F^2.$$

Where $\eta = 1$, $e\xi/\chi \geq -1$. Elsewhere, a negative $e\xi$ can occur only if $s \leq |\mathrm{d}\eta|_h$, and then

$$\frac{|e\xi|}{\chi} \leq \frac{|\mathrm{d}\eta|_h s}{s_0 \eta} \leq \frac{|\mathrm{d}\eta|_h^2}{s_0 \eta} \leq C_1.$$

By the quadratic pressure upper bound,

$$E \geq \chi^2 \{(1/2 - 2C_1\varepsilon)F^2 - 2C_1 b_0 t^2\} \geq -C_2 \chi^2 t^2. \quad (3.27)$$

Since $\theta\chi^2 \leq 2a^2 s_0^2$, the loss in Equation (3.23) is at most $2C_2 a^2 t^2$. Also $-\beta \leq \theta \leq a^2 s_0^2$ and $t \geq 1$, so $t_u^2 - t^2 \leq 3a^2 t^2$. The choice in Equation (3.10) pays both terms with $(d/2)t^2$. The trace penalty is favorable and may be omitted in this regime. This proves Equation (3.21) wherever $s > 0$. At a zero of du it follows by continuity unless du vanishes on a neighborhood; on such a neighborhood the raw output equals the input and the inequality is immediate. $\square$

3.6 Cutting off the deformation and obtaining distance

Define on the original manifold

$$h' = h + \zeta(h_u - h), \quad t' = t + \zeta(t_u - t), \quad (3.28)$$

and extend by (h, t) outside the auxiliary neighborhood. The result is smooth, $h' \geq h$, $t' \geq t$, and its changes have compact support in $\{r_A < W\}$. We now choose r small enough to verify both required estimates.

First, the scalar inequality must use the trace of the actual cutoff metric. On the fixed compact annulus $\mathcal{A}_\zeta$, define the residual for any positive definite metric H and function z by

$$\mathfrak{E}(H, z) = \mathcal{D}(H, z) - \mathcal{D}(h, t) - (z^2 - t^2) + \delta(\text{tr}_h h_0 - \text{tr}_H h_0)z^2 + dz^2.$$

This is a continuous function of the C^2 jets of (H, z) on a compact set of positive definite metrics. By Equation (3.21),

$$\mathfrak{E}(h_u, t_u) \geq dt_u^2 - (d/2)t^2 \geq (d/2)T^2.$$

The annulus lies compactly in the fixed region $\psi = 0$, $\eta_1 = 0$. By Theorem 3.3, $u_r \rightarrow 0$ there in C^3 ; hence both the raw increments and their products with ζ tend to zero in C^2 . Choose r so small that

$$|\mathfrak{E}(h', t') - \mathfrak{E}(h_u, t_u)| < dT^2/4$$

on this annulus. Then $\mathfrak{E}(h', t') > 0$ there. Off the annulus the output is either raw or unchanged, and the same desired inequality holds directly. This proves Equation (3.1). In particular, no rank-one trace identity for the uncut metric has been substituted for the trace of the cutoff metric.

It remains to prove the distance gain, including for curves that revisit the cutoff annulus or leave the auxiliary neighborhood. Introduce the globally defined effective height

$$v_{\text{eff}} = \zeta u,$$

extended by zero outside the neighborhood. On A the height barrier and Equation (3.11) give $v_{\text{eff}} > W/5 - 1$, whereas on B it is zero. Decrease r further, if necessary, so that

$$|dv_{\text{eff}}|_h \leq .01 \quad (3.29)$$

throughout $\mathcal{A}_\zeta$; locality justifies this choice. Consider a piecewise smooth curve from A to B of h' length at most L_*W . Its h length is at most L_*W . Partition its parameter interval according to $0 < \zeta < 1$, $\zeta = 1$, and $\zeta = 0$. At points where ζ equals either endpoint, $d\zeta = 0$ because ζ is smooth and valued in $[0, 1]$. The total variation of v_{eff} on $\{0 < \zeta < 1\}$ is at most $.01L_*W$, counting all repeated visits. On the region where $\zeta = 1$ and $s \leq 5s_0 = .05$, its variation is at most $.05L_*W$. Where $\zeta = 0$ it has zero variation. Therefore the variation on

$$\Omega_{\text{graph}} = \{\zeta = 1, s \geq .05\}$$

is greater than

$$(1/5 - .06L_*)W - 1 > .11W.$$

The strict last inequality holds for our fixed sufficiently large W . On this region $h' = h + du^2$, while everywhere $h' \geq h$. If γ is parametrized on an interval I , the integral triangle inequality in $\mathbb{R}^2$ yields

$$\begin{aligned} \text{Length}_{h'}(\gamma) &\geq \int_I \sqrt{|\dot{\gamma}|_h^2 + \mathbf{1}_{\Omega_{\text{graph}}} |(u \circ \gamma)'|^2} dq \\ &\geq \left\{ \text{Length}_h(\gamma)^2 + \left(\int_{\gamma^{-1}(\Omega_{\text{graph}})} |(u \circ \gamma)'| dq \right)^2 \right\}^{1/2} \\ &> W\sqrt{1 + .11^2} > L_*W. \end{aligned}$$

Here $\text{Length}_h(\gamma) \geq W$ follows from $h \geq h_0$ and the original separation of A, B . This contradiction proves Equation (3.2); curves with infinite length are irrelevant. Finally, a smooth metric dominating a complete metric is complete: an h' -Cauchy sequence is h_0 -Cauchy, and local equivalence of the two smooth metrics supplies convergence in h' . The proof of Theorem 3.1 is complete.

4 Wide separations at a fixed original scale

The local estimate pays for a metric change by a decrease in a trace. That decrease can be summed over arbitrarily many labels. We first carry out this summation, then explain why different labels may require different finite numbers of rounds.

Fix $L_* = 1.004$, and choose $\delta > 0$, depending only on n , so small that

$$2(n+1)\delta < 1 - L_*^{-2}. \quad (4.1)$$

Let W, T be the constants in Theorem 3.1. Enlarge T if necessary to make $T \geq 1$, and choose

$$R_0 \geq \sqrt{2}T, \quad D = WR_0. \quad (4.2)$$

All of these constants depend only on n .

Lemma 4.1 (A complete round). *Suppose that h_0 is a complete metric, $t_0 \geq T$ is smooth, and*

$$\mathcal{D}(h_0, t_0) \geq c + t_0^2$$

for a constant c . Let (A_i, B_i) be a finite or countable family of nonempty closed pairs, with each A_i compact and $\text{dist}_{h_0}(A_i, B_i) \geq W$. Suppose there are locally finite compact sets K_i such that the open h_0 -distance- W neighborhood of A_i is contained in $K_i \setminus B_i$. Then there are smooth $h_1 \geq h_0$ and $t_1 \geq t_0$, obtained by a locally finite sequence of compactly supported changes inside the K_i , such that

$$\text{dist}_{h_1}(A_i, B_i) \geq L_*W \quad \text{for every } i, \quad (4.3)$$

$$\mathcal{D}(h_1, t_1) \geq c + (1 - (n+1)\delta)t_1^2. \quad (4.4)$$

The metric h_1 is complete.

Proof. Enumerate the labels and choose $d_i > 0$ with $\sum_i d_i < \delta$. Apply Theorem 3.1 in this order, keeping h_0 as the background metric throughout. Earlier changes increase distances, so every pair still meets the separation hypothesis when it is processed. If $(h^{(j)}, t^{(j)})$ denotes the data after j changes, write

$$Q_j = \text{tr}_{h^{(j)}} h_0, \quad A_j = 1 - \delta(n - Q_j) - \sum_{i \leq j} d_i.$$

Thus $Q_0 = n$, $0 < Q_j \leq n$, and $A_j \leq 1$. We claim that

$$\mathcal{D}(h^{(j)}, t^{(j)}) \geq c + A_j(t^{(j)})^2. \quad (4.5)$$

It holds at $j = 0$. In a single step, abbreviate the old coefficient by A , the new one by A' , and the old and new functions by t, t' . The local estimate gives

$$\begin{aligned} \mathcal{D}(h', t') &\geq c + At^2 + (t'^2 - t^2) - (\delta(Q - Q') + d)t'^2 \\ &= c + A't'^2 + (1 - A)(t'^2 - t^2) \geq c + A't'^2. \end{aligned}$$

The last inequality uses both $A \leq 1$ and $t' \geq t \geq T > 0$. This proves (4.5). Its coefficient is greater than $1 - (n+1)\delta$, which proves (4.4). Each processed separation satisfies (4.3), and later changes preserve it.

The supports lie in the locally finite family (K_i) . Hence every point has a neighborhood on which the sequence is eventually constant; its limit is smooth and satisfies the same inequalities. Completeness follows from $h_1 \geq h_0$: an h_1 -Cauchy sequence converges in h_0 , and near its limit the two smooth metrics are equivalent. $\square$

The passage from a positive scalar-curvature form to circle warps uses torical symmetrization; see Gromov [21, Section 2]. The following identity fixes the finite-dimensional convention used here.

Lemma 4.2 (Scalar curvature of diagonal circle warps). *For smooth functions $p_1, \dots, p_k$ on (M, h) , the metric $\hat{h} = h + \sum_{\alpha=1}^k e^{2p_\alpha} d\theta_\alpha^2$ on $M \times \mathbb{T}^k$ has scalar curvature*

$$\text{Scal}_{\hat{h}} = \text{Scal}_h - 2\Delta_h P - |dP|_h^2 - \sum_{\alpha=1}^k |dp_\alpha|_h^2, \quad P = \sum_{\alpha=1}^k p_\alpha.$$

Proof. Let $E_\alpha = e^{-p_\alpha} \partial_{\theta_\alpha}$ be the unit vertical fields. For horizontal fields X, Y , the Koszul formula gives

$$\hat{\nabla}_X Y = \nabla_X^h Y, \quad \hat{\nabla}_X E_\alpha = 0, \quad \hat{\nabla}_{E_\alpha} X = (Xp_\alpha)E_\alpha, \quad \hat{\nabla}_{E_\alpha} E_\alpha = -\text{grad}_h p_\alpha,$$

and $\hat{\nabla}_{E_\alpha} E_\beta = 0$ for $\alpha \neq \beta$. Therefore, for a horizontal unit vector X , the mixed sectional curvatures are

$$K(X, E_\alpha) = -\text{Hess}_h p_\alpha(X, X) - (Xp_\alpha)^2, \quad K(E_\alpha, E_\beta) = -\langle dp_\alpha, dp_\beta \rangle_h.$$

Summing twice over all orthonormal two-planes gives the identity. $\square$

Theorem 4.3 (Prescribed widths for a locally finite family). *Let (M, g) be complete, connected and smooth, with $\text{Scal}_g \geq 1$. There is a countable set of centers (x_i) such that*

$$A_i = \overline{B}_g(x_i, D), \quad B_i = M \setminus B_g(x_i, 2D)$$

have the following properties: the A_i cover M , and the closed balls $\overline{B}_g(x_i, 2D)$ are locally finite. Suppose that every B_i is nonempty. Given arbitrary finite numbers $L_i > 0$, there exist a complete smooth metric $h \geq g$ and a countable family of smooth functions (p_α) , whose supports form a locally finite family of compact sets, for which

$$\text{dist}_h(A_i, B_i) > L_i \quad (i \geq 1), \tag{4.6}$$

$$\text{Scal}_h - 2\Delta_h \sum_{\alpha} p_\alpha - \left| d \sum_{\alpha} p_\alpha \right|_h^2 - \sum_{\alpha} |dp_\alpha|_h^2 \geq \frac{2}{5}. \tag{4.7}$$

Every sum in (4.7) is locally finite. If some B_i is empty, then $\text{diam}_g M \leq 4D$.

Proof. A complete connected Riemannian manifold is proper. Choose a maximal D -separated set of centers. Maximality gives the covering by closed D -balls. A compact set contains only finitely many centers; more generally, centers whose closed $2D$ -balls meet a fixed compact set lie in a fixed compact neighborhood of that set. This proves local finiteness of the outer balls and countability of the net. The triangle inequality gives $\text{dist}_g(A_i, B_i) \geq D$. The last assertion follows immediately if $M = B_g(x_i, 2D)$.

We keep a scalar reserve of $1/2$ in the original units. Put $R_j = L_*^j R_0$, start with $h^{[0]} = g$, $t^{[0]} = T$, and work in round j with the metric $h^{[j]}/R_j^2$. At the start of that round we require

$$\mathcal{D}(h^{[j]}/R_j^2, t^{[j]}) \geq \frac{R_j^2}{2} + (t^{[j]})^2. \quad (4.8)$$

For $j = 0$, this follows from $\text{Scal}_g \geq 1$ and $R_0^2/2 \geq T^2$.

A label remains active until its separation exceeds its prescribed L_i . At the start of round j , every active pair has separation at least WR_j in $h^{[j]}$; this is initially true and follows inductively from (4.3). Apply Theorem 4.1 to the active pairs in the working metric, with reserve $c = R_j^2/2$. The required support condition follows from $\text{dist}_{h^{[j]}/R_j^2}(A_i, B_i) \geq W$: the open distance- W neighborhood of A_i misses B_i , and is therefore contained in the original outer ball $B_g(x_i, 2D)$. We may take $K_i = \overline{B}_g(x_i, 2D)$.

Return the output metric to original units and denote it by $h^{[j+1]}$. It dominates $h^{[j]}$. Scalar curvature, the Laplacian and squared gradient norms all multiply by R^2 when a metric is divided by R^2 . Consequently, on passing from working scale R_j to R_{j+1} , the round estimate becomes

$$\mathcal{D}(h^{[j+1]}/R_{j+1}^2, t^{[j+1]}) \geq \frac{R_{j+1}^2}{2} + L_*^2(1 - (n+1)\delta)(t^{[j+1]})^2.$$

By (4.1), the coefficient of the final square is greater than one. Thus (4.8) is restored. In original units the reserve remains exactly $1/2$; it has not been spent by the change of scale.

Each fixed label becomes inactive after finitely many rounds, since its separation is at least WR_j as long as it remains active and $R_j \rightarrow \infty$. A compact set meets only finitely many of the original outer balls. Each of their labels is processed only finitely many times. Thus there are only finitely many changes on a neighborhood of any compact set over the *entire* construction, not merely in each round. The limiting data (h, t) are smooth, satisfy

$$h \geq g, \quad t \geq T, \quad \mathcal{D}(h, t) \geq \frac{1}{2},$$

and satisfy (4.6). The same comparison argument as in Theorem 4.1 proves completeness of h .

It remains to realize the drift term by ordinary finite-dimensional warped products on each compact window. Enumerate the nonzero compactly supported increments of t as $\tau_\nu \geq 0$, so that $t - T = \sum_\nu \tau_\nu$. The supports are locally finite. For each ν , choose an integer $m_\nu \geq 1$ such that

$$\frac{1}{m_\nu} \sup_M |d\tau_\nu|_h^2 \leq \frac{2^{-\nu}}{10}.$$

The supremum is finite because the support is compact. Use m_ν fresh log warps, all equal to $-\tau_\nu/m_\nu$. Listing these functions as (p_α) , we obtain

$$\sum_\alpha p_\alpha = -(t - T), \quad \sum_\alpha |dp_\alpha|_h^2 \leq \frac{1}{10}.$$

On a neighborhood meeting only finitely many supports, equip a finite product of circles with metric $h + \sum_\alpha e^{2p_\alpha} d\theta_\alpha^2$, omitting all unused unit factors. Its scalar curvature is the left side of (4.7); this is the identity in Theorem 4.2. Substitution gives $\mathcal{D}(h, t) - \sum_\alpha |dp_\alpha|_h^2 \geq 1/2 - 1/10 = 2/5$. This proves (4.7). $\square$

Remark 4.4. The list of warps is bookkeeping for compatible finite-dimensional calculations. We never construct an infinite-dimensional torus. Adding unit factors changes neither the weight nor the scalar formula. After the radial partition in Theorem 6.6, only finitely many initial warp supports occur on any whole packet; later stages add finite groups of factors. This stronger finiteness is used in the one-dimensional obstruction, not just local finiteness of a series on the original manifold.

5 Finite cuts with compatible ends

The stretching construction supplies separated sets and a scalar-curvature reserve. We now describe the second input: a cutting procedure that retains both sides of every cut and controls how later cuts meet them. Its output is a family of smooth walls and complete manifolds obtained by attaching cylinders to the closed sides. The definitions below make its hypotheses and output explicit; Appendices A–E give the proofs. Section 6 will separately justify its use with countably many labels.

Let $X = M$ denote the original complete manifold, let g be its original metric, and write $d_0 = d_g$ for its distance. The metric used in an analytic packet can change; the map to X always records its position in the original manifold. Fix finitely many of the supplied ball pairs

$$A_i = \overline{B}_{d_0}(x_i, D), \quad B_i = X \setminus B_{d_0}(x_i, 2D), \quad 1 \leq i \leq \ell,$$

with A_i, B_i nonempty. There is one label for each ball pair; these labels are also called *colors*. An additional first label is radial. Thus $N_* = \ell + 1$, and the order is radial, then $1, \dots, \ell$. Properness of (X, g) makes each closed outer ball compact. No covering-space or quotient hypothesis is used.

The analytic input at stage zero is a complete smooth metric $h \geq g$ on X and finitely many smooth positive circle warps for which the stabilized scalar curvature is at least .4. Each prescribed distance $\text{dist}_h(A_i, B_i)$ can be required to exceed any given finite number. Theorem 4.3 supplies these data. For finitely many requested pairs its construction uses only finitely many compactly supported increments and hence finitely many circle factors. The countable version has a locally finite list of nonunit warps and is used in Theorem 6.6; the geometric statements below apply to each finite collection of the relevant factors. No uniform geometry bounds on h , or uniform packetwise constants, are assumed.

For a single smooth cut, let u be a smooth function on an m -manifold E without boundary, with 0 a regular value, and put $L = u^{-1}(0)$. Attach $L \times [0, \infty)$ to each closed side $\{\pm u \geq 0\}$ along its boundary, and also retain L as a separate space. The two extended sides have dimension m , while the wall has dimension $m - 1$. The cylinders replace the new boundaries by ends. The following definition records these spaces and their incidences after repeated cuts.

Definition 5.1 (Packet diagram). At stage r , a status is $\alpha \in \{+, -, *\}^r$. Its entries record the first r processed cuts. A sign $+$ or $-$ selects the corresponding retained side of a cut, and a star $*$ selects the wall, called a *hit*. The space E_α is a smooth second countable manifold without boundary, possibly empty or disconnected; when nonempty its base dimension is $m(\alpha) = n - \#\{i : \alpha_i = *\}$. Its connected components are called *packets*. Replacing signs by stars gives the face order. If B is a packet with d additional stars relative to a coarser status, then for that coarser status it has exactly one incident packet E and one corresponding occurrence of a full product tail

$$\iota_{BE} : B \times (T_1, \infty) \times \cdots \times (T_d, \infty) \longrightarrow E. \quad (5.1)$$

The normal coordinates are labelled by the changed status entries. Thresholds may depend on the incidence. For a single entry, tails from distinct face components are disjoint. Intersections of sufficiently far tails for different entries are exactly the tails over their common further faces. The identifications commute: composing face maps in any order gives the same occurrence and the same individually labelled normal coordinates. After slightly raising the entry thresholds, each closed tail is properly embedded.

Each packet E is equipped with a continuous map $q_E : E \rightarrow X$. After the radial stage we require the following additional properties. Every packet has only finitely many incident faces and cofaces,

including through chains; the complement of its sufficiently far elementary tails is compact. On each sufficiently far face product these maps satisfy $q_E(\iota_{BE}(b, \mathbf{t})) = q_B(b)$. A metric, scalar function, warp, or set is *compatible* if, in every sufficiently far tail, it is exactly the product or pullback of the corresponding face datum. For metrics the normal factors are $dt_1^2 + \dots + dt_d^2$; all warps are independent of the normal coordinates there. The thresholds for different data may differ.

There is no boundary in an analytic packet. Its face is represented by an entire cylindrical end, not by a boundary subset. Finite boundaries appear only when we truncate these ends.

A *regular split* applies the preceding construction using compatible smooth defining functions whose zero sets are regular on every status space. Each zero set is the new hit space. In inherited ends the operation is the corresponding face operation times the old normal factors. The proof that these operations preserve the packet diagram is Theorem A.1.

For the diagrams constructed here, these maps are defined recursively, starting with $q_X = \text{Id}_X$. In the split of Theorem A.1, let A be a new stage- $(r+1)$ packet with stage- r parent E . Define $p_A : A \rightarrow E$ by inclusion on a hit space or the bulk of a joined side, and by $p_A(x, t) = x$ on each newly attached cylinder. Set $q_A = q_E \circ p_A$. Thus q_A is pulled back from the base on each fresh cylinder. In a sufficiently far inherited tail of A , let C be the corresponding new face with old parent B , and let I denote the old normal intervals. Under the identifications of these tails with $C \times I$ and $B \times I$, the map p_A is $p_C \times \text{Id}_I$; explicitly,

$$p_A \circ \iota_{CA} = \iota_{BE} \circ (p_C \times \text{Id}_I), \quad q_A(\iota_{CA}(c, \mathbf{t})) = q_C(c).$$

Thus the recursion preserves compatibility. The proper maps q_r on finite reconstructions are constructed in Theorem A.5.

Figure 1 depicts one extended side and the separate hit space.

The label set and the status depth are finite in this section. The family of connected packets may be countable; finiteness is required only for the directed face and coface chains incident to each packet. The downward diagram of a packet consists of that packet and all its further faces. Numerical requests are allocated downward through these chains before actual data are constructed upward from faces. This does not require finite components of the undirected incidence graph.

We now specify the analytic data that must survive each split. For the finite-label statements in

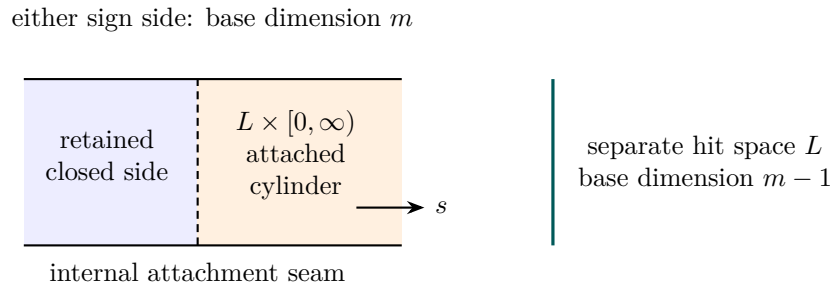


Figure 1: A regular cut retains each closed sign side with a full cylinder attached, and also retains the wall as a separate hit space of one lower base dimension. The attachment seam is internal to the extended side. In old product ends this entire operation is a product. The raw metric and its later smoothing are defined in Appendices B and E.

this section and the appendices we set $\sigma = 1$ and use the following constants:

$$\begin{aligned} \lambda_r &= \frac{1}{4} - \frac{r}{8N_*}, & b_r &= 1.12 - \frac{.05r}{N_*}, & a_r &= .60 + \frac{.03r}{N_*} \quad (0 \leq r \leq N_*), \\ \pi_0 &= \frac{1}{10N_*}, & k &= 399, & C &= \frac{4k}{k+1} = 3.99, & e_0 &= \frac{\lambda_r - \lambda_{r+1}}{100} = \frac{1}{800N_*} \quad (r < N_*). \end{aligned} \quad (5.2)$$

Here k is the number of circle factors added at each split; the initial number may be different. In particular $\lambda_r \geq 1/8$, $a_r \leq .63$, and $1.07 \leq b_r \leq 1.12$. The common floor π_0 and the equally spaced stage parameters are bookkeeping for a finite list, not uniform lower bounds for the countable construction.

Definition 5.2 (Analytic invariant). At stage r , write J_r for the unprocessed labels. On a packet E of base dimension m , retain its base metric and all circle warps. Write R_E for the scalar curvature of that full warped metric and ω_E for the product of its circle lengths, suppressing fixed circle-volume constants. Gradients of invariant functions are base gradients; their integrals use $\omega_E d\text{vol}_E$, and their Laplacian is the corresponding weighted base Laplacian. For every $j \in J_r$ the data comprise a compatible continuous profile $\mu_j : E \rightarrow [-\infty, \infty]$ with locally Lipschitz finite truncations, a compatible smooth potential $\Pi_j \geq \pi_0$, and compatible open cores U_j^+, U_j^- satisfying

$$\overline{U_j^\pm} \subset \text{int}_E \{\mu_j = \pm\infty\}. \quad (5.3)$$

On the open finite band the profile obeys, almost everywhere,

$$2|\text{grad } \mu_j| \leq a_r \mu_j^2 + \Pi_j. \quad (5.4)$$

Set

$$G_j = E \setminus (U_j^+ \cup U_j^-), \quad D_E = \{x : \#\{j \in J_r : x \in G_j\} \geq m-1\}. \quad (5.5)$$

The required scalar bound is

$$R_E \geq \lambda_r + b_r \sum_{j \in J_r} \Pi_j \quad \text{on } D_E. \quad (5.6)$$

For an unprocessed color i , the cores contain $q_E^{-1}(A_i) \subset U_i^+$ and $q_E^{-1}(B_i) \subset U_i^-$. Enlarging cores must preserve these conditions. The sets D_E are closed: they are finite unions of finite intersections of the closed sets G_j (and equal E when the count threshold is nonpositive). All data are exact products in the sense of Theorem 5.1; at stage zero the single space is X , with no product-end requirement.

The threshold $m-1$ is preserved by the two kinds of output of a split. A side keeps its dimension and loses one unprocessed label; a hit loses both one dimension and that label. The lower threshold $m-2$ defines a guard containing the projections of all new required sets, even after the remaining cores have been enlarged.

Lemma 5.3 (The two count inclusions). *If B is a d -fold further face of E , the product projection sends D_E in that tail into D_B . At a split of label j_0 on a parent of dimension m , suppose the remaining cores have been enlarged, giving $G'_j \subset G_j$ for $j \neq j_0$, and define the closed guard*

$$H_E^* = \{x : \#\{j \in J_r \setminus \{j_0\} : x \in G'_j\} \geq m-2\}. \quad (5.7)$$

Suppose a regular zero wall L lies in $\text{int } G_{j_0}$. Pull the remaining cores to a new side or hit packet A by bulk inclusion, wall restriction, and projection of the attached cylinder. Define D_A using these cores, the remaining labels, and the dimension of A , as in (5.5). Then D_A projects into both H_E^ and the parent's old set D_E . Guards likewise project into guards on further faces.*

Proof. In an exact product tail all label memberships pull back unchanged, whereas the face dimension is $m - d$. A count at least $m - 1$ is at least $(m - d) - 1$, proving the first assertion. The same argument with thresholds $m - 2$ and $(m - d) - 2$ proves the guard assertion. On a new side, whose dimension is m , a point of D_A has at least $m - 1$ remaining undecided labels. This puts its projection in the guard and, because $G'_j \subset G_j$, in the old required set. On a hit packet, whose dimension is $m - 1$, the required count is $m - 2$. Its projection lies on $L \subset G_{j_0}$, so the processed label contributes one further old undecided label. The old count is therefore at least $m - 1$. The cylinder projection has exactly the same remaining memberships as its base, so the side argument also covers all cylinder points. These are pointwise count statements and use no scalar estimate away from D_E . $\square$

Lemma 5.4 (Initialization and radial confinement). *Let the stage-zero stabilized scalar curvature be at least .4, and suppose that every color pair has separation*

$$\text{dist}_h(A_i, B_i) > T_0 + 4, \quad T_0 = 2\pi\sqrt{2/\pi_0}.$$

Then the data of Theorem 5.2 exist at stage zero with $\Pi_j = \pi_0$. On a noncompact X the radial cores can be chosen as alternating separating subbands of a smooth proper function, with compact smooth barrier domains around their closures lying in the same pole plateaus. Every regular radial partition respecting these signs has locally finite compact wall components and compact closed-side components before collar attachment. On a compact X a decided radial label with empty wall has the same properties.

Theorem 5.5 (Finite compatible cutting). *Fix a finite number N_* of labels and the stage constants in (5.2). Suppose a stage- r packet diagram is obtained by the regular splitting construction of Theorem A.1, with compatible smooth base metrics and a finite common list of positive circle warps, and satisfies Theorem 5.2. After the radial stage, require the complete compact-part/product-end geometry of Theorems A.2 and 5.1. Let j_0 be the next label, and suppose every nonempty parent packet has base dimension at least two. Then j_0 can be processed as follows.*

On every parent there is a compatible smooth regular defining function with the prescribed signs on the current forced core closures. Keep its zero wall as the new hit space. To each closed sign side attach the full product cylinder over its boundary. These pieces form a compatible smooth stage- $(r + 1)$ packet diagram with the continuous projections described above. The wall on a parent E lies in $\text{int } G_{j_0}$. The remaining profiles may first be modified, their cores enlarged, and their potentials increased in the parents; these data are then carried to the new pieces by restriction and cylinder projection. There are compatible smooth metrics and potentials, and exactly 399 new positive circle factors on each new packet, for which the entire stage- $(r + 1)$ invariant holds. Earlier forced signs are preserved. The new packets have complete compact-part/product-end geometry. This assertion includes a possible one-dimensional output, on which its scalar lower bound holds everywhere.

The operation fixes every old warp identically equal to 1. For a remaining label that is passive on the finite downward source diagrams and the face copies used in their comparisons—constant pole profile, full corresponding core, and spatially constant potential c_j agreeing on compared faces—the construction can preserve the pole and core and choose its new potential to be exactly $(1 + 3\eta_r/2)c_j$, where $\eta_r = 1/(200N_)$.*

At stage zero the single parent may instead be an arbitrary complete smooth manifold with no product-end or bounded-geometry assumption. Subject to the same scalar and separation assumptions, if its data are the radial initialization of Theorem 5.4, the same conclusion holds for the first split. Every resulting wall component and every closed-side component is compact before the cylinders are attached, and these components form locally finite families.

Proof. Realize the processed profile by a smooth finite pressure and take a local absolute minimizer of weighted perimeter minus its pressure-volume term, as in Theorems B.1 and B.3. The phase has the forced signs of Theorem B.4; its boundary, called the true cut, has the regular-part geometry of Theorem B.6 but may still have singularities. After the radial cut, the generic pressure choice in Theorem B.5 makes these true cuts converge to the corresponding face products at packet ends (Theorem B.7). On the initial radial parent use shift zero and a phase from compactness; compact barriers replace this post-radial growth and uniqueness argument.

The wall, and each retained side with its attached cylinder, carry the quadratic-form lower bounds of Theorem B.10. These bounds include a positive gradient reserve and, on a joined side, the scalar contribution of its internal attachment seam. They are not yet pointwise scalar bounds on smooth new packets. Keep these true cuts and their pressures fixed. The constructions of Theorems C.1, C.5 and D.6 change only the remaining profiles, enlarge their cores, and add nonnegative potentials. Their form cost is paid by the gradient reserve and the gap $\lambda_r - \lambda_{r+1}$, and the updated guard H_E^* of (5.7) misses the singular set. Thus the future required regions can be controlled through regular portions of the true cut. When the parent dimension is at most seven the singular set is empty, and no singular forcing is needed; compactness on the resulting smooth windows is the smooth case of Theorem D.2. More generally this smooth-window case applies whenever the limiting wall is smooth, including products of smooth faces with Euclidean factors even when their total dimension is at least eight. At the first radial split the same arguments are applied on the locally finite compact transition blocks, as specified in Theorem D.6.

The guarded approximation in Theorem E.2 now replaces the true cut by a smooth regular wall. Its defining functions are exact products in the old ends. The comparison with the true cut on the guard, followed by the seam smoothing of Theorem E.4, transfers the form bound to the smooth split. The positive supersolutions of Theorems E.7 and E.9 then supply 399 compatible new circle factors; the scalar identity in Theorem E.10 converts that form bound into the pointwise stage- $(r+1)$ scalar inequality. Throughout this transfer the parent scalar inequality is used only over D_E : Theorem 5.3 places every new required region inside that old region as well as inside the guard.

Theorems A.1 and A.2 give the new geometry; Theorem 5.4 gives the radial confinement. Unit entries are fixed by restriction, product extension, and seam smoothing. For a remaining label passive in the sense stated above, we omit its forcing updates, and Theorems E.5 and E.6 reproduce its chosen constant potential exactly. The choice of η_r satisfies the strict stage margins in (E.3). $\square$

This theorem is a finite-label statement. Its common positive floor and finite list of warp factors do not provide a common floor or a global finite list for the construction that follows. The next section proves the stronger locality property actually needed there: on each whole packet and each directed comparison diagram, only finitely many labels and warp factors are nonconstant, and the remaining potential sum is a spatial constant reproduced exactly by the construction.

6 Countably many compatible partitions

The separation widths in Theorem 4.3 may depend on the label. We now use this freedom to construct the partitions. The analytic construction is performed at finite stages. Its essential finiteness property concerns each *whole packet*, including all its infinite product ends; local finiteness merely on compact subsets would not suffice. The finite-label procedure is stated in Theorem 5.5, and its detailed proof is in Appendices A to E. We adapt that proof using individual positive floors and finite exceptional data on each whole packet. This section establishes their hypotheses for the countable construction and proves that those hypotheses persist.

6.1 Initialization and the inductive data

Let D , the centers x_i , and the pairs (A_i, B_i) be as in Theorem 4.3. We assume every B_i is nonempty, since otherwise $\text{diam}_g M \leq 4D$. There is one label for each center, and an additional label 0 used first for radial confinement. The set of labels is finite or countable; write it as $I = \{0, 1, 2, \dots\}$, stopping the enumeration if necessary. Choose

$$p_i^* > 0, \quad \sum_{i \in I} p_i^* = \frac{1}{20}. \quad (6.1)$$

Use Theorem 4.3 with

$$L_i = 10 + 2\pi\sqrt{2/p_i^*} \quad (i \neq 0). \quad (6.2)$$

We henceforth use the resulting complete metric $h \geq g$, its locally finite list of log warps p_α , and its stabilized scalar lower bound $2/5$.

The initial weighted base measure is globally well defined:

$$\omega \, d \text{vol}_h, \quad \omega = \exp\left(\sum_{\alpha} p_{\alpha}\right) > 0.$$

The sum is locally finite. On a buffered compact set, only finitely many nonunit factors e^{p_α} occur, so ordinary finite warped-product formulas compute the stabilized scalar, weighted variations, and invariant test forms there. These formulas agree on overlaps: adjoining a unit factor changes none of them. In particular, the initial problem is a weighted problem on the n -dimensional base M .

Fix numbers

$$0 = t_0 < t_1 < t_2 < \dots < 1, \quad t_r \longrightarrow 1,$$

and use as many stages as there are labels. These are budget parameters, unrelated to a spatial coordinate. Put

$$\begin{aligned} a_r &= .60 + .03t_r, & b_r &= 1.12 - .05t_r, & \lambda_r &= \frac{1}{4} - \frac{t_r}{8}, \\ \Delta_r &= t_{r+1} - t_r, & e_r &= \frac{\Delta_r}{800}, & \eta_r &= \frac{\Delta_r}{200}, & k &= 399, & C &= \frac{4k}{k+1} = 3.99. \end{aligned} \quad (6.3)$$

Thus $\lambda_r \geq 1/8$, $.60 \leq a_r < .63$, and $1.07 < b_r \leq 1.12$. No positive infimum of the Δ_r is required. In a stagewise appendix lemma, its symbol e_0 denotes the current e_r .

We use the packet notation and regular splitting operation introduced in Section 5. The countable invariant below replaces the common finite-label floor by the individual positive floors p_j^* . At stage r , the processed labels are $0, \dots, r-1$ and the remaining set is J_r . A packet with s hit entries has base dimension $m = n - s$. Its continuous collapse map to M is denoted by q_E . For each $j \in J_r$ retain an extended-real continuous profile μ_j with locally Lipschitz finite truncations, a smooth potential $\Pi_j \geq p_j^*$, and disjoint open cores U_j^+, U_j^- with

$$\overline{U_j^\pm} \subset \text{int}\{\mu_j = \pm\infty\}, \quad 2|\text{grad } \mu_j| \leq a_r \mu_j^2 + \Pi_j \quad \text{almost everywhere on the finite band.} \quad (6.4)$$

For unprocessed center labels require $q_E^{-1}(A_j) \subset U_j^+$ and $q_E^{-1}(B_j) \subset U_j^-$. All data are compatible products in the packet ends. With

$$G_j = E \setminus (U_j^+ \cup U_j^-), \quad D_E = \{x : \#\{j \in J_r : x \in G_j\} \geq m-1\}, \quad (6.5)$$

the scalar requirement is

$$R_E \geq \lambda_r + b_r \sum_{j \in J_r} \Pi_j \quad \text{on } D_E. \quad (6.6)$$

Here R_E denotes the stabilized scalar expression, and integrals and Laplacians use the product of the old warp lengths as weight. The finiteness statement below makes these meanings unambiguous. A nonpositive counting threshold means the whole packet.

Lemma 6.1 (Initial profiles and radial blocks). *The data in (6.4)–(6.6) exist at stage zero with $\Pi_i = p_i^*$. The radial forced signs can be chosen so that every regular radial split preserving them has locally finite compact wall and closed-side components before cylinder attachment. On every compact radial block and buffer, only finitely many center labels fail to be identically the negative pole with full negative core, and only finitely many initial warps fail to be one.*

Proof. For a positive number p , the decreasing crossing

$$2|y'| = .5y^2 + p$$

from $+\infty$ to $-\infty$ has length

$$T(p) = \int_{-\infty}^{\infty} \frac{2 \, dy}{.5y^2 + p} = 2\pi\sqrt{2/p}.$$

Extend it by its poles. For center i , begin this crossing at stretched distance 2 from A_i and compose it with $\text{dist}_h(A_i, \cdot)$. The distance is 1-Lipschitz, so the resulting profile obeys the inequality with .5 in place of $a_0 = .60$. Take cores

$$U_i^+ = \{\text{dist}_h(A_i, \cdot) < 1\}, \quad U_i^- = \{\text{dist}_h(A_i, \cdot) > T(p_i^*) + 3\}.$$

Their closures lie in the respective pole interiors. The stipulated width puts B_i strictly inside the negative core. In particular, on the complement of the original closed ball $\overline{B}_g(x_i, 2D)$, the profile is the negative pole and the negative core is full.

There exists a smooth proper function $\rho : M \rightarrow \mathbb{R}$, bounded below, with $|\rho - \text{dist}_h(o, \cdot)| < 1$ and $|\text{grad } \rho|_h \leq 2$. Here no uniform geometric assumption is necessary. To construct it, use relatively compact coordinate balls in a locally finite cover, convolve the distance on slightly larger balls, and blend with a smooth partition of unity. The metric distortion and local convolution errors can be made arbitrarily small on each ball. In differentiating the blend, write each local approximation as the distance plus its error; the terms involving derivatives of the partition sum to zero on the common distance. Choose the value errors so that the remaining partition-derivative terms have sum less than a fixed small number at each point. A locally finite refinement and successively smaller positive errors achieve this, while keeping the total value error below one. Properness follows from the bounded value error and properness of the complete metric h .

Choose alternating positive and negative plateaus in the ρ -axis, with transition intervals wider than $2T(p_0^*)$. Rescale the preceding crossing to these intervals, reversing its sign on alternate intervals. The gradient bound for ρ gives the profile inequality with coefficient .5 and potential p_0^* . Choose the core closures strictly inside the plateaus. Regular values in the remaining margins give compact smooth barrier annuli, and a compact central barrier, whose closures stay in the same pole plateaus. Their value bands tend to infinity and are locally finite.

A wall respecting the signs cannot cross a separating plateau band; each component is closed in one compact radius block and hence compact. A component of either closed side is trapped between two separator bands of the opposite sign, or in the initial compact block. Regularity of the defining function gives manifold-with-boundary charts for these sides. A compact block therefore contains only finitely many components. The same charts prove local finiteness of the component families. When M is compact, use a radial label with constant positive pole and full positive core; its split has no wall and its only side is compact.

The scalar requirement initially holds everywhere because

$$\lambda_0 + b_0 \sum_i p_i^* = .25 + 1.12/20 = .306 < .4.$$

Finally a compact block with a compact buffer meets only finitely many original closed outer balls and finitely many initial warp supports. Every other center label and warp has the asserted constant data there. $\square$

6.2 Finite exceptional data on entire packets

A *downward diagram* of a packet consists of it and all its further faces. An *upward diagram* consists of it and all its cofaces. Each is finite at a fixed stage, by Theorem A.1. A finite union of either type is finite. The constructions below make finitely many directed passes through such diagrams. They never ask for closure under arbitrary alternating face and coface steps.

Radial confinement gives the geometric reason to expect finite data on a whole packet. At the first split, a compact radial side or wall component is prolonged by cylinders; its entire collapse image stays in that original compact component. At every later split a child A of E has

$$q_A = q_E \circ p_A, \quad p_A = \begin{cases} \text{inclusion,} & \text{on the wall or bulk,} \\ \text{projection to the wall,} & \text{on a fresh cylinder.} \end{cases} \quad (6.7)$$

Hence $q_A(A) \subset q_E(E)$, including in all simultaneous normal directions. A finite incidence diagram therefore has image in a finite union of compact original radial blocks. Only finitely many original outer balls and initial nonunit warp supports meet that image. Shielding changes no collapse map, and regularization chooses new walls inside old parents, so both preserve this confinement.

This argument controls the initial supports. Later smooth data need not factor through q_A : they also use data copied from faces. We must therefore retain finite exceptional lists through those directed dependencies, as well as preserve every other entry exactly.

Definition 6.2 (Finite exceptional data). At stage $r \geq 1$, on each whole packet E there are finite sets $I_E \subset J_r$ of label indices and F_E of old warp indices such that, for $j \notin I_E$,

$$\mu_j \equiv -\infty, \quad U_j^- = E, \quad U_j^+ = \emptyset, \quad \Pi_j \equiv c_r p_j^*. \quad (6.8)$$

For every old warp index outside F_E , its length is identically one. The number c_r is the same on all packets and is prescribed by

$$c_0 = 1, \quad c_{r+1} = (1 + 3\eta_r/2)c_r. \quad (6.9)$$

For any finite diagram used in a comparison, take the union of its finite exceptional sets, adding the current label explicitly. This one finite union then works simultaneously on all its whole packets. At stage zero the analogous property is required on every compact radial block and buffer, with $c_0 = 1$.

The sets I_E need not consist exactly of the nonconstant entries: an entry may be retained in the exceptional list even if it happens to be constant. This convention avoids any need to discard previous exceptions. Only the existence of a finite list matters.

Lemma 6.3 (Consequences and directed dependencies). *Suppose the packet geometry and Theorem 6.2 hold at a finite stage. Then the following assertions hold.*

- (i) *The sum of potentials on a whole packet is a smooth function with bounded derivatives of every fixed order. The undecided counts, required sets, and guards use only a finite list and are closed.*
- (ii) *On each whole packet and each finite comparison diagram, the old weight, stabilized scalar, stability forms, and weighted Laplacian are computed with one finite list of warps, after padding with unit factors. The bounded geometry and polynomial weighted growth conclusions of Theorem A.2 apply, including on any finite Euclidean product.*
- (iii) *Compatible choices with positive packetwise tolerances can be made in the following order: fix individual requests, propagate numerical requests down the finite coface chains, and then construct data upward from deeper faces. Enlarging a coface's exceptional list by the union over its downward source diagram suffices for the last pass. No data are propagated back from a newly constructed coface to an already constructed face.*

Proof. On E , the sum of potentials is exactly

$$\sum_{j \in J_r} \Pi_j = \sum_{j \in I_E} \Pi_j + c_r \sum_{j \in J_r \setminus I_E} p_j^*. \quad (6.10)$$

The second term is a finite spatial constant. Indeed

$$1 \leq c_r \leq \exp\left(\frac{3}{2} \sum_{s < r} \eta_s\right) \leq e^{3/400}.$$

The first term is a finite sum; its derivative bounds follow from compatible smoothness and the compact-part/product-end decomposition in Theorem A.2. On a finite comparison diagram choose common product thresholds for its finite exceptional list. Every other entry is already a spatial constant with the same value on each copy. The full sum is therefore an exact face pullback beyond those thresholds; no supremum of infinitely many thresholds is taken. Passive entries have empty undecided set. Thus a set requiring at least $a \geq 1$ undecided labels is the finite union, over a -element subsets of I_E , of the intersections of their closed G_j 's. This proves closedness, including for the guards with threshold $m - 2$. Threshold zero gives E .

For (ii), take the union of the finitely many relevant old warp indices on the finite diagram and set every missing entry to one. Writing $u_\alpha = \log f_\alpha$ and $U = \sum_\alpha u_\alpha$, the scalar expression is

$$R_E = \text{Scal}_{h_E} - 2\Delta_{h_E}U - |dU|_{h_E}^2 - \sum_\alpha |du_\alpha|_{h_E}^2.$$

A padded logarithm is zero; the product weight is unchanged. The weighted Laplacian is $\Delta_{h_E} + \langle dU, d \cdot \rangle_{h_E}$ and is unchanged as well. Weighted perimeter and its first and second variations depend on the same weight. Equivalently, a flat unit factor adds zero second fundamental form and zero curvature to the invariant variation calculation. Thus the wall and seam forms also agree. The finite-diagram geometric lemma now gives completeness, bounded geometry, positive lower and finite upper bounds for the weight, and center-uniform polynomial volume growth. It gives the same statements after taking a finite Euclidean product. The regularity problem for a minimizing phase remains on the base, with singular dimension at most $m - 8$ by Theorem B.6; padding does not alter that dimension.

For (iii), a particular face receives requests from only its finitely many cofaces and coface chains. Allowing for the finitely many fixed product and cutoff bounds, the minimum of the requested positive errors is positive. Settle all these numbers before constructing the approximations. A

request only decreases an error tolerance: it does not copy any coface value into the face. During the subsequent upward pass, the inputs on a packet are its own data and already constructed copies from its finite downward diagram. Their exceptional sets have a finite union. Outside that union every copy in (6.8) agrees with every other copy. A newly constructed coface is not used to reconstruct its sources. The same argument applies to finitely many successive directed passes, and proves the stated dependency rule. With these numerical tolerances fixed, at each upward pass the output on E depends only on its original data and the outputs on its strict faces. Its source set for copied functions is therefore contained in the finite downward diagram of E . The preceding downward pass changes only positive numbers, not the input functions. In particular two packets sharing a face do not copy one another's newly constructed data through that face. $\square$

6.3 One countable invariant step

Proposition 6.4 (Preservation of the invariant). *Suppose that all nonempty parents at stage r have base dimension at least two and satisfy (6.4)–(6.6). At $r = 0$ use the compact radial blocks of Theorem 6.1; at $r \geq 1$ suppose also that the packet geometry and Theorem 6.2 hold. Processing label r produces the compatible smooth split, preserves the forced signs, and gives the same assertions at stage $r + 1$. Exactly 399 new warp factors, with common indices, are appended at this stage. Each whole resulting packet has a finite relevant old warp list.*

Proof. We verify the choices in their order of dependence.

The current pressure and minimizing phases. The pressure construction of Theorem B.1 uses a positive floor only for the label being processed. Replace its finite-family floor by p_r^* . Compatible clipping heights can first be chosen above the local barrier bounds, with a uniform extra barrier margin of one. Then use local smoothing errors to obtain

$$2|\text{grad } p_0| \leq .65p_0^2 + 1.01\Pi_r.$$

All smoothing requests involve only this one profile and finite incidence diagrams. The elementary inequality

$$.67(y + c)^2 - .65y^2 \geq -21.775c^2$$

shows that one common nonempty shift interval, for example

$$|c| < \min\{.01\sqrt{p_r^*}, 1/2\}, \tag{6.11}$$

preserves both $2|\text{grad}(p_0 + c)| \leq .67(p_0 + c)^2 + 1.02\Pi_r$ and the barrier inequalities. Floors for other labels do not enter this interval.

After the radial stage, Theorem 6.3 supplies all bounded geometry and weighted polynomial-growth hypotheses of Theorem B.5. There are countably many packets, Euclidean product dimensions, and tests in that lemma's exhaustion. Remove the countable exceptional shift sets and choose a single c in (6.11). The resulting true minimizers have the common product limits of Theorem B.7; choose them before any shielding update. At the initial radial step no uniqueness or polynomial growth at spatial infinity is used. The smooth global positive weight and locally bounded smooth pressure give local BV existence and strict compactness by Theorem B.3. The relatively compact radial barriers force the phases by compact comparisons. All variations and tests lie in buffered compact windows with finite warp lists, so the formulas in Theorem B.10 apply. The GMT and shielding dimensions are the base dimensions at this first step as well.

Anchors, guards, and shielding. On each post-radial packet retain its finite exceptional list, enlarged by its finite downward source diagram and by the current label. Labels outside it are

already forced everywhere and require no shell addition. Choose compatible smooth clipped anchors as in Theorem C.1, retaining the exact value -1 for a passive profile. For each packet of dimension $m \geq 8$ and each set of $m - 7$ remaining labels, the image of its singular set under the corresponding smooth anchor map has $(m - 7)$ -dimensional Lebesgue measure zero. There are countably many such conditions, even if all finite subsets of the countable label set are included. In the countable product of uniform probability intervals $(-.3, .3)$, each bad cylinder has measure zero by its finite-dimensional marginal. A threshold vector outside their union therefore exists. At most $m - 8$ anchors equal their thresholds at any singular point. The sign exclusions at poles follow from the .1 anchor approximation, exactly as in Theorem C.1.

Process the finitely many face depths from deepest upward, keeping the true cuts fixed. Suppose all strict faces of E have been shielded. Its remaining risky set

$$K_E = \Sigma_E \cap \{x : \#\{j \in J_r \setminus \{r\} : x \in G'_j\} \geq m - 2\}$$

is closed. It is compact: an escaping sequence would, by Theorem A.3, have a product limit over a further face with compact convergent base coordinates. The finite exceptional list and closed core complements pass the undecided count to that limit. The face has a smaller dimension, so the limit lies in its guard. The product convergence of true cuts forces the limiting point to be singular, since regular-point graphical convergence excludes nearby singular points. This contradicts the completed shielding of that face. This argument also treats escape in several normal coordinates.

Apply Theorems C.2, C.3 and C.5 on this compact risky set, only for the finite list which can still be undecided. Enumerate the signed compact flat pieces. Choose each shell scale by Theorem D.3, hold it fixed, and then send its coface fronts far enough out by Theorem D.4. In the cap construction of Theorem D.5, the crossing for label j satisfies

$$2|w'| \leq .58w^2 + p_j^*.$$

Its length is finite; the inequality is paid by $\Pi_j \geq p_j^*$. The minimum-of-branches formula in that lemma adds no simultaneous normal derivative cost. Its cutoff finishes before the cap crossing starts, so whenever the shell branch is active the whole shell potential is available. Opposite signs remain in disjoint strict anchor regions, and cores only enlarge.

Only finitely many additions are required on E . Indeed, every point of the initial compact K_E eventually belongs to an open forced core for each of its strict anchors among the finite exceptional list. At most $m - 8 < m - 2$ labels can then remain undecided there. The residual risky subsets of the initial K_E are nested and closed, and their intersection is empty. Compactness gives a finite index at which the residual is empty. Stop then. Each specified coface receives additions from finitely many source faces through finitely many chains. Thus only finitely many additions and finite minima or maxima affect any whole packet. In particular no infinite limiting profile is taken on that packet.

To control all the countably many packets at once, enumerate each possible local addition and assign positive allowances with total less than .12. Split each allowance between its shell-scale choice and its simultaneous finite coface extension choices. Starting the bookkeeping form bound at .78, the maximum estimate and the extension estimate give one $\theta_* < .9$ such that

$$\int 1.16 \sum_{j \neq r} D_j v^2 \leq \theta_*(\mathcal{N}(v) + e_r \|v\|_2^2). \quad (6.12)$$

Here $\mathcal{N}$ has coefficient 1.7 on a wall and coefficient .8 on bulk plus 1.7 on its attached cylinder. These are the same coefficients on every packet and product. Only a finite list of additions occurs on each post-radial packet, so the bound persists after a whole depth has been treated.

Initially, instead work in separated compact transition shells with compact buffers between radial barriers. Each has finitely many exceptional labels and warps. Its risky set is compact, and the same finite termination applies; there is no coface extension yet. Choose supports within the shell buffers. Their local finiteness makes all updates locally finite on M , and a compact test meets only finitely many of them. The summable allowances prove (6.12) for every such test. Every compact radial side block meets only finitely many shell buffers. This supplies the whole-packet finite lists after the first attachment.

Counts and the quantitative transfer. Core enlargement gives $G'_j \subset G_j$. Only the finite exceptional list can contribute to the counts on a post-radial packet; the analogous statement holds on buffered compact blocks at the radial step. Thus Theorem 5.3 places every new required set in both the old required set and the shielded guard under projection, including along attached cylinders. The finite lists and closed core complements also pass these count inclusions to all further-face product limits.

The guarded tube and comparison constructions of Theorems E.1 and E.2 use these closed guards, regular graphical convergence on product limits, and finite packetwise derivative bounds. All are now available. Numerical errors are allocated down coface chains before the defining functions are constructed upward.

Apply the quantitative transfer of Theorem E.3 with the current t_r , $\Delta_r = t_{r+1} - t_r > 0$, and $e_0 = e_r = \Delta_r/800$. Its stage margins (E.3) were proved for these general adjacent parameters, so they do not require equally spaced stages. The potential calculation also applies: on each finite comparison diagram, (6.10) expresses the old potential sum as finitely many smooth functions plus one spatial constant. Only finitely many shielding additions occur there. Thus, with $\tilde{\Pi}_j = \Pi_j + D_j$, the same raw form and comparison estimates control

$$\lambda_{r+1} + (1 + 3\eta_r)b_{r+1} \sum_{j \neq r} \tilde{\Pi}_j \quad (6.13)$$

near the new required set, with the positive mass and gradient reserves proved there. The constant tail has no derivative error; potential approximation errors concern only the finite exceptional list. At the radial step these estimates are made on buffered compact blocks, as in that proof. The seam and potential smoothing can consequently retain an e_r local mass gap. This positive number is fixed for the current stage; all geometric tolerances may depend on the packet. No infimum over future stages or over all packets is taken.

Exact passive constants in regularization. Restrictions, bulk inclusions, and cylinder projections preserve identically constant functions. Seam smoothing of an old identically unit warp leaves it one. For the remaining potentials, smooth $(1 + 3\eta_r/2)\tilde{\Pi}_j$ as in Theorems E.5 and E.6, so that

$$(1 + \eta_r)\tilde{\Pi}_j \leq \Pi_j^{\text{new}} \leq (1 + 2\eta_r)\tilde{\Pi}_j. \quad (6.14)$$

On a finite exceptional list, allocate the required relative errors separately, using $p_j^* > 0$ for its j th entry. If an aggregate error is prescribed, subdivide it among that finite list. If an entry is passive on the whole source diagram, set its output *exactly* to $c_{r+1}p_j^*$. This is compatible with the smoothing procedure: local convolution of a constant is that constant, partition weights sum to one, and every copied end value is the identical constant. Equivalently this entry can simply be omitted from all smoothing and restored as that constant afterward. We only impose this rule on whole constant inputs and their agreeing copies, so it makes no claim about preserving the edge of an arbitrary nonsmooth plateau.

Before blending on a coface, enlarge its exceptional list by the union over its entire finite downward source diagram. Thus a face-active entry is recorded even if the coface's individual

input was passive. The remaining copies agree identically. Coface requests merely adjust numeric tolerances; they do not introduce new source values on faces. This proves exact passive reproduction on the completed diagram and precludes a cascade through infinitely many same-depth neighbors. At the radial step choose a compact buffer for each compact radial block, enlarged to contain the finitely many chart closures and shell supports used by its approximations. Only finitely many original outer balls and initial nonunit warp supports meet this buffer. Enlarge the exceptional list by precisely these indices. Every other entry is the same constant on every averaging neighborhood and is therefore reproduced exactly before cylinder attachment.

The local-slope argument of Theorem E.6 justifies restriction of profiles to the wall: an almost-everywhere gradient bound by a continuous function gives the corresponding local Lipschitz slope bound at every point of each finite band. Restriction and cylinder projection then preserve it. The small metric change increases slopes by at most $1 + \eta_r$, and the first inequality in (E.3) together with (6.14) gives the next profile inequality. Continuous pullback and core enlargement preserve all pole-neighborhood and forced-preimage conditions.

Corrections, supersolutions, and the new warp group. The sum in (6.10) is now smooth, bounded, and exactly compatible, as are its finite-order derivatives. On every required finite diagram, pad the old warps by units. In a product end over B the operator of Theorem E.7 therefore satisfies exactly

$$L_A = L_B - C\Delta_{\text{normal}}, \quad L_A = -C\Delta_{\text{old}} + R_{\text{old}} - V_A^* + K_A, \quad V_A^* = \lambda_{r+1} + b_{r+1} \sum_{j \neq r} \Pi_j^{\text{new}}.$$

In particular the potential formula does not depend on the base dimension or on how many unit factors were padded. The correction is zero near the new required set and produces a positive form gap on each whole packet. The proof of Theorem E.7 uses finite minima of the gaps on the packet's face diagram. Its bounded resolvent and the compatible positive supersolutions in Theorems E.8 and E.9 apply with the packetwise bounded geometry and polynomial weighted volume already proved. Their tolerances are chosen after the operators and their individual exact solutions are fixed, and before the final upward blending pass. Thus their finite-dependency hypothesis is precisely (iii) of Theorem 6.3.

Let $v_A > 0$ be the resulting supersolution with $L_A v_A \geq 1/2$. Append the common group of $k = 399$ circle lengths $f = v_A^{2/(k+1)} = v_A^{1/200}$. The scalar identity in Theorem E.10 gives

$$R_{\text{new}} = R_{\text{old}} - C \frac{\Delta_{\text{old}} v_A}{v_A} = V_A^* - K_A + \frac{L_A v_A}{v_A} \geq V_A^* \quad \text{on } D_A.$$

Only the number of *new* factors enters the coefficient C . The old finite list may vary from packet to packet; unit padding has no effect. Each stage adds just this one finite group, with the same indices in every status. All its lengths are positive and bounded above and away from zero on a whole packet, and exactly compatible. The initial lists and their finite propagated exceptions remain finite, so the complete old list at stage $r + 1$ is still finite on each whole packet and each finite comparison diagram.

Finally the regular split preserves compact truncations and finite incidence by Theorem A.1. Initially the radial separators make all components compact before prolongation. Compatible smooth metrics and finite relevant warps then give the complete packet geometry by Theorem A.2. The recursion (6.7) preserves confinement in original radial blocks. All exceptional-list enlargements just described are finite, and all passive constants have multiplier c_{r+1} . This completes the induction step, including the first radial step. $\square$

6.4 Termination in dimension and finite-stage walls

Lemma 6.5 (The one-dimensional obstruction). *No nonempty one-dimensional packet can satisfy the invariant at a finite stage of the construction.*

Proof. Such a packet is complete by Theorem A.2 and has only finitely many relevant positive smooth warps $f_1, \dots, f_q$ on its whole base. Its arclength universal cover is $\mathbb{R}$, including when the packet itself is a circle. On this line put $u_\alpha = \log f_\alpha$ and $P = \sum_{\alpha=1}^q u_\alpha$. The warped-product scalar formula reduces to

$$R = -2P'' - (P')^2 - \sum_{\alpha=1}^q (u'_\alpha)^2.$$

The required count in dimension one is zero, so the invariant holds everywhere and gives $R \geq \lambda_r > 0$. Consequently

$$-2P'' \geq \lambda_r + (P')^2, \quad \frac{d}{ds} \arctan \frac{P'}{\sqrt{\lambda_r}} \leq -\frac{\sqrt{\lambda_r}}{2}.$$

Integration over a line interval longer than $2\pi/\sqrt{\lambda_r}$ contradicts the range of the arctangent. If $q = 0$, the scalar formula is simply $R = 0$. The argument takes place at one finite stage and uses an ordinary finite sum. $\square$

Proposition 6.6 (The countable compatible construction). *All labels can be processed successively. At every finite stage the packet diagram has the invariant above and no nonempty packet of dimension at most one. After the radial stage it has finite exceptional data on whole packets; at stage zero this assertion holds on buffered compact radial blocks. Its finite reconstruction X_r has a proper collapse $q_r : X_r \rightarrow M$ and closed signed sides $P_i^{(r)}, N_i^{(r)}$ for each processed center label, with*

$$P_i^{(r)} \cup N_i^{(r)} = X_r, \quad P_i^{(r)} \cap N_i^{(r)} = W_i^{(r)},$$

where $W_i^{(r)}$ is its wall. The forced preimages satisfy

$$q_r^{-1}(A_i) \subset P_i^{(r)} \setminus W_i^{(r)}, \quad q_r^{-1}(B_i) \subset N_i^{(r)} \setminus W_i^{(r)}.$$

These inclusions hold in open phases and persist at every subsequent finite stage. No $n - 1$ distinct walls, even including the radial wall, have a common point. Each center wall lies over its original open outer ball, and its components are compact and locally finite.

Old horizons can be enlarged by identifications preserving collapse, all walls, and all signed statuses. For any prescribed sequence $\varepsilon_r > 0$ with $\sum_r \varepsilon_r < 1$, the next finite collar compression can be chosen with collapse displacement less than ε_r , preserving the previously decided wall strata and signed sides. These are finite-stage conclusions; the limiting fixed-copy construction is made in the next section.

Proof. Initialization is Theorem 6.1. If the construction has reached stage r , its parents have dimension at least two: a first nonempty output of smaller dimension would have dimension exactly one, since a hit decreases dimension by one and a sign preserves it. Theorem 6.4 produces the next invariant, and Theorem 6.5 excludes that first output. Thus the induction continues through every finite stage. There is no passage to an infinite-stage analytic metric, weight, or torus.

Apply Theorem A.5 to each finite initial segment. Its hypotheses are the regular compatible splits and the compact radial components already established. Its status pieces are exactly the corresponding compact packet truncations. If $n - 1$ walls met, assign those labels hit status and choose a closed sign status for each other processed label containing the point. The corresponding

nonempty truncation would have dimension one, a contradiction. Forced signs persist under the continuous collapse recursion and enlargement of the cores. In particular

$$q_r(W_i^{(r)}) \subset M \setminus (A_i \cup B_i) \subset B_g(x_i, 2D).$$

Properness of q_r and compactness of the original closed outer ball make each wall component compact; regular local wall charts give local finiteness, as in Theorem A.5.

The horizon identifications in Theorem A.5 preserve collapse, walls, and signed statuses. Its collar compressions preserve the old strata and their sides, and Theorem A.6 allows their collapse oscillations to be bounded by the prescribed ε_r . These choices are made separately on the locally finite compact component family; no uniform collar width or horizon at infinity is required. The next section chooses these finite-stage homeomorphisms coherently on one fixed copy of M . $\square$

7 Reconstruction and the map to a complex

All distances in this section are measured in the original metric g ; write d_0 for that distance. Put $D = WR_0$, as in Theorem 4.3. The centers x_i form a maximal D -separated set, and the color pairs are

$$A_i = \overline{B}_0(x_i, D), \quad B_i = M \setminus B_0(x_i, 2D).$$

The A_i cover M . Every family of balls with these centers and a fixed finite radius is locally finite: a compact set has a compact neighborhood of that radius by completeness, and a compact metric space contains only finitely many D -separated points. The family is countable, since M is second countable. If some B_i is empty, $\text{diam}_g M \leq 4D$ and a constant map proves the theorem. We therefore assume that all B_i are nonempty.

7.1 A limit on the original manifold

The analytic packets have infinite product ends. Their finite reconstructions in Theorem A.5 are manifolds homeomorphic to M ; only finite intervals of the product ends are used in these reconstructions. We now choose those homeomorphisms coherently. This is the point at which local finiteness of the color supports is essential.

Proposition 7.1 (Reconstructed partitions). *The compatible cuts of Theorem 6.6 give a continuous proper map $q : M \rightarrow M$, and for every color i closed sets P_i, N_i, W_i such that*

$$d_0(q(x), x) < 1, \tag{7.1}$$

$$P_i \cup N_i = M, \quad P_i \cap N_i = W_i, \quad W_i \subset q^{-1}(B_0(x_i, 2D)), \tag{7.2}$$

$$q^{-1}(A_i) \subset P_i \setminus W_i, \quad q^{-1}(B_i) \subset N_i \setminus W_i. \tag{7.3}$$

The wall family (W_i) is locally finite, and no $n - 1$ distinct walls have a common point.

Proof. Choose positive numbers $\varepsilon_0, \varepsilon_1, \dots$ with $\sum_{r \geq 0} \varepsilon_r < 1$. Here ε_0 is assigned to the single radial stage; the other numbers are assigned to successive color stages. In this proof only, index the post-radial reconstruction by $r = 0$. For a finite reconstruction write

$$q_r : X_r \longrightarrow M, \quad H_r : X_r \longrightarrow M, \quad Q_r = q_r \circ H_r^{-1}.$$

The map q_r is the proper collapse of Theorem A.5, whereas H_r is a homeomorphism to a fixed copy of the original manifold. We maintain

$$d_0(Q_r(x), x) \leq E_r, \quad E_r \leq \sum_{j=0}^r \varepsilon_j < 1. \quad (7.4)$$

Walls and their two closed sides will also be viewed on this fixed copy using H_r .

For the radial stage, the cut has a locally finite family of compact components. Choose pairwise disjoint, locally finite collars of these components on which the collapse varies by less than ε_0 in each normal segment. Compactness of each component and continuity of the collapse permit the collars to be this small, independently from one component to another. Compress the finite inserted cylinders into these collars. Since their supports are disjoint, the displacement bound is ε_0 , rather than a sum over components. This constructs H_0 and proves (7.4) initially. The finite length of an inserted cylinder may be arbitrarily large; its length does not impose a lower bound on the width of the collar into which it is compressed.

We describe a subsequent color stage i . First raise any old finite horizons needed for the new compatible defining functions. Denote the resulting reconstruction by $\hat{X}_r$. The horizon reparameterization

$$\Phi_r : \hat{X}_r \longrightarrow X_r$$

preserves collapse and every old wall and signed side. Indeed, on an old product end the collapse is independent of the normal coordinate, and an increasing change of that coordinate preserves its sign and its zero section. Compatible products of these changes agree at corners. Thus we may set

$$\hat{q}_r = q_r \circ \Phi_r, \quad \hat{H}_r = H_r \circ \Phi_r, \quad \hat{q}_r \circ \hat{H}_r^{-1} = Q_r. \quad (7.5)$$

In particular, increasing a horizon contributes no displacement on the fixed copy.

The new wall lies over $B_0(x_i, 2D)$. Choose its collar strip S_i in $\hat{X}_r$ so that

$$S_i \subset \hat{q}_r^{-1}(B_0(x_i, 2D + 1)), \quad (7.6)$$

and so that collapse oscillation along every segment used by the compression is less than the error assigned to this stage. These choices are possible because the relevant wall components in a finite reconstruction are compact and locally finite. More specifically, their union lies in the inverse image of the compact ball $\overline{B}_0(x_i, 2D)$ under the proper finite collapse; hence only finitely many components are involved in this color stage.

The strips can be chosen to preserve every old stratum, including the radial strata. To check this assertion, work near a point of a new wall on an old stratum, and denote the new defining function by u . The horizons were chosen beyond the product thresholds, so the terminal collars glued at each old hit wall lie in the product range. In the resulting product chart, u is pulled back from the corresponding face and is independent of the old signed normal coordinates. Its restriction to that face is regular at zero. In the corresponding product chart choose a vector field V in the face directions with $V(u) = 1$, and with zero component in every old normal direction. The chart may avoid the old walls not passing through the point. A partition of unity combines these fields: the linear identity $V(u) = 1$ and tangency to every old wall both survive. Its short flow supplies a collar preserving the old walls and their sides. On a fresh cylinder, each old stratum is the product of its intersection with the new wall and the cylinder interval. The compression uses the same increasing normal reparameterization on all these products. It consequently preserves every old signed side as well as all their intersections.

Let

$$c_r : X_{r+1} \longrightarrow \widehat{X}_r \quad \text{and} \quad h_r : X_{r+1} \longrightarrow \widehat{X}_r$$

be, respectively, the new cylinder collapse and the homeomorphism that compresses those cylinders into the chosen strips. They agree off the inverse image of S_i . Figure 2 shows the two maps schematically in a normal coordinate: c_r collapses the inserted interval, while h_r compresses it into the chosen old collar. Define

$$q_{r+1} = \widehat{q}_r \circ c_r, \quad H_{r+1} = \widehat{H}_r \circ h_r, \quad Q_{r+1} = \widehat{q}_r \circ c_r \circ h_r^{-1} \circ \widehat{H}_r^{-1}. \quad (7.7)$$

The oscillation choice gives $d_0(Q_{r+1}(x), Q_r(x)) < \varepsilon_{r+1}$ wherever they differ, proving (7.4) at the next stage. Their difference is supported in $\widehat{H}_r(S_i)$. If $x = \widehat{H}_r(z)$ belongs to this set, then (7.5), (7.6), and (7.4) imply

$$d_0(x, x_i) \leq d_0(x, Q_r(x)) + d_0(\widehat{q}_r(z), x_i) < 1 + (2D + 1) = 2D + 2. \quad (7.8)$$

This support radius is independent of the stage.

The balls in (7.8) form a locally finite family. Consequently, on every relatively compact neighborhood in the fixed copy, the maps Q_r are exactly unchanged after finitely many stages. Their locally stationary limit q is continuous and satisfies (7.1). It is proper: the inverse image of a compact set is a closed subset of its compact closed 1-neighborhood in (M, d_0) .

At each finite stage the signed sides are the closed reconstructed sides, with their intersection equal to the central wall. The stratum-preserving identifications just constructed preserve every previously installed side and wall exactly. Newly installed color walls are confined to the fixed balls in (7.8). Thus all the partition data are locally eventually unchanged. Their final closed sets satisfy (7.2) and (7.3), since these are the finite-stage forced-sign properties of Theorem 6.6, and both collapse and statuses are stationary near each point. The wall family is locally finite by the same support bound. Finally, a point on $n - 1$ distinct final color walls would, on a neighborhood where these data have stabilized, lie on the same $n - 1$ finite-stage walls. This contradicts Theorem 6.6. $\square$

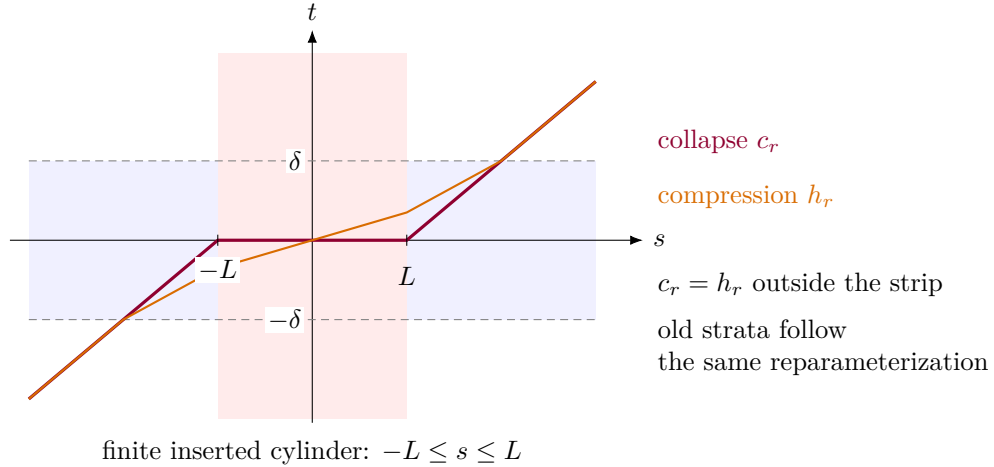


Figure 2: The two maps used in a finite reconstruction, shown schematically in local normal coordinates: s on the reconstructed domain X_{r+1} and t on the old target $\widehat{X}_r$. The target strip $|t| < \delta$ represents the chosen collar S_i . The collapse is constant on the inserted cylinder; the homeomorphism squeezes it into an arbitrarily narrow old collar. Where the maps differ, both images lie in the target strip. Infinite analytic ends are truncated before this topological operation.

7.2 Thickening the walls without increasing their multiplicity

Put $k = n - 2$. We need open neighborhoods of the whole walls, rather than independently chosen neighborhoods of their individual compact pieces.

Lemma 7.2 (Simultaneous wall neighborhoods). *There are locally finite open sets $T_i \supset W_i$ such that*

$$T_i \subset q^{-1}(B_0(x_i, 2D)) \setminus q^{-1}(A_i \cup B_i), \quad \#\{i : x \in T_i\} \leq k \quad (x \in M). \quad (7.9)$$

Proof. Set $O_i = q^{-1}(B_0(x_i, 2D))$ and $F_i = q^{-1}(A_i \cup B_i)$. By Theorem 7.1, W_i is disjoint from the closed set F_i and lies in the open set O_i . At a point x , local finiteness and closedness of the walls give an open neighborhood meeting only walls that contain x . Shrink it to a neighborhood U_x contained in $O_i \setminus F_i$ for every such wall. Therefore U_x meets at most k walls, and

$$U_x \cap W_i \neq \emptyset \implies U_x \subset O_i \setminus F_i. \quad (7.10)$$

Choose a locally finite open star refinement $\mathcal{V}$ of this cover: the union of the members of $\mathcal{V}$ meeting any fixed member is contained in some U_x . For completeness, such a refinement can be obtained directly with the metric d_0 . The function

$$\rho(x) = \sup_U \min\{1, d_0(x, M \setminus U)\}, \quad U \in \{U_x : x \in M\},$$

is positive and 1-Lipschitz, with distance to the empty set truncated to 1. The balls $V_x = B_0(x, \rho(x)/8)$ star-refine the original cover. Indeed, if V_y meets V_x , the Lipschitz inequality gives $d_0(x, y) < 2\rho(x)/7$, and hence $V_y \subset B_0(x, 25\rho(x)/56)$. The latter ball is contained in $B_0(x, \rho(x)/2)$, which lies in a member of the original cover. Fix $o \in M$. Obtain a locally finite open refinement of (V_x) by covering each compact annulus $\overline{B}_0(o, m) \setminus B_0(o, m-1)$ with finitely many open subsets of members of (V_x) lying in $\{m-2 < d_0(o, \cdot) < m+1\}$, for $m \geq 1$. These sets cover M and are locally finite. A refinement of a star refinement is still a star refinement of the original cover, which proves the assertion about $\mathcal{V}$.

Define

$$T_i = \bigcup \{V \in \mathcal{V} : V \cap W_i \neq \emptyset\}.$$

It contains W_i , and (7.10) implies the containment in (7.9). If z belongs to several T_i , choose $V_0 \in \mathcal{V}$ containing z . For each such i there is a $V_i \in \mathcal{V}$ containing z and meeting W_i . All the V_i meet V_0 , so their union lies in a single U_x . That U_x meets each of these walls, and therefore there are at most k of them. Finally, (T_i) is locally finite because $T_i \subset O_i$ and the family (O_i) is locally finite. $\square$

7.3 Coordinates, triangulation, and entire fibers

For each i the closed sets

$$C_i^+ = P_i \setminus T_i, \quad C_i^- = N_i \setminus T_i$$

are disjoint and cover $M \setminus T_i$. For a closed set C write $\delta_C(x) = \min\{1, d_0(x, C)\}$, with $\delta_\emptyset = 1$, and put

$$t_i(x) = \frac{\delta_{C_i^-}(x)}{\delta_{C_i^-}(x) + \delta_{C_i^+}(x)}. \quad (7.11)$$

The denominator is positive because the two closed sets are disjoint. These continuous coordinates have the following properties:

- (i) $t_i = 1$ on $q^{-1}(A_i)$ and $t_i = 0$ on $q^{-1}(B_i)$;

- (ii) $0 < t_i(x) < 1$ implies $x \in T_i$;
- (iii) $\text{supp } t_i \subset q^{-1}(\overline{B}_0(x_i, 2D))$, so the coordinate supports are locally finite;
- (iv) at every point at least one coordinate equals 1, and at most k coordinates lie strictly between 0 and 1.

Here the last assertion uses the inner-ball cover and Theorem 7.2.

Consider the cubical set of finitely supported coordinate vectors described by these properties. Its cubes are

$$Q(S, A) = \{t : t_i = 1 \ (i \in S), \ 0 \leq t_i \leq 1 \ (i \in A), \ t_i = 0 \ (i \notin S \cup A)\},$$

where S, A are disjoint finite sets of labels, $S \neq \emptyset$, and $|A| \leq k$. Triangulate each cube by the staircase triangulation: for every ordering $a_1, \dots, a_j$ of A , take the simplex with vertices

$$\mathbf{1}_S, \ \mathbf{1}_{S \cup \{a_1\}}, \ \dots, \ \mathbf{1}_{S \cup A}. \quad (7.12)$$

The triangulations agree on faces: setting a free coordinate to zero removes it, while setting it to one moves it to S . They therefore define a simplicial complex K of dimension at most k .

We use the usual polyhedral topology on $|K|$. To verify continuity without placing a product topology on the infinite coordinate space, fix a point of M and a neighborhood meeting only finitely many coordinate supports. On that neighborhood all other coordinates are identically zero. The vector (t_i) thus takes values in the finite cubical complex using this finite set of labels, and is a continuous map to that complex. Its staircase triangulation is a finite geometric simplicial complex, whose inclusion in $|K|$ is continuous. The coordinate map consequently defines a continuous map

$$F : M \longrightarrow |K|, \quad \dim K \leq n - 2. \quad (7.13)$$

The triangulation retains the entire coordinate vector: the affine map from each simplex in (7.12) to its coordinate simplex agrees on faces and is injective on their union. One can see injectivity by restricting any two finite supports to their finite union; there it is the usual triangulation of a cubical complex. Thus $F(x) = F(y)$ implies $t_i(x) = t_i(y)$ for every i .

Choose, for any two points of the same fiber, a coordinate that equals 1 at the first point. It also equals 1 at the second. By (i), both collapse images lie in the same open outer ball, and hence

$$d_0(x, y) \leq d_0(x, q(x)) + d_0(q(x), x_i) + d_0(x_i, q(y)) + d_0(q(y), y) < 4D + 2.$$

This applies to every pair of points in the fiber, regardless of its number of components. Consequently

$$\sup_{z \in |K|} \text{diam}_g F^{-1}(z) \leq 4D + 2 = 4WR_0 + 2. \quad (7.14)$$

Together with Theorems 4.3 and 6.6, this proves Theorem 1.1.

8 Universal covers and the uniform simplex metric

We use the continuous macroscopic dimension of a metric space X : $\dim_{\text{mc}} X \leq d$ means that a continuous map from X to a simplicial complex of dimension at most d , with its uniform simplex metric, has uniformly bounded entire fibers. Here each simplex is a standard Euclidean simplex with the same edge length. This is the continuous convention in Dranishnikov [13, Definitions 1.1–1.2]. We first convert the map already constructed to this target metric. We then supply the two lower dimensions using the classical geodesic index argument and the width theorem of Liokumovich and Wang.

Lemma 8.1 (Changing the simplex metric without changing fibers). *Let $k \geq 0$ be an integer. Suppose continuous functions $t_i : Y \rightarrow [0, 1]$ have locally finite supports, at least one equals 1 at each point, and at most k are strictly between 0 and 1. Their coordinate map admits an injective conversion to a simplicial complex of dimension at most k with its uniform simplex metric. The converted map is continuous, and two of its values agree if and only if all the original coordinates agree.*

Proof. Use the cubes $Q(S, A)$ and staircase simplices of (7.12), where S, A are disjoint finite sets, $S \neq \emptyset$, and $|A| \leq k$. Realize the abstract simplices as standard simplices, and send each coordinate simplex to its new realization with the same barycentric coordinates. The rules agree on common faces. Equality of two images identifies their minimal simplex and its positive barycentric coordinates; the affine combination of the corresponding finite indicator vectors recovers the same original coordinate vector. Thus the conversion is injective.

For $k \geq 1$, give each simplex edge length 1 and define d_K as the infimum of the total Euclidean lengths of finite piecewise straight paths in the simplices. This infimum is finite: vertices $\mathbf{1}_S$ and $\mathbf{1}_T$ are joined by adding the coordinates in $T \setminus S$ and then removing those in $S \setminus T$, always retaining a nonempty set of coordinates equal to 1. Every point is joined to a vertex in its simplex. For a vertex v , its barycentric coordinate b_v , set to zero on simplices not containing v , agrees on faces and is $\sqrt{2}$ -Lipschitz on each unit-edge simplex. Indeed such a simplex is realized by the vectors $e_v/\sqrt{2}$ in a Euclidean coordinate space. Hence

$$|b_v(x) - b_v(y)| \leq \sqrt{2} d_K(x, y).$$

The barycentric coordinates distinguish points, so d_K is a genuine finite metric. For $k = 0$ use the discrete vertex metric.

Near a point of Y , only a finite set F of coordinates can be nonzero. The coordinate map there lands continuously in the finite cubical subcomplex with all other coordinates zero. The conversion is continuous on each of its closed simplices and hence on the finite complex. Inclusion into the full uniform realization is continuous on a simplex because distance there is at most the length of its straight segment. Finite pasting therefore proves continuity on the entire finite subcomplex, and thus near every point of Y . The same local finite argument applies for $k = 0$. No equality between two topologies on an arbitrary infinite complex is required. $\square$

Lemma 8.2 (Level components of a proper Morse function). *Let $u : N \rightarrow \mathbb{R}$ be a proper Morse function on a connected smooth manifold without boundary. There is a locally finite simplicial graph K and a continuous map $q : N \rightarrow K$, with K carrying its uniform simplex metric, whose point fibers are precisely the connected components of the levels of u .*

Proof. Choose an auxiliary smooth Riemannian metric on N to define gradients. The critical set is closed and its points are isolated. Properness therefore makes the number of critical points in any compact slab $u^{-1}([a, b])$ finite. Thus the critical values form a locally finite subset of $\mathbb{R}$, even if several critical points have the same value. Every level is compact and locally path connected: regular charts give a hypersurface, while the Morse normal form at a critical point gives the zero set of a nondegenerate quadratic form, which is a cone or an isolated point. A compact locally path connected space has finitely many components, since its components are open and form an open cover.

Let S consist of the critical values and the integers. It is locally finite. At each $a \in S$ put a vertex for every component of $u^{-1}(a)$. Between consecutive values $a < b$ of S , the flow of $\nabla u/|\nabla u|^2$ identifies $u^{-1}((a, b))$ with a regular level times (a, b) . The flow exists for each compact

subinterval because its trajectory stays in a compact slab. Each of the finitely many connected product components defines an open edge, parametrized by its u -value. The image of a proper map to $\mathbb{R}$ is closed; any finite endpoint of $u(N)$ is an attained extremum and hence a critical value. Thus no further finite endpoints are missing.

An edge component E has a unique attachment at each end. At its lower end, the nonempty compact connected sets

$$\overline{E \cap u^{-1}((a, a + \epsilon])}, \quad 0 < \epsilon < b - a,$$

are nested as $\epsilon \downarrow 0$. Their intersection equals $\overline{E} \cap u^{-1}(a)$: a point of this latter set is approached by points of E whose heights tend to a , and therefore belongs to every displayed closure. The intersection is nonempty, connected and contained in $u^{-1}(a)$, hence in exactly one of that level's components. Attach the edge to its vertex. The same argument applies at b . There are finitely many edges in either adjacent slab, so every vertex has finite valence. The resulting graph is countable and locally finite. Its points correspond exactly to the connected components of all levels, using full level components as vertices when critical values coincide.

Define q by this correspondence. On an open edge it is continuous by the product description. At a vertex corresponding to a compact level component $C \subset u^{-1}(a)$, an adjacent edge not attached to C has no sequence approaching C : such a sequence would put a point of C into its endpoint intersection, contrary to its unique attachment. The finitely many other components of $u^{-1}(a)$ are disjoint compact sets. A neighborhood of C can therefore exclude them and the finitely many nonincident edges. Requiring also that $|u - a|$ be sufficiently small places its remaining points in any prescribed short tails of the finitely many incident edges. This proves continuity at the vertex and shows that its whole inverse image is C . The graph is connected, since q is a continuous surjection and N is connected.

Subdivide every edge once with a different midpoint vertex for each edge. This removes parallel edges from the abstract realization; loops do not occur because the endpoint heights differ. The result is a simplicial graph. Give each new edge the same length. Finite valence implies that its ordinary graph topology equals the topology of this uniform simplex metric: vertex neighborhoods contain short segments on each of finitely many incident edges. The subdivision and change of edge lengths preserve points and fibers. Thus q has the asserted continuity and exact fibers. $\square$

Proof of Theorem 1.3. Put $s = \min_M \text{Scal}_h > 0$ and $X = \widetilde{M}$. The lifted metric is complete. If X is compact, use a constant map to a vertex, whose only nonempty fiber has finite diameter.

For $n \geq 4$, the complete metric $s\tilde{h}$ has scalar curvature at least 1. Apply Theorem 1.1 using the coordinates of (7.11), then Theorem 8.1. Local finiteness of their supports was proved in Section 7, and the conversion preserves their entire fibers. Scaling back gives the bound $C_n/\sqrt{s}$ in the original metric.

For $n = 2$, the Gaussian curvature of the cover is at least $s/2$. If a unit-speed minimizing geodesic had length $L > \pi\sqrt{2/s}$, take a parallel unit normal E along it and the endpoint-zero variation field $V(t) = \sin(\pi t/L)E(t)$. Its index form obeys

$$I(V, V) = \int_0^L (|\nabla_t V|^2 - K|V|^2) dt \leq \frac{\pi^2}{2L} - \frac{sL}{4} < 0,$$

contradicting minimality. Completeness supplies minimizing geodesics, so the cover has diameter at most $\pi\sqrt{2/s}$ and is compact by Hopf–Rinow. The constant map therefore proves this case.

Finally let $n = 3$ and X be noncompact. Apply Liokumovich and Wang [26, Theorem 1.1, arXiv version 1] to the complete metric $(s/6)\tilde{h}$, whose scalar curvature is at least 6. It gives a proper Morse function such that every connected component of every level, including critical levels, has

ambient diameter less than 18π in that metric. Apply Theorem 8.2. Each whole fiber of its graph map is one such level component, so in the original lifted metric it has diameter less than $18\pi\sqrt{6/s}$. This is a continuous map to a one-dimensional uniform simplicial complex, as required. $\square$

The low-dimensional conclusions use their stated classical and public inputs. The quantitative theorem in dimensions at least four has the separate proof given above. All map statements here are continuous; changing the simplex metric does not supply a Lipschitz bound.

A Packets, finite horizons, and the induction invariant

This appendix gives the geometry and bookkeeping for a finite number of partitions. The minimization, shielding, and regularization arguments in the following appendices construct the regular cuts to which this geometry applies. The passage to countably many labels is proved in Theorem 6.6.

The packet diagrams, finite stage constants, analytic invariant, and radial initialization are stated in Section 5. We prove here their geometric properties and the compatibility of finite reconstructions.

A.1 Product ends and regular splitting

Lemma A.1 (Compatible splitting and finite truncations). *Suppose a packet diagram after the radial stage has the properties in Theorem 5.1. On every packet let g_E be a smooth compatible function, with 0 a regular value, including on every face. Keep $L_E = g_E^{-1}(0)$ as the new hit space. To each closed side $\{\pm g_E \geq 0\}$ attach $L_E \times [0, \infty)$ along its boundary. Use smooth normal collars across the attachment, chosen as products in all sufficiently far old tails. Connected components of these spaces, with the appended statuses, form a new packet diagram.*

It has finite face and coface incidence, properly embedded closed tails, and compact remainders. Each packet admits a compact truncation which is a smooth manifold with corners and is a strong deformation retract of the full packet. Truncations can be chosen simultaneously on every finite downward diagram, with their faces equal to the corresponding truncated packets. If thresholds are in the product range of g_E , the old retractions preserve its zero set and both closed sides. These conclusions also hold at the first radial stage provided that, before attachment, its wall and closed-side components are compact and form a locally finite family in X .

Proof. First work in the finite downward diagram of one old packet. Raise all entry thresholds until every overlap is in a commuting quadrant, and until all the functions in this diagram are products there. For an elementary normal coordinate t_i choose a larger horizon h_i . On its tail use $t_i \mapsto \min(t_i, h_i)$ and the homotopy $t_i \mapsto (1-s)t_i + s\min(t_i, h_i)$; use the identity elsewhere. The formulas agree continuously at $t_i = h_i$. In an overlap the other normal coordinates are unchanged and the formulas commute. One may therefore perform these clamps in any order. Their joint image is compact: remove the far elementary tails, and then inductively truncate each of the finitely many face tails. This gives a finite union of compact products. Its boundary strata have precisely the claimed product descriptions. Each clamp fixes the cutting function, since that function is a face pullback throughout the moving region.

The regular zero set in this compact manifold with corners is transverse to every corner stratum. Indeed on a far old face the derivative in face directions is the derivative of the regular face function. Consequently the compact zero set and the two compact closed sides are manifolds with corners. Each has finitely many components: every point has a connected neighborhood in the relevant manifold with corners, and compactness supplies a finite such cover. Clamping shows that a full zero or side component has a nonempty connected intersection with the truncation. Conversely a

connection in the full set clamps to a connection in the truncation. Thus no additional components or connections occur only at infinity.

On an old tail the whole operation is literally the same operation on its face times the old normal intervals. In particular the cylinder attached to a face zero set is the same subset in every order of passing to that face. For the new sign-to-hit relation the tail is exactly the newly attached cylinder. These observations prove all product and commutation rules, including the unique occurrence for a prescribed coarser status: the outgoing product of a connected face packet is connected and belongs to a unique coarser component. A new packet has finitely many faces by the compact truncation just constructed. A face has at most one incident packet for each coarser status, of which there are finitely many. Applying these assertions through the finite status depth proves finiteness in both directions.

For clarity, properness of tails is part of this argument, not an inference from a metric completeness assertion. Fresh closed cylinder tails are proper by the attachment topology. The continuous collapse of a new space to its old parent sends a compact set to a compact set. On an inherited tail, old properness then bounds its old parameters in a compact subset of the old tail; the fresh cylinder length is also bounded on the given compact set. The remaining zero and side conditions are closed. Thus the inverse image of that compact set in the inherited closed tail is compact. Equivalently, it is contained in the union of closed old pieces and bounded fresh cylinders over compact zero pieces. This also shows that the complement of all new far tails is compact. Clamp the fresh cylinder coordinate as well to obtain the asserted new compact truncations and retractions.

At the radial stage there are initially no old tails. The hypotheses give compact wall and side pieces, with finite incidence on each compact piece, and the fresh cylinders start the preceding induction. All diagrams remain countable, because all manifolds involved are second countable and the radial family is locally finite.

For later metric uses, normal collars can be obtained from any compatible smooth parent metric. In an old product tail, both the regular defining function and the metric are pulled back from the face, so the unit normal and its normal geodesics split off the old normal coordinates. On a compact truncation of the zero set choose a positive tubular radius by finitely many normal-coordinate charts. Induction through its finite face diagram extends this radius to a positive uniform radius on that parent packet: any failure would be in a product over a face where the radius was already chosen. Thus these collars are products on all old ends; at the radial step the same choice is made separately on each incident compact block. $\square$

A.2 Completeness, growth, and product limits

Lemma A.2 (Metric geometry of a packet). *Equip a packet diagram after the radial stage with compatible smooth base metrics and finitely many compatible smooth positive warp functions. Each connected packet E , separately, is complete and has bounded geometry: positive injectivity radius and bounds on curvature and every covariant derivative. Every warp and all its covariant derivatives are bounded on E , and every warp has a positive lower bound. For ω_E the product of the warps there are constants $A_E < \infty$ and an integer $d_E \leq \dim E$ such that*

$$\int_{B_E(x,R)} \omega_E \, d \text{vol}_E \leq A_E (1+R)^{d_E} \quad (x \in E, \, R \geq 0). \quad (\text{A.1})$$

The constant is uniform over centers in this packet. The same assertions hold on $E \times \mathbb{R}^a$, with the product metric, pulled-back warps, and growth exponent at most $d_E + a$, for every finite a .

Proof. Induct upward from packets with no faces, which are compact. A full elementary product tail over a complete face has no finite-distance exit through its face coordinate. More precisely, a

finite-length curve in that product whose normal coordinate stays above the entry level has Cauchy face and normal coordinates. Completeness of the face and properness of the closed tail keep its limit in the product. It follows that, after fixing a closed-tail entry level T_i , the function

$$\rho_i = \max(t_i - T_i, 0) \quad \text{on that tail}, \quad \rho_i = 0 \quad \text{outside it}$$

is continuous and globally 1-Lipschitz. The only exit is through $t_i = T_i$, and its differential has norm at most one wherever defined. A Cauchy sequence has all the finitely many ρ_i bounded. The compact-truncation assertion of Theorem A.1 puts such a sequence in a compact set. Hence the packet is complete.

Recursively subdividing each coordinate into a bounded interval and its far range gives a finite cover of E by compact sets and sectors

$$K_B \times [T_1, \infty) \times \cdots \times [T_d, \infty), \tag{A.2}$$

where K_B is a compact subset of a further face and the metric and warps are products. The recursion terminates because a face decreases base dimension. On such a sector each displayed normal coordinate is also the corresponding elementary coordinate on E , by commutation. In an R -ball its range therefore has length at most $2R$, by the Lipschitz functions ρ_i (enlarge by a fixed bounded interval at an entry if necessary). The sector's weighted volume in that ball is bounded by its compact base volume times a constant multiple of $(1 + R)^d$. Summing the finitely many bounds proves Equation (A.1), for all centers, including centers outside the sector. Notice that $d \leq \dim E$.

Bounds on smooth data follow from the same finite sector decomposition: on compact pieces they are finite, and on a product they are precisely face bounds. For bounded geometry one can equivalently use uniform charts. Any failure of a uniform chart radius or of a fixed-order chart bound has an escaping sequence; a subsequence lies in a far product over a compact part of a further face, as detailed in Theorem A.3. Fixed-radius neighborhoods there are product neighborhoods with uniform smooth chart bounds and a positive injectivity radius. Compact remaining regions have the same properties by a finite chart cover. This proves bounded geometry and all warp bounds. The product assertions follow either by the same proof or from $B_{E \times \mathbb{R}^a}((x, z), R) \subset B_E(x, R) \times B_{\mathbb{R}^a}(z, R)$. $\square$

Lemma A.3 (Escaping sequences and product windows). *If a sequence escapes all compact sets of a fixed packet E , a subsequence has coordinates $(x_\nu, t_{1,\nu}, \dots, t_{d,\nu})$ in one further-face product, where $d \geq 1$, x_ν converges in a compact subset of the face B , and every $t_{i,\nu} \rightarrow \infty$. After translating these normal coordinates, the product charts contain each fixed metric neighborhood of the sequence for all sufficiently large ν . They exhaust $B \times \mathbb{R}^d$ locally. All compatible smooth data equal the corresponding product data on these windows.*

Proof. Choose a sector from the finite cover in Equation (A.2) containing a subsequence. Pass to a subsequence so each of its normal coordinates either converges finitely or tends to infinity. The compact base coordinate also converges. Absorb the bounded normal coordinates into the base of the intermediate face specified by the remaining, unbounded coordinates. Commutation identifies this with the corresponding full product occurrence. At least one unbounded coordinate remains, since otherwise the sequence lies in a compact truncation. A path of length at most R changes each normal coordinate by at most R . It cannot escape through a face coordinate, by completeness of the face as in Theorem A.2. Thus a fixed-radius neighborhood lies in the product once all its normal coordinates exceed the entry levels by more than R . Translating yields the stated exhaustion. This argument uses only the completeness part of Theorem A.2, which was established before its bounded-geometry conclusion. $\square$

The initial metric and warps need not have bounded geometry on X . At the radial split each bulk piece and each wall piece is compact before cylinder attachment. The smooth data have finite derivative bounds there, the positive warps have positive minima, and the cylinders freeze those data. This initializes Theorem A.2. For a locally finite initial warp list, a compact radial block and its compact buffer meet only finitely many nonunit factors; the same argument applies after omitting the unit factors. Polynomial growth is a conclusion for a single post-radial packet, never an assumption on the original complete manifold.

A.3 Compatible choices

Lemma A.4 (Compatible choices and finite tolerance allocation). *On a packet diagram after the radial stage, choices of smooth cutoffs and approximations can be made from deeper faces to cofaces, retaining exact products outside finite thresholds. More precisely, fix a finite derivative order and, on each packet, positive error tolerances for any finite collection of smooth data or approximations. A datum on a face may additionally be required to meet the tolerances of all its incident cofaces, including through chains. These requirements can be replaced in advance by strictly positive face tolerances, and then met by compatible blending, provided the individual local approximations and their required convergence to the face data are available. In particular there exist compatible smooth positive functions arbitrarily small in any prescribed finite derivative norm on each packet. Also, if compatible closed sets F_E and compatible open sets O_E satisfy $F_E \subset O_E$ on every packet, then $\text{dist}_E(F_E, E \setminus O_E) > 0$ on each fixed packet whenever both sets are nonempty.*

Proof. There are only finitely many requests on a particular face, because there are only finitely many coface chains. Starting with the given requests, propagate stricter tolerances down these chains, allowing for the finitely many product, derivative, and cutoff constants. Take finite minima; no minimum over an infinite adjacency component is taken. This finishes all tolerance choices before construction begins. A countable collection of packets causes no problem, since each one has finitely many faces and cofaces and the depth is finite.

For the construction on one packet, enumerate its elementary tails. Choose smooth cutoffs $f_i(t_i)$ rising from zero to one sufficiently far out, with bounded derivatives through the chosen order, and extend them by zero off their tails. Write u_0 for an individual approximation and u_i for the already chosen face copy. The weights

$$w_0 = \prod_i (1 - f_i), \quad w_i = f_i \prod_{a < i} (1 - f_a), \quad w_0 + \sum_i w_i = 1 \quad (\text{A.3})$$

produce $w_0 u_0 + \sum_i w_i u_i$. Put the transitions beyond all relevant intersection thresholds. On an overlap the participating face copies then agree by their own product rules. In particular, wherever some $f_i = 1$, every nonzero term is the same face copy. The result is an exact copy there. Leibniz' rule bounds the derivative error by a finite multiple of the preassigned errors. A perturbation on the remaining compact region changes none of these end identities. This proves the conditional approximation assertion. It does not assert convergence for a temporary nonproduct object; whenever such an object is used below, its convergence is proved separately.

For positive small functions start with positive constants on the compact deepest packets, take positive small individual choices on compact parts, and use the positive weights above. Decrease the preassigned tolerances to cover all coface derivative requests. This gives strict positivity at every point and any stipulated finite norm bounds, including the C^2 bounds used for seam widths. For the separation assertion, a sequence of distances tending to zero either stays in a compact region, where closedness and inclusion in the open set give a contradiction, or escapes. In the latter case

Theorem A.3 supplies a common product window over a compact part of a deeper face. The two sequences project to the same limit point, lying both in F_B and in $B \setminus O_B$, a contradiction. $\square$

A.4 Finite reconstruction

Proposition A.5 (Finite-horizon reconstruction). *Starting with X , perform finitely many compatible regular splits as in Theorem A.1. For the first split require locally finite compact radial wall and side components. At the later split for color i , require the defining functions to be positive on the inverse image of A_i and negative on the inverse image of B_i , with those inverse images contained in open regions of the stipulated signs. After any $r \geq 1$ of these splits, the diagram has a reconstruction X_r , obtained by finite collar truncations and gluing at hit faces, with the following properties:*

- (i) X_r is a smooth manifold homeomorphic to X . Its r labelled walls are closed smooth hypersurfaces with simultaneous transverse intersection charts. Each wall component is compact, the component families are locally finite, and the closed sign/hit pieces are exactly the compact cornered truncations of the analytic packets, by components.
- (ii) The packet maps glue to a proper continuous map $q_r : X_r \rightarrow X$. There is a homeomorphism $H_r : X_r \rightarrow X$ such that
$$d_0(H_r(z), q_r(z)) \leq r \quad (z \in X_r). \quad (\text{A.4})$$
- (iii) Horizons can be enlarged to exceed any already specified compatible product thresholds on the full analytic packets. A horizon is constant for all occurrences of a normal coordinate belonging to one connected global wall component. Such enlargement has an identification with the previous reconstruction which preserves q_r , every old wall, and every old side status.
- (iv) Once color i has been processed, $q_r^{-1}(A_i)$ remains in its open positive phase and $q_r^{-1}(B_i)$ remains in its open negative phase at every subsequent stage. In particular its wall misses $q_r^{-1}(A_i \cup B_i)$.

Proof. We prove the assertions in their order of dependence. At stage zero use $X_0 = X$ and $q_0 = H_0 = \text{Id}$. At the radial split, the assumed compact pieces give the first finite reconstruction by inserting cylinders of finite length on the two sides of each wall component. The properness and homeomorphism argument below applies to this locally finite compact family. Its compact pieces also initialize the analytic diagram by Theorem A.1.

Suppose the reconstruction through stage r has been made. Fix all product thresholds for the next cutting functions and their collar metrics in the *full analytic coordinates* first. Consider a connected component W of an existing global wall. A packet with the corresponding hit status has a nonempty connected truncation, by the commuting retractions of Theorem A.1; it therefore lies in a single such W . The same assignment holds at all incidences through further faces, since the corresponding truncations glue along those faces. Compactness of W and local finiteness of the reconstructed pieces imply that only finitely many such pieces meet W . Thus there are only finitely many threshold requests for its normal coordinate. Choose one horizon exceeding all of them, with a positive buffer. Do this for each existing wall component and coordinate.

This is a direct choice from previously fixed finite lists. Increasing a horizon cannot introduce a new analytic threshold: the functions and the incidence diagram were fixed on the full infinite tails. On a tail where topology and q_r are already products, choose an increasing smooth reparameterization from the enlarged finite normal interval to the old one, equal to the identity before the old product range and sending the new endpoint to the old endpoint. Choose the same reparameterization at all occurrences of W , on both sides of its central hit wall. It preserves those sides and the hit wall.

At corners the separately labelled normal reparameterizations commute. Their product gives an identification $\Phi_r : \widehat{X}_r \rightarrow X_r$ preserving every wall and sign status. The glued position maps satisfy

$$\widehat{q}_r = q_r \circ \Phi_r, \quad \widehat{H}_r = H_r \circ \Phi_r, \quad d_0(\widehat{H}_r, \widehat{q}_r) \leq r,$$

because each position map is independent of the moved normal variables. Properness is preserved by this homeomorphism. Thus horizon enlargement introduces no error or change of position data. For the rest of the step, relabel these enlarged representatives as X_r, q_r, H_r .

The compatible cutting functions now glue to a single smooth function on this enlarged X_r . Near any old wall stratum the function is a pullback from the hit piece, independent of all its signed normal coordinates. Its zero set L is therefore regular and transverse to every old wall stratum. More explicitly, in a chart $(b, s_1, \dots, s_d)$ at an old stratum, the old walls are $s_j = 0$ and the new function is $g_B(b)$. The face hypothesis says $dg_B \neq 0$ at its zeros. Hence the old normal covectors $ds_1, \dots, ds_d$ together with dg_B are independent. This also covers a stratum involving the radial wall. At this point L has been formed without choosing a new horizon. For a color split its forced signs give

$$q_r(L) \subset X \setminus (A_i \cup B_i).$$

In the present one-ball-per-label setup, $L \subset q_r^{-1}(B_{d_0}(x_i, 2D))$. The latter set has compact closure because q_r is proper and the original closed outer ball is compact. Since L is closed in X_r , the whole color wall L is compact. In particular each component is compact, and there are only finitely many components. More generally, a closed regular hypersurface with compact components has a locally finite component family: a regular hypersurface chart meets only its own component, and compact subsets of the hypersurface have finite such covers. The first radial wall need not be compact as a whole; its stated locally finite compact-component hypothesis gives exactly what is needed for the same construction.

Only now assign a finite new horizon to each component of L . Cut along L , attach a cylinder on each side, and identify the two outer cylinder boundaries as the new central hit wall. Near an old wall this is the product of the operation on its hit face with the old signed normal coordinate. The same statement holds at multiple old intersections. Thus old walls extend smoothly through the new cylinders, and all walls have transverse intersection charts. The resulting pieces are exactly the truncations of the newly appended statuses: the old clamps preserve L and its signs and extend across fresh cylinders keeping their new coordinate fixed; the fresh-coordinate clamp commutes with all of them. In particular every full new packet retracts onto one nonempty connected piece in the reconstruction. This verifies the component assertion needed to assign its coordinate at the next stage.

Let $c : X_{r+1} \rightarrow X_r$ collapse the inserted cylinders to L and be the identity on the bulk. This map is proper. A compact subset of X_r meets only finitely many new wall components; its inverse image is contained in its compact bulk copy together with finitely many finite cylinders over compact subsets of these compact components. The glued map is $q_{r+1} = q_r \circ c$, and so is proper. Every old compact wall meets finitely many new components, and its transverse intersection with them is compact. Its extension adds only finitely many compact finite cylinders; the old wall therefore remains compact. Local finiteness of all pieces follows locally from the finite collection of transverse wall charts, together with the compact truncation descriptions. This proves (i) and the properness assertion without using any later wall.

We next choose collars that preserve all old strata. The compatible packet metrics glue on the enlarged reconstruction: near an old hit stratum they have the form $h_B + \sum_j ds_j^2$. On a neighborhood

of L , write u for the glued cutting function and use the transverse field

$$V = \frac{\text{grad } u}{|\text{grad } u|^2}, \quad V(u) = 1.$$

In each old product chart, $u = g_B(b)$ and this field has no old normal component. Thus it is tangent to every old wall meeting that chart and is the pullback of the corresponding face field. Equivalently, one can patch such local fields with a compatible partition of unity: $V(u) = 1$ and tangency are linear conditions, so they survive patching. The flow supplies product collars at old intersections. It preserves every old wall and each side, by uniqueness of the flow and continuity from the identity. This explains why a transverse ordinary collar chosen without the product condition would not suffice here.

Choose pairwise disjoint locally finite strips around the compact components of L using this flow. A positive strip width exists on each component by compactness; it may be decreased to keep strips disjoint. For each component, continuity of q_r and compactness permit the width to be so small that q_r -images of any two points on the same flow segment have d_0 -distance less than one. A monotone normal reparameterization compresses the two finite inserted intervals and the old strip into that old strip, fixing its outer boundary. It produces a homeomorphism $h : X_{r+1} \rightarrow X_r$, equal to the identity off the strips, with

$$d_0(q_r(h(z)), q_r(c(z))) < 1.$$

Different strip widths and cylinder lengths are allowed on different components. Local finiteness makes these definitions a global homeomorphism. Set $H_{r+1} = H_r \circ h$. The triangle inequality and Equation (A.4) give the new bound $r + 1$. The compression follows the stratum-preserving collar, so it also preserves every old wall and its two side statuses. The old signs are pulled back under c , including across the cylinders, and the newly imposed signs hold on the prescribed inverse images. Pullbacks of their open phases stay open. This proves (iv) and finishes the induction. $\square$

Remark A.6 (Prescribed reconstruction errors). The number 1 in the last collar comparison can be replaced at stage r by any prescribed $\varepsilon_r > 0$. Each compact wall component has its strip chosen separately, with the same error bound ε_r on every component. Disjoint strips give a supremum bound, rather than a sum over components. The resulting error after finitely many stages is at most $\sum_{j < r} \varepsilon_j$. This freedom concerns the original position map and is independent of the analytic lengths of the inserted finite collars. The spatial localization and limit of such reconstructions are treated in the main reconstruction argument.

A.5 Initialization

Proof of Theorem 5.4. The decreasing solution of $2|y'| = .5y^2 + \pi_0$, from $+\infty$ to $-\infty$, has finite crossing length

$$\int_{-\infty}^{\infty} \frac{2 \, dy}{.5y^2 + \pi_0} = 2\pi\sqrt{2/\pi_0} = T_0.$$

Extend it by $+\infty$ before the crossing and by $-\infty$ after it. This is continuous as an extended-real-valued function, and all its finite truncations are locally Lipschitz. Composing the transition, begun at distance 2 from A_i , with the 1-Lipschitz function $d_h(A_i, \cdot)$ gives

$$2|\text{grad}_h \mu_i| \leq .5\mu_i^2 + \pi_0 \leq a_0\mu_i^2 + \pi_0$$

almost everywhere on the finite band. Take

$$U_i^+ = \{d_h(A_i, \cdot) < 1\}, \quad U_i^- = \{d_h(A_i, \cdot) > T_0 + 3\}.$$

Their closures lie strictly inside the respective pole interiors; they contain A_i and B_i . The requested finite separations are available from Theorem 4.3 after N_* , and hence T_0 , are fixed. Their size does not change the original radius D .

We give the radial construction without a bounded-geometry hypothesis. Assume first that X is noncompact, fix $o \in X$, and put $d(x) = d_h(o, x)$. Choose a countable locally finite cover by relatively compact coordinate neighborhoods on which the metric oscillation is small, and a subordinate smooth partition of unity $\{\chi_j\}_{j \geq 1}$ whose supports are compact in those neighborhoods. Convolution in a slightly larger coordinate neighborhood gives smooth d_j with

$$|d_j - d| < \epsilon_j, \quad |dd_j|_h \leq \frac{3}{2} \quad \text{near } \text{supp } \chi_j.$$

The derivative estimate follows from the local Lipschitz bound for d and the small metric oscillation; the convolution radius can then be reduced to make its value error arbitrarily small. Let $C_j = \sup |d\chi_j|_h < \infty$, and choose $0 < \epsilon_j < \min\{1/2, 2^{-j-2}/(1 + C_j)\}$. The locally finite sum $\rho = 2 + \sum_j \chi_j d_j$ is smooth and $|\rho - (2 + d)| < 1/2$. Moreover, since $\sum_j d\chi_j = 0$,

$$d\rho = \sum_j \chi_j dd_j + \sum_j (d_j - d)d\chi_j, \quad |d\rho|_h \leq \frac{3}{2} + \sum_j 2^{-j-2} < 2.$$

Completeness makes d proper, so ρ is proper and bounded below. Thus a smooth proper radius with bounded gradient has been constructed using only local smoothness and completeness.

Choose successive plateau intervals of the values of ρ , with alternating signs and positive-width cores strictly inside each plateau. Separate successive plateaus by transition intervals of width greater than $2T_0$. On such an interval, a copy of the crossing y , parametrized by $(\rho - a)/L$ with $L > 2$ and with sign reversed on alternate intervals, obeys the same profile inequality because $|d\rho|_h/L \leq 1$. Include an initial plateau extending below the minimum of ρ . Core closures have positive value margins to both ends of their plateaus, except at the harmless lower end of the initial plateau. Sard's theorem supplies regular values in those margins. The sublevel domain at the initial plateau, and the domains between two such regular levels at later plateaus, are compact smooth barriers enclosing the corresponding core closures and lying in the same pole interiors. These barriers and the transition blocks form locally finite families, since their value intervals tend to infinity. Connectedness and properness imply that the image of ρ contains all sufficiently large values, so the separating subbands are nonempty.

A zero wall respecting the signs misses every separating core subband. On any connected wall component, the continuous function ρ therefore takes values in one bounded intervening block. The component is closed in X , and properness of ρ makes it compact. For a fixed sign, successive core subbands of the opposite sign bound the ρ -values of every connected component of the closed side; the central component is bounded by the first opposite separator. These closed-side components are compact as well. In a compact block, regularity gives a finite cover by charts in which the wall is a single sheet and each closed side is a half-space or the full chart. Only finitely many components can meet that block. The blocks are locally finite, which proves the required local finiteness of wall and side components.

If X is compact, set the radial profile identically $+\infty$, with $U_{\text{rad}}^+ = X$ and $U_{\text{rad}}^- = \emptyset$. The constant positive defining function realizes its empty wall; the single nonempty closed side is X , which is compact. All radial profile and confinement assertions then hold directly.

Finally, the initial scalar requirement holds everywhere, since

$$\lambda_0 + b_0 \sum_{j \in J_0} \Pi_j = \frac{1}{4} + 1.12N_* \frac{1}{10N_*} = .362 < .4 \leq R_X.$$

The profile, core, and forced-preimage conditions have already been verified. This proves the invariant at stage zero. $\square$

The geometric lemmas require compatible regular defining functions. The minimizing boundaries used to obtain them may initially be singular. The remaining appendices first construct these minimizing partitions, enlarge the remaining cores until the required guards avoid the singular sets, and then produce the regular compatible functions to which Theorem A.1 applies. The curvature restriction to the required sets is respected precisely by Theorem 5.3.

B Weighted minimizing cuts and the forms on retained sides

Prescribed-curvature perimeter minimization and convergence of its boundaries have classical precedents in Massari and Massari–Pepe [28, 29]. For scalar-curvature applications, prescribed-mean-curvature minimization is the μ -bubble method described by Gromov [18, §5 $\frac{5}{6}$]. We prove here the weighted, varying-metric and product-end versions required by the cutting construction, including the forms on both retained sides.

Fix a finite stage r and the label j_0 to be processed. The packet and profile hypotheses are those of Theorems 5.1 and 5.2. For the finite-horizon formulation in this appendix, only finitely many labels and finitely many nonunit old warps occur in each incident diagram under consideration. The processed potential has a positive lower bound

$$\Pi_{j_0} \geq \pi_{\text{cut}} > 0.$$

A common lower bound for all labels at all stages is not required. The constants satisfy $a_r \leq .63$, $b_r \geq 1.07$, and $\lambda_r > \lambda_{r+1}$. The extension to a countable collection of labels is proved in the main text; every argument below uses only the current pressure and the indicated finite lists.

Throughout, E denotes a parent packet of base dimension $m \geq 2$. After the radial split it has the compact truncations, finite incidence, and exact product ends of Theorem 5.1. At the initial split we instead work on the complete original base, and use the locally finite compact radial barriers of Theorem 5.4. We assume no volume growth or bounded geometry on that initial base.

On a finite incident diagram, insert unit warps where necessary so that the old warp lists agree. Their metric is

$$\hat{g} = g + \sum_{i=1}^q f_i^2 d\theta_i^2, \quad \omega = \prod_{i=1}^q f_i, \quad d\mu = \omega d \text{vol}_g.$$

Empty products are one. Fiber volume constants are suppressed throughout. Adding a unit warp changes neither ω nor any formula below. At the initial radial stage the list need only be finite on each compact window; the compatible local expressions then agree on overlaps. No infinite-dimensional torus is used. In particular all minimizing sets, BV compactness, tangent cones, and singular-dimension bounds concern the m -dimensional base, regardless of the length of the warp list. Write $R = \text{Scal}_{\hat{g}}$ and $\text{div}_{\omega} Y = \omega^{-1} \text{div}_g(\omega Y)$. For a set A of locally finite perimeter define its weighted perimeter by

$$P_{\omega}(A; U) = \int_{U \cap \partial^* A} \omega d\mathcal{H}_g^{m-1}.$$

The reduced-boundary representation is the framework developed by De Giorgi [11, 12]; the current formulation and slicing originate in Federer and Fleming [15]. We use the precise representation and slicing statements in Simon [36, Chapter 3, Section 4, and Chapter 6, Section 4]. All set equalities and inclusions involving measurable phases are understood up to μ -null sets. The phase representative used for geometry will be specified below. Constants in local estimates may depend

on a compact window; after the radial split they can be chosen uniformly on each fixed packet, without any uniformity between packets. No scalar-curvature lower bound is used to construct the cuts or to prove their local regularity. The parent scalar bound enters only at the end of this appendix, and only on the required region D_E of Equation (5.5).

B.1 Smooth finite pressures and absolute minimizers

Lemma B.1 (Realizing the processed profile). *There is a smooth finite compatible function p_0 and a number $c_* > 0$ such that every $p = p_0 + c$, $|c| < c_*$, satisfies*

$$2|\operatorname{grad} p| \leq .67p^2 + 1.02\Pi_{j_0}. \quad (\text{B.1})$$

There are compatible smooth barrier domains D^+, D^- whose interiors contain the corresponding forced-core closures and whose closures lie in the respective pole interiors. Each admits a smooth vector field $Y^\pm$ with $|Y^\pm| \leq 1$, equal to the outward unit normal on its boundary, such that, for all these shifts,

$$p \geq 1 + \operatorname{div}_\omega Y^+ \quad \text{on } D^+, \quad -p \geq 1 + \operatorname{div}_\omega Y^- \quad \text{on } D^-. \quad (\text{B.2})$$

After the radial split all this data has bounded derivatives on each packet. At the initial radial split the barriers are the separate relatively compact domains in the radial pole bands, and the choices are locally finite with locally prescribed bounds.

Proof. The nesting of the core closures inside the pole interiors and the compatible approximation construction of Theorem A.4 give smooth separating cutoff functions. A level which is regular on every packet gives the required domains: there are only countably many packets, so Sard's theorem permits a common level. Their far collars are exact copies of the face domains. On a fixed packet, compact parts and the finite face diagram give a positive tubular radius and bounded geometry for the resulting boundary. Extend its outward normal in a smaller tube and multiply by a cutoff of absolute value at most one. This supplies $Y^\pm$ and a bound for its weighted divergence. The radial case uses the relatively compact pole barriers individually.

Fix the barriers and their vector fields first. By Theorem A.4, choose a compatible positive smooth function h with sufficiently small value and first derivative, and put $M = h^{-1}$. Require

$$M \geq 3 + |\operatorname{div}_\omega Y^\pm| \quad \text{on } D^\pm, \quad 2|\operatorname{grad} h| \leq a_r.$$

Only finitely many coface requests reach a given packet; at the radial step the requests are locally finite on compact sets. The choices therefore have the stated compatibility. The derivative inequality implies $2|\operatorname{grad} M| \leq a_r M^2$. The clipped function

$$z = \max(-M, \min(\mu_{j_0}, M))$$

is finite and locally Lipschitz. At almost every point its derivative is that of one active branch, and consequently

$$2|\operatorname{grad} z| \leq a_r z^2 + \Pi_{j_0}.$$

This assertion also holds near the original poles, since the clipping levels are locally bounded and finite truncations of μ_{j_0} are locally Lipschitz. On $D^\pm$ one has $z = \pm M$.

Here is the quantitative smoothing fact being used. If a locally Lipschitz function z obeys $|\operatorname{grad} z| \leq F$ almost everywhere for a continuous positive F , then, for arbitrary positive continuous error targets, local convolution in sufficiently small metric charts followed by a partition of unity gives a smooth z_1 with prescribed value error and $|\operatorname{grad} z_1| \leq F$ plus the prescribed error. Indeed

convolution averages the gradient inequality, the oscillation of F and the metric on its support tends to zero, and differentiating the partition of unity produces only the value errors times its fixed derivatives. A locally finite cover permits the errors to be assigned summably at each point. For compatible smoothing, start on deepest faces. On overlaps in far collars the clipped functions and all face copies agree. Blend with the cutoffs in Theorem A.4; the same value-error estimate pays the additional cutoff derivatives. Preallocate smaller errors on faces to meet their finitely many coface requests. Thus the gradient estimate is always compared with the single continuous function $(a_r z^2 + \Pi_{j_0})/2$, rather than with unrelated bounds for the pieces.

Choose the value error at most $\min\{1/4, 10^{-3}\sqrt{\Pi_{j_0}}\}$ and the gradient error so that

$$2|\text{grad } p_0| \leq a_r z^2 + 1.005\Pi_{j_0}.$$

These are positive continuous error targets. For $e = p_0 - z$,

$$.65(z + e)^2 - .63z^2 \geq -20.475e^2,$$

so $a_r \leq .63$ gives

$$2|\text{grad } p_0| \leq .65p_0^2 + 1.01\Pi_{j_0}.$$

On $D^\pm$ the value error and the earlier choice of M give $\pm p_0 \geq 11/4 + |\text{div}_\omega Y^\pm|$. Finally set

$$c_* = \min\{1/2, .01\sqrt{\pi_{\text{cut}}}\}.$$

The elementary identity

$$.67(y + c)^2 - .65y^2 \geq -21.775c^2 \quad (y, c \in \mathbb{R})$$

and $\Pi_{j_0} \geq \pi_{\text{cut}}$ prove (B.1) for every $|c| < c_*$; the barrier margin also persists. Thus the domains, clipping height, smoothing, and open shift interval have all been fixed before selecting a minimizing phase. On a post-radial packet every smooth datum just constructed has bounded derivatives, by its compact parts and exact products. At the radial step the construction gives the corresponding bounds on each compact window, which is all that is used. $\square$

Definition B.2 (Local absolute minimizer). For smooth p a phase A is a local absolute minimizer if for every relatively compact open $U \subset E$ and every locally finite-perimeter set B with $A \triangle B \Subset U$,

$$P_\omega(A; U) - \int_{A \cap U} p \, d\mu \leq P_\omega(B; U) - \int_{B \cap U} p \, d\mu. \quad (\text{B.3})$$

The notation $A \triangle B \Subset U$ means that the discrepancy is contained, modulo null sets, in a compact subset of U .

Lemma B.3 (Existence and strict local compactness). *Every smooth positive weight and smooth locally bounded pressure on a smooth manifold admit a local absolute minimizer. Suppose on exhausting coordinate windows that metrics and positive weights converge smoothly and pressures converge locally uniformly. Any sequence of corresponding local absolute minimizers has a subsequence whose indicators converge in L^1_{loc} to a local absolute minimizer of the limit data. Along that subsequence the weighted perimeter measures converge weakly locally, as do the unweighted perimeter measures and their associated unoriented perimeter varifolds. The assertion includes smooth pointed rescalings when the rescaled data have the stated convergence.*

Proof. We first establish the comparison formulas that will show both minimality of a limiting phase and absence of perimeter loss. We then construct exhaustion minimizers and apply these formulas to locally convergent sequences.

Comparison formulas. Local L^1 compactness under local perimeter bounds and lower semicontinuity are Simon [36, Chapter 2, Theorem 2.6]; positive smooth weights and smooth metrics on a relatively compact chart give equivalent variation norms. The further BV inputs are the perimeter lattice inequality and the trace and gluing formulas, both across a fixed smooth face and across almost every admissible level of a Lipschitz slicing function. The lattice inequality is

$$P_\omega(A \cap B; U) + P_\omega(A \cup B; U) \leq P_\omega(A; U) + P_\omega(B; U). \quad (\text{B.4})$$

For smooth functions u, v , the gradients of $\min(u, v)$ and $\max(u, v)$ are the two original gradients, possibly interchanged, almost everywhere. Strictly approximate the two characteristic functions, apply this identity, and use lower semicontinuity on the minimum and maximum. This gives (B.4), with its smooth metric and weight.

We record the trace argument for a fixed smooth comparison face, since such a face need not be an almost-everywhere selected level. Work first in a flattened coordinate collar $Q \times (-a, a) \subset \mathbb{R}^{m-1} \times \mathbb{R}$, with the face $T = Q \times \{0\}$ and the plus side given by $t > 0$. All collars used here are buffered inside a relatively compact ambient chart. The functions under consideration are integrable and have finite total variation on each smaller collar reaching T ; for the restrictions of our finite-perimeter phases this follows from local finite perimeter in the ambient chart. Local BV regularity only on sets bounded away from T would not suffice.

Let w be such a BV function on the plus half-collar, and let $D_t w$ denote its distributional derivative in the t direction. Choose a compact tangential box K and an open box K' with $K \Subset K' \Subset Q$, and fix $a_0 < a$. Mollify on the open half-collar, using kernels small enough to remain inside it on each fixed strip. Fix a countable exhaustion by buffered tangential boxes and compact height intervals contained in $(0, a)$. Choose one subsequence whose local L^1 errors are summable on every member of this exhaustion, by making its k th error at most 2^{-k} on each of the first k members. Fubini then gives convergence in L^1 on horizontal slices for all heights in one common set of full measure, simultaneously on the exhaustion boxes. Restriction gives the same convergence on K' . For a smooth mollification the fundamental theorem of calculus, followed by integration over K , bounds the difference of two slices by the integral of the absolute vertical derivative on the intervening strip. Convolution bounds this integral by the variation of $D_t w$ on a slightly enlarged strip; the tangential enlargement lies in K' when the kernel is small enough. In the limit the enlarged tangential box lies in K' and the height interval is contained in $[s, t]$; thus the limiting bound is at most $|D_t w|(K' \times [s, t])$. Excluding heights at which $|D_t w|(K' \times \{t\}) > 0$, gives

$$\int_K |w(y, t) - w(y, s)| dy \leq |D_t w|(K' \times (s, t)), \quad 0 < s < t < a_0, \quad (\text{B.5})$$

for all s, t in a common set of full measure. There are at most countably many excluded atom heights, since the variation on the buffered collar is finite.

The right-hand side of (B.5) tends to zero uniformly for good heights $s, t \downarrow 0$: the sets $K' \times (0, \delta)$ decrease to the empty set inside the open half-collar and have finite variation measure. The good slices are therefore Cauchy in $L^1(K)$ and have a limit, denoted $\text{Tr}_+ w$. Traces obtained on nested tangential boxes agree by uniqueness of the L^1 limit, giving a locally integrable trace on T . Reflection gives the analogous trace $\text{Tr}_- w$ from the minus side. For characteristic functions these traces are themselves characteristic functions almost everywhere, since an L^1 limit of characteristic functions has an almost-everywhere convergent subsequence. In particular the phase traces take values in $\{0, 1\}$.

The trace also gives the required integration-by-parts formula. On the plus side, multiply a smooth vector field, compactly supported in the tangential directions and away from the outer end of the collar, by a cutoff which is zero for $t \leq \varepsilon$ and one for $t \geq 2\varepsilon$. Apply the distributional derivative identity inside the open half-collar and let $\varepsilon \downarrow 0$. The cutoff derivative term tends to the pairing of $\text{Tr}_+ w$ with the vector field's t component at T : its derivative has integral one, and the good slices converge in L^1 uniformly as their heights approach zero. The other terms converge by integrability and finite variation. This gives the boundary term with outward normal $-\partial_t$ on the plus side; on the minus side the outward normal is ∂_t . Undoing the smooth coordinate change gives the corresponding Riemannian formula. More explicitly, the smooth coordinate volume density is included in the divergence test; its boundary flux is exactly $\langle X, \nu \rangle d\mathcal{H}_g^{m-1}$. Thus, if $D_g w$ denotes the intrinsic distributional gradient measure, with convention $\int w \text{div}_g X d\text{vol}_g = -\int \langle X, dD_g w \rangle$ for tests supported in the open side, then

$$\int_{U_\pm} w \text{div}_g X d\text{vol}_g = -\int_{U_\pm} \langle X, dD_g w \rangle + \int_T \text{Tr}_\pm w \langle X, \nu_{\text{out}, \pm} \rangle d\mathcal{H}_g^{m-1}.$$

The vector field here has compact support in the collar, so no other boundary contribution occurs.

Now orient ν_T from the minus side to the plus side, and glue a BV function u on the minus side to a BV function v on the plus side. Adding their integration-by-parts identities shows that the gradient measure of the glued function is

$$(D_g u)[U_- + (D_g v)[U_+ + (\text{Tr}_+ v - \text{Tr}_- u)\nu_T \mathcal{H}_g^{m-1}[T.$$

These three vector measures are concentrated on the disjoint measurable sets U_- , U_+ , and T . Their variations therefore add. Multiplying the variation measures by the smooth positive weight ω gives the weighted formula, whose extra interface cost is precisely

$$\int_T \omega |\text{Tr}_+ v - \text{Tr}_- u| d\mathcal{H}_g^{m-1}.$$

We are weighting the variation, not differentiating ω times the glued function. For phases the last integrand is the indicator of the trace-disagreement set, multiplied by ω .

We will also glue across almost every level of a locally Lipschitz slicing function f , whose level sets need not be smooth. For this distinct use, the reduced-boundary representation identifies a phase locally with its integer-multiplicity integration current and identifies boundary mass with perimeter [36, Chapter 3, Section 4, Equation (4.1) and Theorem 4.3]. In a locally oriented chart the phase current has locally finite mass and locally finite boundary mass. Lipschitz coarea and the current slicing and restriction formulas therefore apply [36, Chapter 3, Section 2, Equations (2.8)–(2.9); Chapter 6, Section 4, Lemmas 4.1 and 4.5 and Definition 4.4]. In particular, writing J for either phase current, for almost every level τ one has

$$\langle J, f, \tau \rangle = \partial(J[\{f < \tau\}]) - (\partial J)[\{f < \tau\}].$$

Apply this identity to both phases, excluding also the null set of levels charged by their local volume or perimeter measures. For a top-dimensional phase current the slice has multiplicity one on the sliced phase and zero elsewhere. Indeed the slicing definition restricts its phase indicator to the noncritical part of the level; coarea shows that critical points and any changes of the ambient almost-everywhere representative affect almost no level in positive hypersurface measure. The two phase slices have the same orientation induced by f in the locally oriented chart, so their difference has multiplicity equal to the difference of the two sliced indicators. The boundary of their gluing has the two restricted boundary currents and the difference of their level slices as its interface term.

Its extra weighted mass is bounded by the weighted hypersurface measure of the set where the sliced phase values differ. This is the almost-everywhere interface-cost bound used in the annular comparisons, distance-ball comparisons, and level-set deletions below. A countable collection of phases, including a converging sequence and its limit, can use one common set of admissible levels by intersecting their full-measure sets. If the phase indicators converge locally in L^1 , coarea on a fixed buffered slicing annulus makes the integrated sliced disagreement tend to zero. Pass to a subsequence for which these integrals are summable. Fubini then makes the disagreement tend to zero at almost every level, still within the common admissible set. This supplies a fixed gluing level with vanishing interface cost in the compactness argument below. All orientations in this argument are local; global orientability is not required.

Exhaustion minimizers and local compactness. Choose a smooth relatively compact exhaustion $U_k \Subset E$. Minimize over phases vanishing outside $\overline{U}_k$, taking the total perimeter, including the interface on ∂U_k , and the pressure integral inside U_k . The empty phase is admissible. Because the pressure is bounded on $\overline{U}_k$, an energy bound gives a perimeter bound. The class is closed under L^1 convergence, the volume term is continuous, and perimeter is lower semicontinuous. The direct method thus gives a minimizer, locally absolute minimizing in U_k .

The same local bound applies to arbitrary local absolute minimizers. In a coordinate ball B_t compactly contained in their comparison domain, comparison with the phase cleared inside B_t gives, for almost every t ,

$$P_\omega(A; B_t) \leq \int_{\partial B_t} \omega \, d\mathcal{H}_g^{m-1} + \int_{B_t} |p| \, d\mu.$$

Choose t in a fixed annulus around the smaller window to be bounded. Coarea bounds the first term for a suitable slice. Smoothly convergent metrics and positive weights and locally uniformly bounded pressures make this a uniform local bound. Diagonal BV compactness therefore produces a locally L^1 convergent subsequence, both for the exhaustion minimizers and for any sequence in the statement.

Minimality and perimeter convergence. The gluing formulas now apply to the locally convergent subsequence. Choose nested relatively compact windows $U \Subset V$ with a Lipschitz foliation of the intervening annulus. The integral over slicing parameters of the trace discrepancy between A_i and A tends to zero by L^1 convergence and coarea. One can therefore choose a slicing hypersurface between U and V whose discrepancy tends to zero along a subsequence. Glue A inside this surface to A_i outside. Minimality of A_i bounds its interior perimeter from above by that of A , the vanishing trace cost, and a vanishing difference of volume terms. The last difference vanishes by the convergence of the data and indicators. First use windows whose limiting perimeter gives zero mass to their boundaries; then shrink the annulus. Together with lower semicontinuity this proves weak local convergence of the perimeter measures. Replacing the interior copy of A by any compact competitor B gives the same upper comparison with B , since $B = A$ on the gluing annulus. Lower semicontinuity on the left proves (B.3) for A . Smooth metric and weight changes make only errors tending uniformly to zero on each fixed window, so the same argument proves the convergent-data assertion.

Changing metrics and varifold convergence. For the varifold assertion, account explicitly for the changing metric. In a coordinate chart let $D_0\chi_{A_i}$ be the Euclidean distributional derivative and let (g_i) denote the positive metric matrices. The vector measures

$$\mathbf{v}_i = \sqrt{\det g_i} \, g_i^{-1/2} D_0\chi_{A_i}$$

have Euclidean total variation equal to the g_i -perimeter measures. The matrices and their derivatives converge smoothly, so distributional convergence and the local mass bounds give $\mathbf{v}_i \rightharpoonup \mathbf{v}$. The perimeter conclusion gives $|\mathbf{v}_i| \rightharpoonup |\mathbf{v}|$ as well.

This strict convergence also controls the unit polar vectors. To see this, on a compact window approximate the limiting polar vector in $L^2(|\mathbf{v}|)$ by a continuous bounded vector field e . Expand

$$\int |\mathbf{v}_i/|\mathbf{v}_i| - e|^2 d|\mathbf{v}_i| = \int (1 + |e|^2) d|\mathbf{v}_i| - 2 \int e \cdot d\mathbf{v}_i.$$

Weak convergence gives its limit, and then the approximation makes that limit arbitrarily small. Uniform continuity consequently gives convergence for every continuous compactly supported integrand of position and polar vector. The unoriented g_i -orthogonal tangent hyperplane is a continuous function of this vector and g_i . This proves varifold convergence. Multiplication or division by the smooth positive weight on a fixed compact window gives the asserted weighted and unweighted versions. $\square$

B.2 Barriers, ordering, and exact product limits

Lemma B.4 (Barriers hold for every minimizer). *Every local absolute minimizer for a pressure of Theorem B.1 contains D^+ and excludes D^- . Its boundary support consequently misses a neighborhood of each current core closure. After the radial split this neighborhood can be chosen with a positive uniform radius on each packet. At the radial split the assertion is local on the corresponding compact radial blocks.*

Proof. We first prove the positive assertion. Put $D = D^+$, $Y = Y^+$ and $G = D \setminus A$. For almost every distance radius R about a fixed point, $U = B_R$ is a finite-perimeter set, its boundary has zero mass for the relevant perimeter measures, and BV traces are available. Write

$$V(R) = \mu(G \cap B_R), \quad S_R = \int_{\partial^* B_R} \text{Tr}(\chi_G) \omega d\mathcal{H}^{m-1}.$$

Distance coarea, with $|\text{grad } d| = 1$ almost everywhere off the center, gives local absolute continuity and $V'(R) = S_R$ for almost every R . Weighted Gauss–Green, $Y = \nu_D$ on ∂D , and $|Y| \leq 1$ give

$$\begin{aligned} P_\omega(D; U) - P_\omega(A \cap D; U) &\leq \int_{G \cap U} \text{div}_\omega Y d\mu \\ &\quad - \int_{\partial^* U} \text{Tr}(\chi_G) Y \cdot \nu_U \omega d\mathcal{H}^{m-1}. \end{aligned}$$

By (B.4) this bounds the interior perimeter increase on replacing A by $A \cup D$ in U . Glue back to A outside U ; the extra perimeter is at most S_R . This is a compact comparison in a slightly larger relatively compact open set. By (B.2) and minimality,

$$V(R) \leq 2S_R = 2V'(R). \tag{B.6}$$

The weighted ball volume is at most polynomial by Theorem A.2, since ω is bounded on the packet. If $V(R_0) > 0$, (B.6) would imply $V(R) \geq V(R_0)e^{(R-R_0)/2}$, a contradiction. Thus G is null. For a relatively compact radial barrier, simply add all of D ; no slicing terms occur, and the same comparison gives $\mu(G) = 0$ directly. The complement of A minimizes pressure $-p$, proving the negative assertion. The domains contain the core closures in their interiors; the packetwise positive separation follows from exact products and the compact-part property. A phase equal to one or zero on an open set has no perimeter support there. $\square$

Proposition B.5 (Generic uniqueness on all required products). *After the radial split one common shift c , with $|c| < c_*$, can be chosen such that there is exactly one local absolute minimizer, modulo*

null sets, on every packet E and every product $E \times \mathbb{R}^d$, $d \geq 0$, for the pulled-back weight and pressure $p_0 + c$. The unique product phase is exactly $A_E \times \mathbb{R}^d$ modulo null sets, where A_E is the unique phase on E .

Proof. On any one such space, consider *arbitrary* minimizers A for $p_0 + c_1$ and B for $p_0 + c_2$, with $c_1 < c_2$. Put $G = A \setminus B$, $\delta = c_2 - c_1$, and use the same distance-ball slicing as in the preceding proof. Replace A by $A \cap B$ inside the ball and B by $A \cup B$ there, gluing each to its own exterior phase. The two gluing costs total at most $2S_R$. Their interior perimeter increments have nonpositive sum by (B.4); their pressure increments sum to precisely $-\delta\mu(G \cap B_R)$. Adding the two minimality inequalities gives

$$\delta V(R) \leq 2V'(R). \quad (\text{B.7})$$

Polynomial weighted growth, also on $E \times \mathbb{R}^d$, forces $V = 0$. Thus *every* minimizer at the smaller shift is contained in *every* minimizer at the larger shift.

Choose countably many nonnegative compactly supported continuous test functions φ_a which separate locally integrable functions. For each a and shift c , let

$$I_a(c) = \left(\inf_{A \text{ minimizes at } c} \int \varphi_a \chi_A \, d\mu, \sup_{A \text{ minimizes at } c} \int \varphi_a \chi_A \, d\mu \right).$$

Existence is supplied by Theorem B.3. The preceding ordering says that the nonempty open intervals $I_a(c)$ for distinct c are disjoint. Each contains a rational number, so only countably many can be nonempty. If minimizers differ, some φ_a distinguishes them and $I_a(c)$ is nonempty. There are therefore at most countably many exceptional shifts on this space. The packet collection and the set of integers d are countable, so their exceptional shifts have a countable union. Choose c in the remaining part of $(-c_*, c_*)$.

A Euclidean translation of the unique minimizer on $E \times \mathbb{R}^d$ is again a minimizer. Uniqueness implies translation invariance of its indicator. It suffices to impose the countable dense group of rational translations: continuity of translations in L^1_{loc} then gives all translations, so its Euclidean distributional derivatives vanish. The indicator is therefore $\chi_F(x)$ for a base phase F . BV slicing shows that F has locally finite perimeter. If a compact base competitor improved its energy by $\eta > 0$, use that competitor over $(-L, L)^d$ and the phase F outside, retaining the original phase outside the compact base discrepancy. The energy gain is $\eta(2L)^d$. The interfaces at the Euclidean box boundary cost at most $O(L^{d-1})$ times the weighted volume of that compact discrepancy. For large L this contradicts minimality. Thus F minimizes on E , and base uniqueness identifies it with A_E . The case $d = 0$ needs no product argument. $\square$

At the radial split set $c = 0$ and choose any minimizer supplied by Theorem B.3. Its compact barriers confine its boundary to the locally finite compact transition shells. No uniqueness or polynomial-growth assertion on the initial manifold is used. We call the selected minimizing boundary a *true cut*; it will later be replaced by a compatible regular wall near the required region.

B.3 The precise local regularity input

Theorem B.6 (Local GMT properties used here). *Let A satisfy (B.3) on a smooth m -manifold, $m \geq 2$, with smooth positive ω and smooth p . Put $S = \text{supp } |D\chi_A|$. Then $S = S_{\text{reg}} \cup \Sigma$, where S_{reg} is a smooth embedded hypersurface, Σ is relatively closed, and*

$$\dim_{\mathcal{H}} \Sigma \leq m - 8. \quad (\text{B.8})$$

A negative dimension bound means that Σ is empty. The outward measure-theoretic normal extends smoothly on S_{reg} . With $B(u, v) = \langle \nabla_u \nu, v \rangle$, its base mean curvature satisfies

$$H_g = p - \partial_\nu \log \omega. \quad (\text{B.9})$$

The unweighted multiplicity-one perimeter varifold has locally bounded generalized mean curvature. Its density exists at every support point; monotonicity gives local uniform area upper bounds and positive area lower bounds at support points. Under the convergences of Theorem B.3, densities are upper semicontinuous at converging points, provided the local generalized-mean-curvature bounds are uniform. The supports converge locally, and on every compact subset of the regular limiting hypersurface convergence is smooth and single-sheeted when the data converge smoothly. On compact subsets of the interiors of the limiting phases there is eventually no boundary.

Every tangent at a point of S is a multiplicity-one locally perimeter-minimizing boundary cone in the Euclidean tangent space. A flat tangent implies regularity. A tangent at a singular point is therefore nonflat, and its translation-invariance spine has dimension at most $m - 8$.

Proof by first variation, strict blowups, and regularity. Put $d = m - 1$. The classical inputs are the bounded-mean-curvature monotonicity theorem and Allard's graphical theorem [36, Chapter 4, Theorem 3.17; Chapter 5, hypotheses 5.1 and Theorem 5.2], and the singular-dimension theorem for *exactly* minimizing boundary currents [36, Chapter 7, Theorem 5.8]. The minimizing-boundary dimension bound follows from the work of Simons and Federer [14, 38]. The last theorem will be applied only to constant-metric blowup limits. We verify the transfer to the present smooth weighted problem, including the singular-set compactness needed for the Hausdorff-dimension conclusion.

First variation and local area bounds. First variation is available before any regularity conclusion. The flow of a compactly supported smooth field Z is an admissible comparison in (B.3), so

$$\int_{\partial^* A} \omega (\text{div}_S Z + \langle \text{grad} \log \omega, Z \rangle) d\mathcal{H}_g^d = \int_{\partial^* A} p\omega \langle Z, \nu \rangle d\mathcal{H}_g^d.$$

Here tangent planes and ν are the measure-theoretic ones. Substituting $Z = W/\omega$ cancels the tangential weight derivatives and gives

$$\int_{\partial^* A} \text{div}_S W d\mathcal{H}_g^d = \int_{\partial^* A} (p - \partial_\nu \log \omega) \langle W, \nu \rangle d\mathcal{H}_g^d.$$

Thus the unweighted perimeter varifold has intrinsic generalized mean-curvature vector $-(p - \partial_\nu \log \omega)\nu$, locally bounded by $|p| + |\text{grad} \log \omega|$. Its multiplicity is one almost everywhere.

On a relatively compact coordinate patch, first pass to a smaller patch and extend its metric to a closed manifold. Explicitly, place its coordinate closure inside a coordinate ball of a flat torus and blend the given positive definite metric with the flat metric outside the smaller patch; the convex combination stays positive definite. The resulting metric is smooth and the ambient manifold is compact, so the smooth isometric embedding theorem applies [33, Theorem 2]. The preceding intrinsic first variation then gives Euclidean generalized mean curvature with the additional bounded term $\text{tr}_{TS} \text{II}$, where II is the second fundamental form of this fixed embedding. Indeed decompose a Euclidean test field into its ambient-manifold tangent and normal parts and differentiate that decomposition along TS . Its normal contribution is exactly the trace of II . Consequently Euclidean monotonicity applies locally. Intrinsic and embedded density limits agree, since the embedding is isometric to first order. Monotonicity gives density existence and upper semicontinuity, including at nonreduced support points; their densities are at least one, by approximation by reduced points. Together with the clearing-ball upper bound from Theorem B.3, it gives uniform upper and positive lower area bounds on compact patches.

A common ambient model for varying metrics. We record carefully the uniform version when the metric changes. Suppose $g_i \rightarrow g$ smoothly on a fixed patch, and all other data have the local bounds in the theorem. Any claimed failure of regular-point persistence or support convergence has a subsequence on which the failure persists. Pass further so that, on a slightly larger compact patch,

$$\|g_i - g\|_{C^i} \leq 2^{-i^2}.$$

Set $t_i = 2^{-i}$ and take disjoint smooth bump functions $\beta_i(t)$, equal to one on a neighborhood of t_i , supported in intervals of width comparable to t_i , and satisfying $|\beta_i^{(b)}| \leq C_b t_i^{-b}$. After omitting finitely many terms,

$$g_t = g + \sum_i \beta_i(t)(g_i - g), \quad G = dt^2 + g_t$$

is a smooth positive metric on a product neighborhood, with $g_0 = g$ and $g_{t_i} = g_i$. For each pair of fixed orders a, b , the summands after a spatial and b temporal derivatives are bounded by $C_b 2^{ib-i^2}$ for all sufficiently large i . These bounds, divided by any fixed power of t_i , still tend to zero. Thus all mixed derivatives extend at $t = 0$. The plateaus also give $\partial_t g_t = 0$ at t_i , so the slices carrying g_i are totally geodesic in G .

Extend this *one* smooth metric G across the edge of a smaller product patch to a closed manifold and isometrically embed it once. The images of the slices $t = t_i$ converge smoothly to the slice $t = 0$, and the second fundamental form of this single embedding is bounded on their common compact neighborhood. The preceding first-variation calculation therefore gives one Euclidean mean-curvature bound for all the embedded perimeter varifolds. Strict perimeter convergence from Theorem B.3 becomes varifold convergence in this same Euclidean space. This construction proves the needed sequential uniformity; it does not assume that independently chosen isometric embeddings have uniform bounds.

Exact minimizing tangent cones. Next consider a blowup of one phase at $x \in S$. In smooth coordinates, at scale ρ , the normalized data are

$$A_{x,\rho} = \rho^{-1}(A - x), \quad g_{x,\rho}(y) = g(x + \rho y), \quad \omega_{x,\rho}(y) = \omega(x + \rho y), \quad p_{x,\rho}(y) = \rho p(x + \rho y).$$

The metric here has been divided by the square of the physical scale: perimeter scales by ρ^d , volume by ρ^{d+1} , and this accounts for the factor ρ in the pressure. On every fixed window the data converge smoothly to $g(x), \omega(x), 0$. The strict compactness lemma therefore identifies every subsequential tangent varifold with the unit-multiplicity perimeter varifold of a phase minimizing perimeter in the constant metric $g(x)$. A constant weight can be divided out. After one linear isometry this is an ordinary Euclidean perimeter-minimizing phase. It is also a minimizing boundary current. Indeed, orient this Euclidean chart. A compactly supported integral current filling a boundary competitor has top dimension and is given by an integer-valued BV function q . The competitor is the boundary of $\chi_A + q$. Clipping that integer-valued function to $[0, 1]$ leaves it unchanged outside the compact discrepancy and gives a characteristic function without increasing variation. Any current competitor with smaller mass would therefore give a smaller-perimeter set competitor. This verifies the precise minimizing-boundary hypothesis of Simon [36, Chapter 7, Theorem 5.8]. All orientations in this argument are local.

The support lower bound makes this limit nonzero at the origin. Monotonicity based at x makes its mass ratio constant on all radii; the equality case, or the stationary radial identity in Simon [36, Chapter 4, Equation (3.6)], makes it conical. Equivalently, after a fixed local isometric embedding, dilations of the embedded ambient manifold converge smoothly to its tangent space, its second fundamental form is multiplied by ρ , and the same blowups give this cone. Its multiplicity is one as a perimeter varifold, although its density at the vertex need not be one.

Regularity and persistence under convergence. We now apply Allard's graphical regularity theorem [2]. In its Euclidean form used here, fix an exponent $q > d$. For a rectifiable d -varifold of multiplicity at least one, on a ball centered at a support point and contained in the first-variation domain, sufficiently small excess of the normalized mass above one and sufficiently small

$$\rho^{1-d/q} \left(\int_{B_\rho} |\mathbf{H}|^q d\mu_V \right)^{1/q}$$

give a single $C^{1,1-d/q}$ graph on a smaller ball [36, Chapter 5, Theorem 5.2]. A unit flat tangent supplies the small mass excess. The bounded mean curvature supplies the other small quantity, since it is $O(\rho)$ under the local area bound. Thus a flat tangent makes the original point regular as a $C^{1,\alpha}$ graph. The graph satisfies the weak prescribed-curvature equation $H_g = p - \partial_\nu \log \omega$; its coefficients and right-hand side are smooth functions of position, graph height, and graph gradient. The bounded slope gives uniform ellipticity. We spell out the passage from a weak graph to a smooth one. In a relatively compact chart write the equation as

$$\partial_i A^i(x, u, Du) = B(x, u, Du).$$

Here A is the gradient, in the slope variables, of the smooth positive weighted graph-area integrand. Thus A_p is symmetric and uniformly positive on the bounded slope range; B is smooth in its arguments. For a compactly supported cutoff η , test translated differences with $\eta^2 \delta_h u$. The difference of A has the form $a_h D \delta_h u + c_h \delta_h u + b_h$, where a_h is uniformly elliptic and all three coefficients are uniformly bounded. Since u is Lipschitz, $\delta_h u$ is bounded. On the right move the difference quotient from B to the test function, using $\|\delta_{-h} \phi\|_2 \leq \|D\phi\|_2$. Ellipticity and Cauchy's inequality then give

$$\int \eta^2 |D \delta_h u|^2 \leq C_\eta$$

independently of small h . Hence $u \in H_{\text{loc}}^2$. For $v = \partial_k u$, weak differentiation now gives

$$\partial_i (a^{ij} \partial_j v) = \partial_i F^i, \quad a^{ij} = A_{p_j}^i, \quad F^i = \delta_{ik} B - A_{x_k}^i - A_u^i v,$$

with coefficients evaluated at (x, u, Du) . Both a and F are $C^{0,\alpha}$ because u is $C^{1,\alpha}$.

To justify the divergence Schauder estimate for this weak H^1 solution, mollify a, F on a slightly larger ball and solve the coercive linear Dirichlet problems on the ball with the same trace as v . Their solutions v_ϵ satisfy

$$\lambda \|D(v_\epsilon - v)\|_2 \leq \|F_\epsilon - F\|_2 + \|a_\epsilon - a\|_\infty \|Dv\|_2,$$

where $\lambda > 0$ is a common ellipticity constant. Thus they converge in H^1 . Smooth-coefficient Sobolev regularity makes each approximant smooth in the interior; the local L^2 - L^∞ estimate and then the a priori divergence Schauder estimate give uniform interior $C^{1,\alpha}$ bounds [37, Lecture 6, Theorem 1 and Corollary 1; Lecture 18, Theorem 2; Lecture 12, Theorem 1]. These bounds use only the common ellipticity and Hölder bounds, not higher derivatives of the mollified coefficients. Passing to the limit gives $v \in C_{\text{loc}}^{1,\alpha}$, hence $u \in C_{\text{loc}}^{2,\alpha}$. Repeating the difference-quotient and weak Schauder argument for differentiated equations makes the graph smooth. The estimates are uniform for families with common $C^{1,\alpha}$ graph bounds and uniformly smooth geometric data.

Likewise, near a regular point of a limiting phase choose a sufficiently small ball with mass ratio as close to one as needed, with strict slack in the Allard threshold, and with small scaled mean-curvature bound. Choose its radius also so that its sphere has zero limiting perimeter measure;

almost every sufficiently small radius has this property. For smoothly convergent data, use the single ambient embedding construction above. Weak perimeter-measure convergence now passes the mass bound on this ball to the approximating phases. To center at a nearby approximating support point, decrease the radius by a fixed arbitrarily small factor: the recentered ball lies in the chosen ball for all large indices, and the strict slack pays the change in the normalizing power of the radius. Allard therefore gives one graph in each smaller ball for all sufficiently large indices. The graph equation and smooth data convergence then give smooth convergence on smaller compact patches. This also proves

$$x_i \in \Sigma_i, \quad x_i \longrightarrow x \quad \implies \quad x \in \Sigma$$

for interior limits: a regular limiting point has just been excluded, and a point outside the limiting support is excluded by the uniform support lower bound and perimeter convergence. The same lower bound and weak convergence prove local support convergence; they also exclude stray support in compact subsets of phase interiors. The regular set is open in the support, so its complement Σ is relatively closed.

Transfer of the singular-dimension bound. For the singular-dimension transfer we use the content argument in Simon [36, Chapter 1, Theorem 3.26(2), and Appendix A, proof of Theorem A.4, step (1)]. Its hypotheses in the present application have all been established: on every compact coordinate patch and at every center, each vanishing scale sequence has a strict tangent subsequence; these tangents are homogeneous minimizing boundaries; singular sets commute exactly with translations and dilations; and, on compact windows of every such convergent sequence, singular sets eventually lie in every open neighborhood of the limiting singular set. For the last assertion, a contrary sequence of points has a convergent subsequence, contradicting regular-point persistence or support convergence. The data on the rescaled windows have uniform local derivative bounds, with derivatives of order j acquiring factors ρ^j and pressure derivatives acquiring ρ^{j+1} . Thus the compactness and regularity argument applies to all these sequences. We use only the cited content-transfer step, and spell it out next; no identification with the scalar-function family of Appendix A is needed.

Suppose $m \geq 8$ and, for some $s > m - 8$, the singular set in an interior window had positive $\mathcal{H}^s$ measure. Let $\mathcal{H}_\infty^s$ denote Hausdorff content, defined by covering with balls and summing their radii to the power s ; normalization constants are immaterial. The Hausdorff-content density selection in the proof of Theorem A.4 provides an interior $x \in \Sigma$, scales $\rho_i \downarrow 0$, and $c > 0$ such that

$$\rho_i^{-s} \mathcal{H}_\infty^s(\Sigma \cap \overline{B}_{\rho_i}(x)) \geq c.$$

Pass to a strict tangent limit C along these scales. The exact minimizing-boundary theorem gives $\mathcal{H}^s(\Sigma_C) = 0$, because its hypersurface dimension is $d = m - 1$ and its bound is $d - 7 = m - 8$. Cover $\Sigma_C \cap \overline{B}_{3/2}$ by finitely many open balls whose sum of s th powers of radii is less than $c/2$; the cover can first be chosen countable with this bound and then reduced to a finite subcover by compactness. Singular-set persistence puts every $\rho_i^{-1}(\Sigma - x) \cap \overline{B}_1$ in that same union for large i : otherwise a convergent sequence in its compact complement would give a singular limiting point outside the cover. Scaling of Hausdorff content now contradicts the preceding lower bound. This proves $\mathcal{H}^s(\Sigma) = 0$ for every $s > m - 8$. It is the density selection and this finite-cover passage, in addition to tangent information, that transfer the dimension bound.

If $m \leq 7$, the exact minimizing-boundary theorem says that every tangent cone is regular at its vertex and hence is a hyperplane. Flat-tangent regularity then makes Σ empty. Finally a nonflat minimizing cone is singular at its vertex, and translation invariance carries that vertex to each point of its spine. The spine is therefore contained in its singular set, giving the asserted spine bound. The preceding monotonicity argument, applied in the common embeddings for convergent data, also gives upper semicontinuity of densities at moving points by first passing to the limit at a

fixed radius and then letting the radius decrease to zero. The smooth graph's coorientation comes from the phase, and its first-variation equation is (B.9). All constructions are local in the base and require no ambient orientability or torus-dimensional regularity theorem. $\square$

Corollary B.7 (True cuts converge at packet ends). *For the common generic shift, let A_E be the true phase on each packet and let A_B be that on a face. On translated windows escaping in a product collar over a compact part of B , the phases A_E converge to $A_B \times \mathbb{R}^d$ in L^1_{loc} , their perimeter measures and varifolds converge, and their regular boundaries converge smoothly on compact sets off the limiting singular set. Their boundary supports converge locally as well. The same conclusions hold after passage to any deeper product furnished by Theorem A.3.*

Proof. The metric, weights, and shifted pressure in the translated expanding windows are exactly the data pulled from B . Local compactness produces a subsequential limiting phase on $B \times \mathbb{R}^d$, and it is an absolute minimizer by Theorem B.3. The uniqueness and factorization of Theorem B.5 identify it with $A_B \times \mathbb{R}^d$. Every subsequence has this same possible limit, which proves convergence of the full sequence. Apply Theorem B.6 for the support and smooth assertions. The deeper-face statement follows by using the commuting product coordinates in Theorem A.3. $\square$

B.4 Stability on a wall and the retained-side estimate

We now choose the representative of each phase which is open away from S and agrees near S_{reg} with the corresponding smooth closed side when a trace is required. Write $S = S_{\text{reg}}$ in integrals below. Every test has compact support away from Σ ; joined-side tests also avoid the cylinder over Σ . The old torus variables remain invariant, and $d\mu_S = \omega d\mathcal{H}^{m-1}_g$ is the wall measure. The intrinsic scalar curvature R_S and the second fundamental form B refer to $S \times \mathbb{T}^q$ with its induced old stabilized metric. Its torus principal curvature in direction i is $\partial_\nu \log f_i$, so (B.9) gives

$$H = \text{tr } B = H_g + \partial_\nu \log \omega = p$$

for the outward normal of the plus phase. The invariant wall Laplacian is $\omega^{-1} \text{div}_S(\omega \text{grad}_S \cdot)$, so its integration by parts uses precisely $d\mu_S$.

Lemma B.8 (The wall stability form). *For every compactly supported smooth invariant test v on S ,*

$$\int_S (2|\text{grad } v|^2 + R_S v^2) d\mu_S \geq \int_S (R + p^2 + |B|^2 + 2\partial_\nu p)v^2 d\mu_S. \quad (\text{B.10})$$

In particular its right side is at least $\int_S (R - 1.02\Pi_{j_0})v^2 d\mu_S$.

Proof. Weighted perimeter is the area of the old torus product for invariant phases. The invariant normal variation with speed v is compactly supported in the smooth regular part and comes from an admissible base variation. Thus its energy has a minimum at parameter zero. For completeness the variation identities hold in every dimension: differentiating the induced metric gives $\dot{g}_{ab} = 2vB_{ab}$, differentiating the unit normal gives $\dot{\nu} = -\text{grad}_S v$, and commuting the two ambient covariant derivatives in the derivative of B_{ab} gives, after tracing and differentiating the inverse metric,

$$\dot{H} = -\Delta_S v - (\text{Ric}_{\hat{g}}(\nu, \nu) + |B|^2)v.$$

The area derivative is $Hv d\mu_S$, while the pressure-volume variation is $pv d\mu_S$. The first energy derivative is therefore $\int_S (H - p)v d\mu_S$. At $H = p$ the derivatives of the area measure and of the chosen speed multiply zero, and $\dot{p} = v\partial_\nu p$. Hence the nonnegative second derivative is

$$\int_S \left[-v \Delta_S v - (\text{Ric}_{\hat{g}}(\nu, \nu) + |B|^2 + \partial_\nu p)v^2 \right] d\mu_S.$$

Weighted integration by parts gives

$$\int_S |\text{grad } v|^2 d\mu_S \geq \int_S (\text{Ric}_{\hat{g}}(\nu, \nu) + |B|^2 + \partial_\nu p) v^2 d\mu_S.$$

Compare the first variation in Chodosh and Li [9, Lemma 13] and the second-variation calculation in the proof of its Lemma 14. The displayed calculation proves the identity in the present stabilized setting; the stability and Gauss-equation mechanism is that of Schoen and Yau [35, Section 1]. The derivation here uses neither Chodosh and Li's dimension restriction for smooth existence nor a regularity assertion for the torus product. Only the already smooth base patch and its invariant variations enter. The Gauss identity

$$R_S = R - 2 \text{Ric}_{\hat{g}}(\nu, \nu) + H^2 - |B|^2$$

gives (B.10). By (B.1),

$$p^2 + 2\partial_\nu p \geq p^2 - 2|\text{grad } p| \geq .33p^2 - 1.02\Pi_{j_0},$$

which proves the final statement. $\square$

For either phase side Ω , put $p_s = p$ on the plus side and $p_s = -p$ on the minus side. The side is itself an absolute minimizing phase for the signed pressure p_s , and its outward stabilized mean curvature is $H_{\text{out}} = p_s$.

Lemma B.9 (Deletion and the seam estimate). *Every nonnegative compactly supported smooth function f on the closed regular side, with support avoiding the singular set, satisfies*

$$\int_S f d\mu_S \leq \int_\Omega (|\text{grad } f| + p_s f) d\mu. \quad (\text{B.11})$$

For every smooth real test v of the same type,

$$3.05 \int_\Omega |\text{grad } v|^2 d\mu + 2 \int_S H_{\text{out}} v^2 d\mu_S \geq -1.02 \int_\Omega \Pi_{j_0} v^2 d\mu. \quad (\text{B.12})$$

Both statements extend to compactly supported Lipschitz tests by approximation.

Proof. For almost every $t > 0$, delete $\Omega \cap \{f > t\}$ from the minimizing phase. The removed boundary area is the area on $S \cap \{f > t\}$; the newly introduced boundary is the level surface of f in the interior of Ω . The pressure-volume term increases by $\int_{\Omega \cap \{f > t\}} p_s d\mu$. Minimality gives the corresponding perimeter inequality. Integrating in t , the layer-cake formula and weighted coarea give (B.11). All comparisons are compact and occur on regular patches; the perimeter slicing formula justifies the same argument at exceptional critical levels by discarding a null set of levels.

Apply (B.11) to $f = 2(-p_s)_+ v^2$, initially by Lipschitz approximation. Its right side is bounded above by

$$\int_{\Omega \cap \{p_s < 0\}} [(2|\text{grad } p_s| - 2p_s^2)v^2 + 4|p_s| |v| |\text{grad } v|] d\mu.$$

Young's inequality gives

$$4|p_s| |v| |\text{grad } v| \leq 3.05 |\text{grad } v|^2 + \frac{4}{3.05} p_s^2 v^2.$$

The pressure estimate holds unchanged for p_s , and

$$2 - \frac{4}{3.05} - .67 = \frac{113}{6100} > 0. \quad (\text{B.13})$$

Discard the resulting nonpositive multiple of $p_s^2 v^2$. Thus

$$2 \int_S (-p_s)_+ v^2 d\mu_S \leq 3.05 \int_\Omega |\text{grad } v|^2 d\mu + 1.02 \int_\Omega \Pi_{j_0} v^2 d\mu.$$

Since $2p_s \geq -2(-p_s)_+$ on the boundary and $H_{\text{out}} = p_s$, this is (B.12). $\square$

B.5 Raw joined forms and the numerical reserve

Attach $S \times [0, \infty)_t$ to the regular boundary of either side, freezing all induced metric and old warp data on the cylinder. This is a raw joined space: it is smooth on its two pieces, and the seam retains its scalar corner contribution. An admissible test is a pair $v = (v_b, v_c)$, smooth separately up to regular seam patches, compactly supported away from singularities and their cylinders, satisfying $v_b|_S = v_c|_{t=0}$. Compact Lipschitz versions are allowed by local approximation in regular seam charts. For the wall alone the tests are compact smooth functions on S away from Σ .

Set $C = 3.99$. On a wall define

$$Q_S(v) = \int_S (C |\text{grad } v|^2 + R_S v^2) d\mu_S.$$

On a joined side define

$$\begin{aligned} Q_\Omega(v) &= \int_\Omega (C |\text{grad } v_b|^2 + R v_b^2) d\mu + 2 \int_S H_{\text{out}} v_b^2 d\mu_S \\ &\quad + \int_0^\infty \int_S (C (|\text{grad}_S v_c|^2 + |\partial_t v_c|^2) + R_S v_c^2) d\mu_S dt. \end{aligned} \quad (\text{B.14})$$

The sign in the seam term can also be checked directly: in outward arclength s the mean curvature changes from H_{out} on the bulk side to zero on the frozen cylinder, and the scalar expression contains $-2\partial_s H$. Its distributional contribution is $+2H_{\text{out}}$ at the seam. The smooth realization and its form convergence will be established in Appendix E.

Proposition B.10 (Raw scalar domination). *Let $V = R - 1.02\Pi_{j_0}$, restricted to the wall and pulled back to the cylinder when necessary. Then*

$$Q_S(v) \geq \int_S V v^2 d\mu_S + 1.99 \int_S |\text{grad } v|^2 d\mu_S, \quad (\text{B.15})$$

$$\begin{aligned} Q_\Omega(v) &\geq \int_\Omega V v_b^2 d\mu + \int_0^\infty \int_S V v_c^2 d\mu_S dt \\ &\quad + .94 \int_\Omega |\text{grad } v_b|^2 d\mu + 1.99 \int_0^\infty \int_S |\text{grad } v_c|^2 d\mu_S dt. \end{aligned} \quad (\text{B.16})$$

Here $|\text{grad } v_c|^2 = |\text{grad}_S v_c|^2 + |\partial_t v_c|^2$.

Proof. The wall assertion follows by adding $(C - 2)|\text{grad } v|^2$ to Theorem B.8. For the bulk use Theorem B.9; the remaining gradient coefficient is $C - 3.05 = .94$. Apply the wall assertion to every cylinder slice and integrate in t . The horizontal coefficient is $C - 2 = 1.99$, while the vertical coefficient is still $C = 3.99$, so lowering the latter to 1.99 gives (B.16). The seam trace in the bulk estimate is precisely the matching trace in (B.14). $\square$

Definition B.11 (Reserved form and test completion). For the wall or a joined side define, respectively,

$$\begin{aligned}\mathcal{N}_S(v) &= 1.7 \int_S |\text{grad } v|^2 d\mu_S, \\ \mathcal{N}_\Omega(v) &= .8 \int_\Omega |\text{grad } v_b|^2 d\mu + 1.7 \int_0^\infty \int_S |\text{grad } v_c|^2 d\mu_S dt.\end{aligned}$$

We write $\mathcal{N}$ when the tested space is understood. The squared norm $\|v\|_2^2$ is the sum of the corresponding piecewise weighted integrals. With

$$e_0 = (\lambda_r - \lambda_{r+1})/100 > 0, \quad \mathfrak{h}(v) = \mathcal{N}(v) + e_0 \|v\|_2^2,$$

the test completion means completion of the admissible compact tests just specified in the norm $\mathfrak{h}^{1/2}$. In particular its seam condition is matching traces, with no zero trace imposed there.

The coefficients have a reserve even after the later shielding charge .9 $\mathfrak{h}$. Specifically,

$$.94 - .9(.8) = .22, \quad 1.99 - .9(1.7) = .46. \quad (\text{B.17})$$

The potential term in Theorem B.10, combined with the parent inequality, satisfies

$$R - 1.02\Pi_{j_0} \geq \lambda_r + b_r \sum_{j \neq j_0} \Pi_j + (b_r - 1.02)\Pi_{j_0} \quad \text{on } D_E. \quad (\text{B.18})$$

Here $b_r \geq 1.07 > 1.02$. The raw form inequalities themselves hold for all admissible compact tests; the replacement of their potential by the right side of (B.18) is permitted only where the test is supported over the old required region. The count inclusions of Theorem 5.3 supply exactly this localization for the later required regions. These are the forms and constants used in shielding and regularization. The minimizing cut, its pressure, and the old weight remain fixed while the other profiles and their cores are changed.

C Local shielding of singular sets

Shielding makes each singular point irrelevant to the next curvature requirement by enlarging enough remaining pole cores near that point. The forcing costs a potential concentrated in a thin annulus. We first pay for this potential from the reserved gradient form; the next appendix makes the choices compatible throughout the diagram.

We work at one fixed stage with a finite set of remaining labels. The countable extension is proved in Theorem 6.4. The local estimates use only $.60 \leq a_r \leq .63$ and the positive reserve of Theorem B.11. Smooth geometric bounds are taken on the specified compact windows, so no bounded-geometry assumption on the original manifold is involved.

Fix a true cut in a parent packet E of base dimension m , as in Theorem B.6. Its boundary support is S , its singular set is Σ , and its regular part is S_{reg} . The old torus volume factor is ω . All measures below include this factor; on a cylinder it is pulled back from S_{reg} . For either phase Ω , write

$$J_\Omega = \overline{\Omega}_{\text{reg}} \cup_{S_{\text{reg}}} (S_{\text{reg}} \times [0, \infty)).$$

Here the notation retains only regular boundary points when forming the join. Tests on J_Ω are compactly supported, piecewise smooth functions with matching values at the seam, whose supports avoid Σ and the cylinders above it. Lipschitz tests with the same properties follow by approximation

in regular charts. Tests on the wall have the analogous meaning. Use the reserved energy from Theorem B.11, namely

$$\mathcal{N}(v) = \begin{cases} 1.7 \int_{S_{\text{reg}}} |\nabla v|^2, & \text{on the wall,} \\ .8 \int_{\Omega} |\nabla v|^2 + 1.7 \int_{S_{\text{reg}} \times [0, \infty)} |\nabla v|^2, & \text{on a joined side.} \end{cases} \quad (\text{C.1})$$

The notation $\|v\|_2^2$ denotes the sum of the corresponding weighted L^2 integrals. An ambient function is restricted to the wall and frozen in the cylinder coordinate. In particular, these conventions specify every potential integral in this section. The true cut remains fixed throughout: only profiles of labels other than the processed label j_0 will change.

Lemma C.1 (Choice of anchors). *There are compatible smooth functions h_j , for the remaining labels, within .1 of $\max(-1, \min(1, \mu_j))$, and common numbers $t_j \in (-.3, .3)$ such that, on the singular set in each parent of dimension m , at most $m - 8$ of the equalities $h_j = t_j$ hold. The assertion is vacuous in dimensions below eight. Moreover*

$$\{h_j > t_j\} \cap \{\mu_j = -\infty\} = \emptyset, \quad \{h_j < t_j\} \cap \{\mu_j = +\infty\} = \emptyset.$$

Proof. Compatible smoothing by Theorem A.4 gives the stated approximations: smooth the continuous clipped functions on compact parts and blend with their already prepared exact face copies. Allocate the error .1 beforehand over the finitely many requests on each face, and reproduce constants exactly on passive copies. For a packet of dimension $m \geq 8$ and a set I of $m - 7$ remaining labels, the smooth map $h_I = (h_j)_{j \in I}$ is locally Lipschitz. Since $\dim_{\text{H}} \Sigma \leq m - 8$, its image of Σ has zero $(m - 7)$ -dimensional Lebesgue measure. This follows on each relatively compact chart from the defining Hausdorff covers and then on the packet by a countable union. The set of threshold vectors for which $t_I \in h_I(\Sigma)$ is null by Fubini. For measurability, intersect Σ with a compact exhaustion first: each image is compact, and their countable union is measurable. Give each interval $(-.3, .3)$ its normalized Lebesgue measure. There are countably many packets and finitely many relevant subsets per packet, so choose a threshold vector outside the union of all exclusions. The same argument also works in a countable product of the threshold intervals: each exclusion is a null cylinder set depending on only $m - 7$ coordinates. If fewer than $m - 7$ labels remain, the asserted count holds without a condition. At a negative pole $h_j \leq -.9 < t_j$, and at a positive pole $h_j \geq .9 > t_j$, proving the last assertions. $\square$

The guard H_E^* in (5.7) requires at least $m - 2$ remaining undecided labels. At $x \in \Sigma$, enlarging the positive core for every label with $h_j(x) > t_j$ and the negative core for every label with $h_j(x) < t_j$ to contain x would leave at most $m - 8 < m - 2$ labels undecided. These pointwise decisions would therefore remove x from the guard.

We need quantitative pieces of Σ , rather than just its Hausdorff dimension. A coordinate chart in the following lemma is chosen with compact buffer and with its metric as close to Euclidean as required. Such charts can be chosen around every point. The plane condition is expressed in Euclidean chart coordinates. Replacing Euclidean balls by metric balls changes only fixed factors, which can be absorbed by requesting a smaller flatness constant.

Lemma C.2 (Compact pieces with one-sided flatness). *Let O be an open subset of a parent packet and let $\eta > 0$. The set $\Sigma \cap O$ is covered by countably many compact sets $F \subset \Sigma \cap O$, each contained in a buffered coordinate chart, with the following property. There is $r_F > 0$ such that, for every $x \in F$ and $0 < r < r_F$, some affine $(m - 8)$ -plane $L_{x,r}$ satisfies*

$$F \cap B_r(x) \subset \{z : \text{dist}_{\text{Eucl}}(z, L_{x,r}) < \eta r\}. \quad (\text{C.2})$$

The plane may depend on both the center and the radius.

Proof. Write $\Theta(x)$ for the unweighted boundary density. First consider a tangent minimizing boundary cone C at a singular point. It is nonflat, since a flat tangent gives regularity. For any $z \in C$, the density ratio centered at z has limit $\Theta_C(0)$ as its radius tends to infinity: balls centered at z are sandwiched between balls centered at zero whose radii differ by $|z|$. Consequently

$$\Theta_C(z) \geq \Theta_C(0) \implies \text{every density ratio centered at } z \text{ equals } \Theta_C(0).$$

Equality in monotonicity says that C is also a cone centered at z . Composing a dilation about z with the inverse dilation about zero shows translation invariance along z . Conversely, translation invariance along z gives equality of the two densities. The translation invariance directions form a vector subspace, the spine. Every spine point is singular: otherwise translation gives regularity at the vertex, and the cone is a plane. The minimizing-cone singular dimension estimate therefore bounds the spine dimension by $m - 8$. These tangent-cone and regularity inputs are precisely those in Theorem B.6.

Fix a singular point x and a small number $\eta' > 0$. There are $d_x > 0$ and $r_x > 0$ such that, at every radius $r < r_x$, the points

$$\{z \in \Sigma \cap B_r(x) : \Theta(z) \geq \Theta(x) - d_x\} \quad (\text{C.3})$$

lie within $\eta'r$ of some $(m - 8)$ -plane through x . If not, choose failing radii $r_i \downarrow 0$ with density tolerances tending to zero. Rescale at x and pass to a tangent cone C . Fix a plane containing its spine. Failure for this particular plane gives points z_i in the rescaled unit ball a fixed positive distance from it. A convergent subsequence has limit z . Upper semicontinuity of densities under these convergent blowups gives $\Theta_C(z) \geq \Theta(x) = \Theta_C(0)$. The preceding paragraph puts z in the spine, a contradiction. Smooth metric errors and bounded mean-curvature errors vanish in this rescaling.

Here is the countable decomposition, including its closure step. Group singular points according to lower bounds $r_x \geq 1/a$ and $d_x \geq 1/b$, with positive integers a, b , and according to a density interval of length less than $1/(2b)$. Any two points x, z in the same group satisfy $\Theta(z) > \Theta(x) - d_x$. Thus the group itself has the plane containment at every one of its centers and every radius less than $1/a$. Work within one buffered chart at a time. Intersect these groups with a countable family of compact subpatches $Q \Subset O$ exhausting that chart's portion of O , and take $F = \overline{Q} \cap \text{group}$. Relative closedness of Σ and the compact buffer ensure $F \subset \Sigma \cap O$. These compact sets still cover $\Sigma \cap O$.

To check the geometric condition at a closure center x , let x_i be group points tending to x and fix $r < 1/(3a)$. Apply the group estimate in $B_{2r}(x_i)$. The planes pass through x_i , so a subsequence converges to a plane through x . Every group point in $B_r(x)$ is in these larger balls eventually. Passing first to the limit of the planes and then to closure points gives containment with width $2\eta'r$. Taking, for example, $\eta' < \eta/3$ proves (C.2); choose r_F below $1/(3a)$ and the buffered chart radius. This argument transfers the containment itself; no density continuity at a closure point is required. $\square$

The next construction fixes the pressure and potential to be estimated. Distances denoted by δ from now on are ambient base distances.

Lemma C.3 (A shell forcing profile). *Let F be a compact subset of $\Sigma \cap \{h_j > t_j\}$. For every sufficiently small $\varepsilon > 0$ there is an extended-real profile u_ε , equal to $+\infty$ for $\delta \leq \varepsilon$ and $-\infty$ for $\delta \geq R_\varepsilon$, where*

$$\delta(x) = \text{dist}(x, F), \quad R_\varepsilon = e^{5\pi}\varepsilon,$$

whose finite truncations are locally Lipschitz and which satisfies

$$2|\nabla u_\varepsilon| \leq .58u_\varepsilon^2 + 2\delta^{-2} \quad (\varepsilon < \delta < R_\varepsilon). \quad (\text{C.4})$$

There is a smooth nonnegative ambient function D_ε with

$$\begin{aligned} D_\varepsilon &\geq 2\delta^{-2} && \text{on } \{\varepsilon \leq \delta \leq R_\varepsilon\}, \\ D_\varepsilon &\leq 2.12\delta^{-2} && \text{everywhere off } F, \\ \text{supp } D_\varepsilon &\subset \{\varepsilon/2 < \delta < 2R_\varepsilon\} \Subset \{h_j > t_j\}. \end{aligned} \quad (\text{C.5})$$

Replacing μ_j by $\max(\mu_j, u_\varepsilon)$, adding D_ε to Π_j , and adding $\{\delta < \varepsilon/2\}$ to U_j^+ preserves the profile inequality with coefficient a_r . There is an identical negative construction in $\{h_j < t_j\}$, using $\min(\mu_j, -u_\varepsilon)$ and enlarging U_j^- . All supports and newly forced regions are compactly contained in the stated strict anchor region.

Proof. The polynomial $q(y) = .29y^2 - y + 1$ is strictly positive, and

$$\int_{-\infty}^{\infty} \frac{dy}{q(y)} = 5\pi.$$

Let $y : (0, 5\pi) \rightarrow \mathbb{R}$ solve $y' = q(y)$, with the two infinite endpoint limits, and put

$$u_\varepsilon(x) = \frac{y(\log(R_\varepsilon/\delta(x)))}{\delta(x)}.$$

At differentiability points of δ , its gradient has norm at most one. Differentiating in δ gives

$$\left| \frac{d}{d\delta} \frac{y(\log(R_\varepsilon/\delta))}{\delta} \right| = \frac{y' + y}{\delta^2} = .29u_\varepsilon^2 + \delta^{-2},$$

which proves (C.4). Extending by the stated poles is continuous in the extended real line. Each finite clipping has a finite Lipschitz constant locally, including across either clipping interface.

Choose a continuous cutoff of δ , equal to one on $[\varepsilon, R_\varepsilon]$ and supported strictly in $(\varepsilon/2, 2R_\varepsilon)$. Multiplying it by $2.06\delta^{-2}$ gives a continuous nonnegative function supported away from F . Smooth it by nonnegative local convolutions and a partition of unity, with sufficiently small value error on this compact annulus and with support retained in a slightly larger annulus of the indicated size. The positive margins .06 on the central annulus and .06 below the upper bound give (C.5). The annuli lie in the strict anchor region for all small ε , by compactness of F .

At almost every point where the new maximum is finite, its gradient is the gradient of an active branch. The old branch obeys the old inequality. The new one obeys it because $.58 < a_r$ and $D_\varepsilon \geq 2\delta^{-2}$ wherever that branch is finite. The same reasoning applies at branch equality, using the elementary weak derivative rule for maxima of locally Lipschitz functions. The new core has closure inside the open positive pole region $\{\delta < \varepsilon\}$. By Theorem C.1 no old negative pole occurs in the support of the positive forcing. Positive and negative additions for the same label have disjoint strict anchor regions, so repetitions introduce no conflict. Indeed any opposite pole introduced later lies in the opposite strict anchor region and therefore cannot meet this support. Original poles and previously introduced opposite cores consequently remain protected throughout. Replacing the profile by its negative proves the other assertion. $\square$

We now prove a Hardy estimate with a positive, dimension-dependent coefficient. Its coefficient must be chosen before the more accurate flatness used later.

Lemma C.4 (Uniform rough Hardy coefficient). *For each $m \geq 8$ there are numbers $\eta_0 > 0$ and $c_0 > 0$ with the following property. Suppose F satisfies (C.2) with $\eta \leq \eta_0$ in a sufficiently accurate buffered chart. There are $d_F > 0$ and $C_F < \infty$ such that, on the wall and on either joined side,*

$$c_0 \int_{\delta < d_F} \delta^{-2} v^2 \leq \mathcal{N}(v) + C_F \|v\|_2^2. \quad (\text{C.6})$$

The constants η_0, c_0 depend only on m and a fixed permitted chart distortion. The constants d_F, C_F may depend on the compact piece, its scale threshold, and the local smooth metric, weight, and pressure. No principal-curvature bound for the wall is assumed.

Proof. Put $s = m - 8$. At scale r , the one-sided plane condition covers $F \cap B_r(x)$ by at most $C_m \lambda^{-s}$ balls of radius λr , with centers in F , where $\lambda = 8\eta_0$ can be made as small as desired. To see the bound, take a maximal λr -separated family in the set. Projection to the approximating s -plane changes pairwise distances by at most $2\eta_0 r$; the projections are separated by a fixed multiple of λr in an s -ball of radius $2r$. Its elementary Euclidean packing bound gives the claim, with harmless changes in C_m and the covering radius. Absorb these fixed changes into the initial choice of λ . Choose η_0 so small that the resulting covering number is at most $\lambda^{-(s+1/2)}$.

Iterating the same covering at centers in F gives, for $0 < t \leq d$ below a fixed fraction of r_F , a covering by at most $C(d/t)^{m-7.5}$ balls of radius t of the part of F controlling a ball of radius d . Here one first centers a ball of radius a fixed multiple of d at a nearby point of F and then iterates; the constants use only the fixed rough choice. Enlarging the final balls by a fixed factor proves the tubular-volume estimate

$$|B_d(x) \cap \{\delta < t\}| \leq C d^{m-7.5} t^{7.5}, \quad 0 < t \leq d, \quad (\text{C.7})$$

whenever $\text{dist}(x, F) \leq d$ and the slightly enlarged ball is in the chart at allowed scales. Metric and Euclidean distances are uniformly comparable here. Dyadically splitting the regions $2^{-k-1}d < \delta \leq 2^{-k}d$, and treating $\delta > d$ separately, gives

$$\int_{B_d(x)} \delta^{-5.5} dx \leq C d^{m-5.5}. \quad (\text{C.8})$$

Indeed the sum of the bounds is a geometric series with ratio 2^{-2} . In particular this density is locally integrable.

Choose a nonnegative smooth chart cutoff χ , equal to one on a neighborhood of F , and, using Euclidean coordinates, define

$$U(x) = 1 + \int \chi(y) \delta(y)^{-5.5} |x - y|^{-(m-4)} dy. \quad (\text{C.9})$$

For small $\delta(x) > 0$, the part $|x - y| < c\delta(x)$ has $\delta(y)$ comparable to $\delta(x)$; it supplies the lower bound for U . The same region supplies the upper bounds for U and its first derivatives. On the successive larger annuli use (C.8); their contributions have orders $(2^k \delta)^{-1.5}$ and $(2^k \delta)^{-2.5}$ respectively. The parts above the allowed local scale are bounded constants depending on F . Shrinking the final distance threshold absorbs them, and gives dimension-dependent constants $c, C > 0$ such that

$$c\delta^{-1.5} \leq U \leq C\delta^{-1.5}, \quad |\nabla U| \leq C\delta^{-2.5}. \quad (\text{C.10})$$

Set $a = m - 4$ temporarily. For any Euclidean d' -plane T , direct differentiation of $K(z) = |z|^{-a}$ gives

$$-\text{tr}_T D^2 K(z) = a|z|^{-a-2} \left(d' - (a+2) |\text{proj}_T(z/|z|)|^2 \right) \geq a(d' - a - 2) |z|^{-a-2}.$$

For $d' = m$ or $m - 1$ the last coefficient is positive. A sufficiently small, dimension-dependent metric distortion preserves a fixed positive fraction of it for traces in metric orthonormal frames. The connection and weight terms in the weighted Laplacian involve only first derivatives of U . On S_{reg} the restriction formula adds its mean-curvature vector paired with ∇U . This vector is locally bounded, since $H_{\text{base}} = p - \partial_\nu \log \omega$ by Theorem B.6. There is no dependence here on individual principal curvatures. Integrating the positive second-derivative kernel over $|x - y| < c\delta(x)$ gives $c\delta^{-3.5}$. Equations (C.10), followed by another reduction of the distance threshold, therefore imply, in the ambient space and on the regular wall,

$$-\frac{\Delta_\omega U}{U} \geq c\delta^{-2}, \quad |\nabla \log U| \leq C\delta^{-1}. \quad (\text{C.11})$$

The leading constants again depend only on the fixed rough data. For example the first-order errors before division have size at most $C_F\delta^{-2.5}$, one power lower than the positive term. The differentiated kernels through order two are locally integrable off F : their strongest singularity is $|x - y|^{-(m-2)}$. Since the density is bounded near any such x and integrable elsewhere on its compact support, ordinary differentiation under the integral, with a vanishing small diagonal ball, proves that U is C^2 off F . No diagonal distributional term is present.

For a weighted domain Q , completing a square and integrating by parts gives the vector-field identity in inequality form

$$\int_Q |\nabla v|^2 \geq \int_Q (\text{div}_\omega A - |A|^2)v^2 - \int_{\partial Q} (A \cdot \nu_Q)v^2. \quad (\text{C.12})$$

Choose a smooth cutoff ρ , equal to one near F and supported in the chart, and set $A_b = -b\rho\nabla \log U$ in the bulk. On the wall use $A_h = -b\rho\nabla_S \log U$. For $0 < b \leq 1$, where $\rho = 1$,

$$\text{div}_\omega(-b\nabla \log U) - b^2|\nabla \log U|^2 = -b\Delta_\omega U/U + b(1-b)|\nabla \log U|^2 \geq bc\delta^{-2}.$$

Outside a smaller neighborhood of F all errors from these fields are bounded by a piece-dependent constant.

On the cylinder take the horizontal field A_h and add the vertical component $a(x)\partial_t$, where

$$a(x) = \frac{.8}{1.7} b\rho(x)(1 + |\nabla \log U|_g^2)^{1/2}|_{S_{\text{reg}}}.$$

It has zero divergence in the vertical direction and satisfies $1.7a \geq .8|A_b|$ at the seam. The outward cylinder normal there is $-\partial_t$. Applying (C.12) with coefficients .8 and 1.7 on the two pieces, their boundary terms sum to

$$\int_{S_{\text{reg}}} [-.8A_b \cdot \nu_\Omega + 1.7a]v(0)^2 \geq 0.$$

This explicitly permits arbitrary matching seam values. The vertical square costs at most $Cb^2\delta^{-2} + C_F$ in the cylinder interior. Choose $b > 0$ sufficiently small, using the universal constants in (C.11), so that this is absorbed by half the positive term of order $b\delta^{-2}$. All remaining localization terms cost $C_F\|v\|_2^2$. This proves (C.6) on a joined side. The wall proof is the same calculation without the seam or the vertical field. Taking the smaller leading coefficient makes one c_0 valid for the wall and both sides. $\square$

Proposition C.5 (The local shielding estimate). *There is a sufficiently small flatness number $\eta > 0$, depending only on m and the fixed numerical coefficients, with the following property. Let F*

be any piece from Theorem C.2 for this flatness number, in a sufficiently accurate chart. For the potential of Theorem C.3, there are $\varepsilon_F > 0$ and $C_F < \infty$ such that, for every $0 < \varepsilon < \varepsilon_F$,

$$1.16 \int D_\varepsilon v^2 \leq .78\mathcal{N}(v) + C_F \|v\|_2^2. \quad (\text{C.13})$$

This holds on the true wall and either joined side, and hence on translated compact tests in a pure wall cylinder. The constants ε_F, C_F may depend on the local smooth data and F , but are independent of ε . No vanishing trace condition is imposed at a joined seam.

Proof. We give the parameter order first. Fix the rough number η_0 and coefficient c_0 from Theorem C.4. Below, constants in partition gradient bounds depend only on dimension and the fixed chart distortion. Choose $\xi > 0$ so small that the rough estimate pays their total squared-gradient error with at most .04 of $\mathcal{N}$. Next choose a large fixed K , then a small fixed $\alpha > 0$, and finally a strong flatness $\eta \leq \eta_0$, as specified below. Only after the piece F has been selected for this η do we choose its small distance threshold and then ε_F . In particular C_F is never used to choose ξ , K , or η .

Put $a_- = 1/2$, $a_+ = 2e^{5\pi}$ and $R_1 = K\varepsilon$. The shell supporting D_ε lies in $a_- \varepsilon < \delta < a_+ \varepsilon$. Choose a smooth angle $\vartheta(\delta)$ equal to zero for $\delta \leq a_+ \varepsilon$ and equal to $\pi/2$ for $\delta \geq \xi R_1$, with

$$|\nabla \vartheta| \leq \xi/\delta.$$

Such a function is obtained by a smooth transition in $\log(\delta/(a_+ \varepsilon))$ of length at least a fixed multiple of $1/\xi$. Thus it suffices to choose $K \geq (a_+/\xi) \exp(C/\xi)$, independently of the piece. Set $q = \cos \vartheta$ and $q_\infty = \sin \vartheta$. They extend as constants across F , and $|\nabla q|^2 + |\nabla q_\infty|^2 \leq \xi^2 \delta^{-2}$.

Take a smooth normalized squared partition $\sum_i \phi_i^2 = 1$ on chart boxes of scale R_1 , with uniformly bounded overlap and $\sum_i |\nabla \phi_i|^2 \leq C_m R_1^{-2}$. One construction translates a fixed compactly supported bump on a Euclidean lattice and divides each bump by the square root of the sum of their squares. Restrict the same ambient functions to the wall and freeze them on the cylinder. Only finitely many boxes meet $\text{supp } q$, which is compact in the chart at the scales in use. The family $q\phi_i, q_\infty$ is a squared partition, and its total gradient square is at most $C_m \xi^2 \delta^{-2}$ on $\{\delta < \xi R_1\}$ and zero elsewhere. The estimate follows from $\delta < \xi R_1$ wherever $q \neq 0$.

For a squared partition $\sum_i \chi_i^2 = 1$, direct expansion gives

$$\sum_i \mathcal{N}(\chi_i v) = \mathcal{N}(v) + \int a(x) \sum_i |\nabla \chi_i|^2 v^2, \quad (\text{C.14})$$

where $a(x)$ is .8 on bulk and 1.7 on wall or cylinder. The mixed terms vanish because $\sum_i \chi_i \nabla \chi_i = 0$. Choose ε_F small enough that $\xi R_1 < d_F$. Equation (C.6) and the choice of ξ now give

$$\sum_i \mathcal{N}(q\phi_i v) \leq 1.04\mathcal{N}(v) + C_F \|v\|_2^2. \quad (\text{C.15})$$

For example it is enough to impose $1.7C_m \xi^2/c_0 \leq .04$; the other partition term is nonnegative and has been discarded. Lipschitz distance cutoffs are allowed by approximation on the regular test supports.

Consider one box meeting the support of q . Choose a nearby point of F and apply (C.2) at a radius $L_m R_1$, where a fixed L_m includes the box and every nearest point of F needed in it. Taking ε_F smaller ensures this is below r_F . Let L be the resulting $(m-8)$ -plane and define

$$\zeta^2 = \text{dist}_{\text{Eucl}}(x, L)^2 + \alpha^2 \varepsilon^2, \quad P = \frac{1}{2} \zeta^2.$$

The Euclidean Hessian of P is an orthogonal projection of rank eight. Its trace on every $(m-1)$ -dimensional subspace is at least seven. Its gradient has size at most ζ , and throughout the box $\zeta \leq C_m R_1$ (after fixing α and K).

For either the ambient or the tangential weighted derivatives, the vector field $A = d_1 \nabla \log \zeta$ has potential

$$(\operatorname{div}_\omega A - |A|^2)\zeta^2 = d_1 \Delta_\omega P - (2d_1 + d_1^2)|\nabla P|^2/\zeta^2. \quad (\text{C.16})$$

Let τ bound the chart's metric distortion. The rank traces have error at most $C_m \tau$, and $|\nabla P|^2/\zeta^2 \leq 1 + C_m \tau$. The connection, weight, and wall mean-curvature contributions to $\Delta_\omega P$ have size at most $C_F R_1$. Thus one may arrange $C_m \tau \leq .005$ and $C_F R_1 \leq .025$, uniformly in the box and its plane. Equation (C.16) then gives

$$\begin{aligned} d_1 = 1 \text{ in bulk : } & \operatorname{div}_\omega A - |A|^2 \geq 4.9\zeta^{-2}, \\ d_1 = 2 \text{ on wall : } & \operatorname{div}_\omega A - |A|^2 \geq 5.9\zeta^{-2}. \end{aligned}$$

Indeed the respective Euclidean lower bounds are $8 - 3 = 5$ and $2 \cdot 7 - 8 = 6$. Only the mean curvature of the wall enters this calculation, so these bounds remain valid arbitrarily near its singular set on the regular test supports. The positive term $\alpha^2 \varepsilon^2$ also makes the fields nonsingular along L ; the estimates hold inside the inner shell as well.

In the bulk use $A_b = \nabla \log \zeta$. On a joined cylinder use $A_c = 2\nabla_S \log \zeta + \beta \zeta^{-1} \partial_t$, where $\beta = (.8/1.7)\sqrt{1 + C_m \tau}$ is chosen so that $\beta/\zeta \geq (.8/1.7)|A_b|$ at the seam. The metric distortion determines this fixed upper factor. The vertical component has zero divergence and is orthogonal to the horizontal component. Its only interior cost is $\beta^2 \zeta^{-2}$. The boundary terms in (C.12) are nonnegative after summing, by exactly the cancellation used in Theorem C.4. After multiplication by the coefficients in (C.1), the limiting interior coefficients are

$$.8 \cdot 5 = 4 \quad \text{in bulk,} \quad 1.7 \cdot 6 - .8^2/1.7 > 9.8 \quad \text{in the cylinder,}$$

and $1.7 \cdot 6 = 10.2$ on the wall alone. With the explicit error bounds just imposed, the coefficients are at least $.8 \cdot 4.9 = 3.92$ in bulk, $1.7 \cdot 5.9 - (.8^2/1.7)1.005 > 9.6$ in the cylinder, and $1.7 \cdot 5.9 > 10$ on the wall. In particular all exceed 3.8. Applying the vector inequality to $q\phi_i v$ gives

$$\mathcal{N}(q\phi_i v) \geq 3.8 \int \zeta^{-2} (q\phi_i v)^2. \quad (\text{C.17})$$

This is a bound throughout the localized support; its nonnegative right side may therefore be restricted to the shell.

We finally compare ζ with δ on that shell. A nearest point $z \in F$ lies in the ball used for the plane condition, so

$$\zeta(x) \leq 1.005\delta(x) + L_m \eta K \varepsilon + \alpha \varepsilon.$$

The first factor is ensured by the fixed chart accuracy and its buffer, after shrinking the scale so that a minimizing short path to F stays in the chart. Since $\delta \geq a_- \varepsilon$ there, choose $\alpha/a_- \leq .005$ and then $L_m \eta K/a_- \leq .015$. These choices give $\zeta \leq 1.03\delta$. Notice that K was already fixed using the rough coefficient, and that the strong flatness was chosen before selecting F . Since $3.8/(1.03)^2 > 3.5$, (C.17) controls $3.5\delta^{-2}$ on the shell. Summing over boxes uses $q = 1$ there and $\sum_i \phi_i^2 = 1$, and then (C.15) gives

$$3.5 \int_{\operatorname{supp} D_\varepsilon} \delta^{-2} v^2 \leq 1.04 \mathcal{N}(v) + C_F \|v\|_2^2.$$

Together with (C.5), this proves the stronger coefficient bound

$$1.16 \int D_\varepsilon v^2 \leq \frac{1.16 \cdot 2.12 \cdot 1.04}{3.5} \mathcal{N}(v) + C_F \|v\|_2^2, \quad \frac{1.16 \cdot 2.12 \cdot 1.04}{3.5} < .731 < .78.$$

This yields (C.13). A compact test on a pure cylinder can be translated to positive height in a joined cylinder and extended by zero onto the bulk, so it obeys the same estimate. $\square$

Remark C.6. The choices can be summarized without a hidden dependence on the compact piece: dimension and chart distortion; rough flatness η_0 and coefficient c_0 ; localization parameter ξ ; logarithmic taper and K ; α and the stronger flatness η ; the compact pieces supplied for that η ; finally their individual distance thresholds and shell scales. Shrinking a threshold absorbs large-scale kernel tails and smooth geometric errors. It never changes the leading coefficient used to select the stronger flatness. All estimates are weighted base estimates, so the number of previously appended circle factors does not replace m in any dimension or rank count.

D Global shielding and extension to cofaces

The local annular cost does not add once for every singular piece: a concentration argument bounds a new shrinking cost by the maximum of the old and local form coefficients. We prove that assertion first, then transport each chosen addition to cofaces, and finally show that only finitely many additions are needed on a whole packet. The finite-label hypotheses in this appendix are extended in Theorems 6.2 and 6.4. Each label may already have its own positive potential floor p_j^* ; none of the estimates uses a positive lower bound common to all countably many labels.

We keep the true cuts fixed throughout this section. Only profiles of remaining labels, their cores, and their potentials are changed. In particular, none of the convergence assertions below concerns minimizers for a changing pressure.

Definition D.1 (The matching form domain). Let E be a parent packet of dimension m , with true wall S and singular set Σ . A raw piece Z means either S_{reg} or one of the two joined spaces

$$Z = \Omega \cup_{S_{\text{reg}}} (S_{\text{reg}} \times [0, \infty)).$$

The tests on a joined space are compactly supported, piecewise smooth functions whose two traces agree at the regular seam and whose support avoids Σ and the cylinders over Σ . The wall has the corresponding compact tests away from Σ . All measures include the old warp weight. Put

$$\mathfrak{h}_Z(v) = \mathcal{N}_Z(v) + e_0 \|v\|_{L^2(Z)}^2, \quad e_0 = \frac{\lambda_r - \lambda_{r+1}}{100} > 0, \quad (\text{D.1})$$

where $\mathcal{N}$ is the reserved gradient form in Theorem B.11: its gradient coefficients are .8 on the bulk and 1.7 on the wall or cylinder. Write $\mathcal{H}_Z$ for the completion of these tests in $\mathfrak{h}_Z^{1/2}$. Pure cylinders $S_{\text{reg}} \times \mathbb{R}$ use the coefficient 1.7 throughout. The definitions also apply after adjoining finitely many Euclidean product coordinates, with the same coefficients and e_0 . Such products are used only to identify limits on translated windows; they do not append further circle warps or change the weight. A compact window means a fixed compact ambient base window together with a bounded cylinder interval, when a cylinder is present.

The trace matching in this definition is essential. There is no Dirichlet condition at the seam. A pure-cylinder test can be translated into the cylindrical part of a joined space, with zero bulk part. Consequently every form inequality on that joined space for potentials pulled back from its base also holds on the pure cylinder.

Lemma D.2 (Compactness on joined windows). *Assume $m \geq 8$. On any fixed compact window, a bounded sequence in $\mathcal{H}_Z$ has a strongly convergent L^2 subsequence. The following moving version*

also holds. Work in fixed smooth ambient geometry on each expanding window, as is the case after translation in an exact product end. Suppose positive weights and pressures converge smoothly there, and the true phases and walls converge as in Theorems B.6 and B.7. Consider normalized tests on the corresponding raw pieces Z_i , all supported in a fixed compact window strictly inside a larger window of convergence. After passage to a subsequence there is a function v on the limiting raw piece Z_∞ such that:

- (i) On every regular compact subwindow, the tests converge weakly in H^1 and strongly in L^2 in the smooth graphical identifications. At a limiting regular seam the two traces agree.
- (ii) No L^2 mass is lost in neighborhoods of the limiting singular set. In particular, integrals of v_i^2 converge to those of v^2 , including integrals against any fixed bounded continuous ambient potential and its pullback to the wall and cylinder.
- (iii) The limit belongs to $\mathcal{H}_{Z_\infty}$ and

$$\mathfrak{h}_{Z_\infty}(v) \leq \liminf_i \mathfrak{h}_{Z_i}(v_i). \quad (\text{D.2})$$

These assertions include pure cylinders, translated cylinder intervals, and Euclidean product limits. Constants are local to the enlarged window; no uniform bound among all packets is asserted. If the limiting wall is smooth on the enlarged window, the same conclusions follow from ordinary Rellich compactness and the bounded trace maps in the smooth graphical charts, in every base dimension.

Proof. We give the estimates for tests first; they extend to the completions by density. On the enlarged window the weights are uniformly comparable to a positive constant, the ambient geometry is uniformly smooth, and the unweighted base mean curvature of the regular walls is bounded. The latter bound follows from the prescribed weighted mean curvature and the bounded first derivative of the logarithm of the weight. The perimeter varifolds have density at least one and no additional boundary term on this interior window, by Theorem B.6. They therefore meet the hypotheses of the Sobolev inequality used below. No bound on the individual principal curvatures is needed.

We use the submanifold Sobolev inequality in the following precise form. For a d -dimensional Euclidean rectifiable varifold of density at least one with generalized mean curvature H , and a nonnegative compactly supported test f ,

$$\left(\int f^{d/(d-1)} \right)^{(d-1)/d} \leq C_d \int (|\nabla f| + |H|f). \quad (\text{D.3})$$

This holds in arbitrary codimension; see Simon [36, Chapter 4, Section 5, Theorem 5.7]. This is the submanifold Sobolev inequality of Michael and Simon [31]. A smooth extension of the fixed ambient chart metric to a compact manifold and a fixed smooth isometric embedding [33] give the corresponding local inequality in our charts. The extra mean curvature of the embedding is bounded on this fixed compact geometry. For the moving-domain assertion the embedding is chosen once in the fixed ambient geometry; its constants are therefore independent of the wall in the sequence. Applying Equation (D.3) to a power of the test gives the usual H^1 Sobolev exponent on the wall. Applied to $S_{\text{reg}} \times \mathbb{R}$, it gives, with

$$q = \frac{2m}{m-2}, \quad b = \frac{2(m-1)}{m-2}, \quad 2(b-1) = q,$$

a local L^q bound on the cylinder. A test on the half-cylinder is reflected evenly at its base before this application. The wall alone has the stronger exponent $2(m-1)/(m-3) > 2$.

Here and below cutoffs are taken on the enlarged window and are pulled back to both seam sides. The cutoff errors are bounded by the local L^2 norm. Normalize $\mathfrak{h}_Z(v) \leq 1$. Choose a cylinder cutoff which is one at $s = 0$ and vanishes at the outer end of a fixed bounded interval. The fundamental theorem of calculus and Hölder give

$$\int_{S_{\text{reg}}} |v(0)|^b \leq C \int_{S \times [0, L]} (|v|^b + |v|^{b-1} |\partial_s v|) \leq C. \quad (\text{D.4})$$

The last step uses the cylinder L^q bound and $2(b-1) = q$.

The bulk need not have a uniform smooth boundary, so we use a BV calculation there. Extend $|v_{\text{bulk}}|^b$ by zero in the ambient chart, including its compact cutoff. For an initial test its support misses the singular set, and the BV jump formula gives precisely the boundary contribution $\int_{S_{\text{reg}}} |v(0)|^b$ in addition to the interior variation. This is also obtained directly by integration by parts against unit vector fields on the regular side. The ambient L^1 Sobolev inequality and Cauchy–Schwarz therefore yield, for $J = \int_{\Omega} |v|^q$,

$$J^{(m-1)/m} \leq C \left(\int_{\Omega} |v|^b + \int_{\Omega} |v|^{b-1} |\nabla v| + \int_{S_{\text{reg}}} |v(0)|^b \right) \leq C(1 + J^{1/2}). \quad (\text{D.5})$$

Here $bm/(m-1) = q$; the first integral on the right is at most $J^{1/2} \|v\|_2$, because $b = (q+2)/2$. Since $(m-1)/m > 1/2$, Equation (D.5) bounds J . Rescaling removes the normalization. Thus all pieces have a local L^p bound for some $p > 2$ controlled by their matching form norm. This argument extends a power as a BV function; it constructs no H^1 extension operator for the bulk.

We next exclude concentration in the moving version. On a slightly larger compact window let F be the compact portion of the limiting singular set. Its ambient volume and its limiting perimeter measure are zero. For a closed δ -neighborhood of F , perimeter measure convergence and the closed-set upper bound for weakly convergent measures imply

$$\limsup_i \mu_{S_i}(\overline{U_{\delta}(F)}) \leq \mu_{S_{\infty}}(\overline{U_{\delta}(F)}),$$

with a fixed outer compact cutoff if necessary. The right side tends to zero as $\delta \downarrow 0$. The same conclusion for bulk measure follows from ambient volume bounds, and for a cylinder follows by multiplying the wall measure by the bounded interval length. Hölder and the local L^p estimates give

$$\int_{A_i} |v_i|^2 \leq C \mu_i(A_i)^{1-2/p}.$$

Thus the order is first $i \rightarrow \infty$ at a fixed singular neighborhood, then $\delta \downarrow 0$. Neither disappearing components nor a thin boundary layer can retain L^2 mass.

Off F , graphical convergence and the absence of stray wall support supply smooth charts for both bulk phases and their seams. Rellich compactness in those charts gives the asserted weak H^1 and strong L^2 limits. The trace map on each fixed regular chart is bounded and linear; weak convergence of both traces preserves their equality. Combining these regular limits with the no-concentration conclusion proves all the stated L^2 and bounded-potential convergence. Exhausting the regular pieces gives lower semicontinuity of their gradient integrals.

It remains to justify membership in the limit completion, rather than assume it. On this one limiting geometry, monotonicity and the local area bound give

$$\mu_S(B_{2a}(x)) \leq Ca^{m-1}$$

for sufficiently small balls in the fixed window. Since $\dim_H \Sigma \leq m - 8 < m - 3$, a compact singular portion has ball covers, inside arbitrarily small neighborhoods, with $\sum_i a_i^{m-3}$ arbitrarily small. For cutoffs equal to one on these balls, supported on their doubles, and with gradients at most C/a_i , the maximum of the cutoffs has squared-gradient integral at most

$$C \sum_i a_i^{m-3}$$

on the wall and on bounded cylinder intervals. The bulk cost is at most $C \sum_i a_i^{m-2}$ and is smaller. A finite subcover suffices. Use the same base cutoff on the bulk and cylinder, preserving seam matching. First truncate the limit function to a bounded function; then multiply it by one minus these cutoffs. The derivative of the cutoff has vanishing energy after truncation, and the other error tends to zero by absolute continuity of the already finite gradient integrals. Its L^2 error tends to zero as well. The resulting functions miss the singular set. Matching traces glue their two H^1 pieces across every regular seam chart; ordinary mollification in such charts, and a partition of unity, approximate them by the defining tests. This proves membership and Equation (D.2). Capacity is used only on the limit; no cheap cutoff on every approximating singular set, and no convergence theorem for the entire moving Sobolev spaces, is needed. The same proof applies to pure cylinders and product limits. For a product with a extra coordinates, use ambient dimension $m + a$ and singular-dimension bound $m + a - 8$ in the preceding exponents; the strict codimension needed for the capacity estimate is unchanged. Thus this argument supplies exactly regular graphical identifications, strong L^2 convergence without singular concentration, matching seam traces, and the lower semicontinuity used in the later form transfers. It does not assert smooth convergence at singular points. $\square$

Lemma D.3 (Maximum of successive form costs). *Fix a packet and its true pieces. Let $V \geq 0$ be a fixed bounded potential, pulled back to cylinders, satisfying*

$$\int_Z V v^2 \leq \theta \mathfrak{h}_Z(v) \quad (\text{D.6})$$

on its wall and both joined sides. Let $F \subset \Sigma$ be a compact piece and let $A_\varepsilon = 1.16D_\varepsilon$ be one of the nonnegative shell potentials of Theorems C.3 and C.5. In particular its base support lies in a fixed compact set and shrinks to F , and

$$\int_Z A_\varepsilon v^2 \leq .78 \mathcal{N}_Z(v) + C_F \|v\|_2^2 \quad (\text{D.7})$$

for all sufficiently small ε , on all three pieces. For every $\eta > 0$, a sufficiently small $\varepsilon > 0$ gives simultaneously

$$\int_Z (V + A_\varepsilon) v^2 \leq (\max\{\theta, .78\} + \eta) \mathfrak{h}_Z(v). \quad (\text{D.8})$$

It suffices that V be bounded on the compact base windows used below. In particular the assertion applies to a finite sum of compact additions on the initial radial parent.

Proof. First $\int A_\varepsilon v^2 \rightarrow 0$ for every fixed $v \in \mathcal{H}_Z$. Indeed Equation (D.7) bounds these forms uniformly by a constant times $\mathfrak{h}_Z$, whereas for a defining test the integral is zero for small ε , since its support misses F . Density and Cauchy–Schwarz for the nonnegative form give the assertion.

We detail the localization needed on noncompact pieces. If $\sum_\alpha \chi_\alpha^2 = 1$ is a locally finite Lipschitz partition which has the same seam traces, then

$$\sum_\alpha \mathfrak{h}_Z(\chi_\alpha v) = \mathfrak{h}_Z(v) + \int_Z c_Z \sum_\alpha |\nabla \chi_\alpha|^2 v^2, \quad c_Z = \begin{cases} .8 & \text{on bulk,} \\ 1.7 & \text{on wall or cylinder.} \end{cases} \quad (\text{D.9})$$

The mass term $e_0\|v\|_2^2$ allows the last integral to be at most $\tau\mathfrak{h}_Z(v)$ for any prescribed $\tau > 0$. For example use a sine/cosine pair of a Lipschitz cutoff angle, equal to $\pi/2$ on a fixed compact base set containing $\text{supp } A_\varepsilon$ for all small ε , and zero outside a sufficiently large compact neighborhood. Its transition width makes its derivative arbitrarily small. The cosine part sees only V . Its bound is the assumed global form inequality for V ; no compact support assumption on V is needed. The sine part has compact base support, where the old potential is bounded. On a joined side also use a squared partition into cylinder intervals of a sufficiently large fixed length. The first member is identically one near $s = 0$ and extends as one on the bulk; all the others vanish near the seam and extend as zero there. Their squared gradients have an arbitrarily small sum. All cutoffs in the base are pulled back to the cylinder.

Choose the partition error so small that a violation of Equation (D.8) forces one localized test to have quotient larger than $\max\{\theta, .78\} + \eta/2$. This follows directly by summing the potential integrals and Equation (D.9); for instance it suffices to have $(\max\{\theta, .78\} + \eta/2)(1 + \tau) < \max\{\theta, .78\} + \eta$. The selected test has compact base support and a bounded cylinder interval. Passing to a subsequence, its interval either stays in a fixed bounded range, or escapes to infinity. In the latter case it is translated to a fixed interval on the pure cylinder. Both the old bound and Equation (D.7) hold there by translating its tests into the original joined side. Thus in all cases the selected sequence lies in one fixed Hilbert space, in a fixed compact window.

Suppose now that $\varepsilon_i \downarrow 0$ gives such violating tests v_i , normalized by $\mathfrak{h}(v_i) = 1$. By Theorem D.2 and weak Hilbert compactness, $v_i \rightharpoonup v$ in this fixed space and $v_i \rightarrow v$ strongly in L^2 . Write $w_i = v_i - v$. Then

$$\mathfrak{h}(w_i) = 1 - \mathfrak{h}(v) + o(1), \quad \|w_i\|_2 \rightarrow 0.$$

The bounded old potential satisfies $\int V v_i^2 \rightarrow \int V v^2 \leq \theta \mathfrak{h}(v)$. The new form vanishes on v , and its mixed term vanishes by Cauchy–Schwarz and the uniform bound on w_i . Consequently Equation (D.7) gives

$$\begin{aligned} \limsup_i \int (V + A_{\varepsilon_i}) v_i^2 &\leq \theta \mathfrak{h}(v) + .78 \limsup_i \mathcal{N}(w_i) \\ &\leq \theta \mathfrak{h}(v) + .78(1 - \mathfrak{h}(v)) \leq \max\{\theta, .78\}. \end{aligned}$$

This contradicts the selected quotient. There are only three raw pieces, so the same sufficiently small scale works for all of them. $\square$

We next extend a single chosen addition. The scale in Theorem D.3 is fixed before this next limit is taken.

Lemma D.4 (Extension with late fronts). *Let B be a packet, let $D \geq 0$ be a fixed smooth compactly supported potential addition on B , and let V denote the old cumulative potential on every packet. Suppose these old potentials are exactly compatible and obey the common form bound $\theta \mathfrak{h}$. Suppose also that*

$$\int (V_B + 1.16D) v^2 \leq \beta \mathfrak{h}_B(v), \quad \beta \geq \theta,$$

on the wall and joined sides over B . In each of the finitely many coface occurrences $B \times (T, \infty)^d \subset E$, introduce a smooth potential D_E with

$$0 \leq D_E \leq D \circ \pi_B, \quad \text{supp } D_E \subset K \times \prod_{i=1}^d [R_i, \infty), \quad K \Subset B. \quad (\text{D.10})$$

Use consistent functions on the normal coordinates identified by commuting face chains, so that these extensions are compatible. For every $\eta > 0$ all starts can be chosen sufficiently large that

$$\int (V_E + 1.16D_E)v^2 \leq (\beta + \eta)\mathfrak{h}_E(v) \quad (\text{D.11})$$

on every affected raw piece. The conclusion concerns the same fixed true cuts; it uses their product convergence from Theorem B.7.

Proof. Only finitely many occurrences are involved by Theorem 5.1. Here and below a source is extended along the directed face-to-coface relations, through their finite depth. Closing under arbitrary alternating face and coface walks is neither required nor performed. For a common normal coordinate take the largest of the finitely many requested starts, so commuting chains give identical occurrences. It is enough to prove the assertion for any sequence of their fronts with $\min_i R_i \rightarrow \infty$, passing to a fixed occurrence and piece if needed. Choose every localization width below using η and e_0 first, independently of the front positions.

Separate the new support from its complement with a sine/cosine pair as in Equation (D.9). More explicitly, an angle can be $\pi/2$ times a product of a compact base cutoff, equal to one on K , and normal cutoffs which equal one on $[R_i, \infty)$ and vanish before $R_i - L$. Enlarge the compact base window and L until the sum of squared derivatives is sufficiently small. Extend the angle by zero outside the product sector. Taking the fronts beyond all collar thresholds makes this extension well defined. The cosine test misses the addition and has the old bound θ . Partition the sine part further into fixed large boxes in every normal coordinate, and into fixed large intervals in cylinder height when applicable. The total localization error is as small a fraction of $\mathfrak{h}$ as required.

If Equation (D.11) failed along fronts tending out, the same quotient selection as in Theorem D.3 would give normalized tests on selected boxes with quotient at least $\beta + \eta/2$. Such a box must meet the new potential support, since otherwise the old bound $\theta \leq \beta$ applies. Its center therefore has

$$t_i \geq R_i - O(L) \longrightarrow \infty \quad \text{for every } i = 1, \dots, d.$$

Its base window is fixed and compact. Recenter every normal coordinate. A bounded cylinder interval stays in a bounded-height joined window, or, after another subsequence, is translated to a pure cylinder. The ambient geometry on the recentered windows is exactly the product geometry, and the true cuts converge to those over B times $\mathbb{R}^d$. The old potential is exactly $V_B \circ \pi_B$ there, because the fronts also exceed its fixed product thresholds. The new potential is bounded above by the fixed bounded function $D \circ \pi_B$.

Apply Theorem D.2. For its limit v ,

$$\limsup \int (V_E + 1.16D_E)v_E^2 \leq \int (V_B + 1.16D) \circ \pi_B v^2, \quad \mathfrak{h}_{\lim}(v) \leq 1. \quad (\text{D.12})$$

The face form bound applies to the right side by slicing the $\mathbb{R}^d$ coordinates. For defining tests this is Fubini's theorem: the gradient terms in the extra directions are nonnegative, and all coefficients $.8, 1.7, e_0$ are the same in every base dimension. Approximation in the limit completion, proved in Theorem D.2, gives the result for v . In a pure-cylinder limit use the face joined bound after translation in cylinder height. Thus the right side of Equation (D.12) is at most β , a contradiction. Finite incidence permits simultaneous choices for all occurrences. This limit compares bounded potential integrals on moving domains; it uses no subtraction of vectors belonging to different form spaces. $\square$

Lemma D.5 (Compatible caps and cores). *A compact forcing update from Theorem C.3 extends to all its cofaces so as to satisfy Equation (D.10), preserve the profile gradient inequality with coefficient a_r , and enlarge only the designated cores. The extension is an exact product sufficiently far in every old face collar and is made once per occurrence, independent of the order of commuting face relations.*

Proof. Consider a plus addition on B , with forcing profile u and shell potential D . Choose a compact base window $K \subseteq \{h_j > t_j\}$ containing both $\overline{\{u > -\infty\}}$ and $\text{supp } D$ in its interior. The first set includes the inner pole region; keeping only $\text{supp } D$, which is annular, would not suffice for this extension. In a coface product over K put

$$u_E(x, t) = \min\{u(x), w_1(t_1), \dots, w_d(t_d)\}, \quad D_E(x, t) = D(x) \prod_{i=1}^d \chi_i(t_i). \quad (\text{D.13})$$

Each cap w_i is $-\infty$ up to a late front, then crosses to $+\infty$, and remains there, with

$$2|w'_i| \leq .58w_i^2 + p_j^*$$

on its finite band. For example integrate $2w'_i = .58w_i^2 + p_j^*$ on the finite band and extend by its poles. The crossing length is $2\pi/\sqrt{.58p_j^*} < \infty$; finite truncations remain locally Lipschitz at both ends. Only this label's positive floor is used. Each smooth cutoff χ_i takes values in $[0, 1]$ and *finishes* increasing to one before w_i begins its crossing. Choose these cutoff fronts to satisfy Theorem D.4 for the already fixed D and shell scale. Outside this product occurrence use the inactive value $-\infty$. The old profile is replaced by its maximum with u_E .

For locally Lipschitz finite truncations, a finite minimum or maximum has the derivative of one active branch almost everywhere (on a tie, the branch derivatives agree almost everywhere on the tie set). If the active finite branch is u , every cap has already left its $-\infty$ plateau; hence every $\chi_i = 1$ and the entire shell potential D is available. If it is a cap, its derivative cost is paid by the preexisting baseline:

$$.58w_i^2 + p_j^* \leq a_r w_i^2 + \Pi_j + D_E.$$

The old branch retains its old inequality. Thus no sum of normal cap derivatives, and no extra cap potential, enters the budget. For a minus update negate u_E and take a minimum with the old profile. The strict anchor regions for opposite signs are disjoint and contain no opposite old poles by Theorem C.1.

Write T_i^+ for the end of the i th cap crossing. Enlarge the plus core by the open set

$$\{x : \delta_B(x) < \varepsilon/2\} \times \prod_i (T_i^+ + b_i, \infty), \quad b_i > 0.$$

The properly embedded closed tails ensure that its closure in the coface has $\delta_B \leq \varepsilon/2$ and $t_i \geq T_i^+ + b_i$. It is therefore contained in the open set $\{\delta_B < \varepsilon\} \times \prod_i (T_i^+, \infty)$, where every branch of the minimum is $+\infty$. This proves closure confinement, including at the cap fronts, with the corresponding statement for a minus core. The front positions are also chosen beyond the exact anchor-product thresholds, so its support retains the designated strict anchor sign.

Use the same caps and cutoffs for the same normal coordinates in every identified occurrence. On a face obtained by hitting one of these coordinates, that coordinate's cap is eventually $+\infty$ and its cutoff is one, and that factor disappears from Equation (D.13). If an unrelated coordinate is hit, an intersection sufficiently far in its collar would require x to escape into a further face of B , which the compact support forbids. Finite incidence allows all relevant intersection thresholds to be exceeded at once. These observations prove compatibility for all chains and show why a commuting product occurrence is not counted more than once. $\square$

Proposition D.6 (The global shielding output). *At a fixed stage r , let j_0 be the current label and keep its true minimizers fixed. There are updates of every remaining profile and core, and compatible smooth additions $D_j \geq 0$, with the following properties on every parent packet E of dimension m :*

- (i) *The updated profiles have locally Lipschitz finite truncations and satisfy, almost everywhere on their finite bands,*

$$2|\nabla \mu'_j| \leq a_r(\mu'_j)^2 + \Pi_j + D_j. \quad (\text{D.14})$$

All these data are exact compatible products. Each core only enlarges, and its closure remains in the interior of its designated updated pole set. The original forced sets and the anchor sign restrictions are preserved.

- (ii) *With $G'_j = E \setminus (U_j'^+ \cup U_j'^-)$, the set*

$$H_E^* = \{x : \#\{j \neq j_0 : x \in G'_j\} \geq m - 2\} \quad (\text{D.15})$$

is closed and misses Σ_E . In an exact collar over a face B , its projection is contained in H_B^ . This holds for every further face and every simultaneous product limit.*

- (iii) *For one common $\theta_* < .9$, all walls and joined sides satisfy*

$$\int 1.16 \sum_{j \neq j_0} D_j v^2 \leq \theta_*(\mathcal{N}(v) + e_0 \|v\|_2^2) \leq .9(\mathcal{N}(v) + e_0 \|v\|_2^2). \quad (\text{D.16})$$

The additions are bounded with their derivatives on each post-radial packet. There are only finitely many local additions on each such packet and only finitely many extensions into it.

For the initial radial parent the same conclusions hold with locally finite additions over its compact transition shells; boundedness is asserted on the corresponding compact blocks and Equation (D.16) holds on every compact test. No convergence assertion at an end of the original manifold is needed.

Proof. First consider the packet system after the radial cut. Process its finite set of depths from the deepest faces upward. The anchors and thresholds of Theorem C.1 are fixed once, before any updates. Initially the cumulative addition is zero. Maintain a single common upper form bound at least .78; it is harmless to start this bookkeeping bound at .78 although the initial form vanishes. Enumerate the countable pairs consisting of a packet and a possible successive local addition number by distinct integers $\nu \geq 1$. Reserve $\gamma_\nu = .12^{-\nu}$ for that pair, allocating at most half to Theorem D.3 and half to the simultaneous extensions of Theorem D.4. The common bound after any collection of completed updates is at most

$$.78 + \sum_{\nu \text{ used}} \gamma_\nu \leq .88 < .9.$$

We may consequently take $\theta_* = .88$. There is no requirement that a shell scale or its extension fronts be uniform in the packet or in the addition number.

Suppose all strict further faces of E have been treated and their updates extended. The remaining risky set

$$K_E = \Sigma_E \cap H_E^*$$

is compact. To prove this, if points in it escaped, the packet escape property Theorem A.3 would give a subsequence over a deeper face B , with compact convergent base coordinates and every escaping normal coordinate tending to infinity. The updated cores are exact products there. There are finitely many relevant labels, so pass to a further subsequence with a fixed subset of $m - 2$ undecided labels. Membership in each of their closed sets G'_j passes to the base limit. Thus the

limit base point has at least $m - 2$ undecided labels. This implies membership in H_B^* , because $\dim B - 2 \leq m - 2$. By Theorem B.7 the translated true cuts converge to the true cut over B times the normal Euclidean factor. Support convergence puts the limiting base point on the true boundary over B . A regular limiting boundary point would have a neighborhood of smooth graphical convergence, which contains no singular points of the approximating walls. The limit must therefore be singular on the face. This contradicts its completed shielding. Thus K_E has no escaping sequence; it is closed, and the compact-part property proves compactness. If $m < 8$ it is already empty.

Denote this initial compact set by $K_E^{(0)}$. For each remaining label and each strict anchor sign, cover $K_E^{(0)}$ in that open region by the countable compact flat pieces of Theorem C.2. These pieces need not themselves be an open cover. Enumerate the signed pieces; their eventual inner forcing neighborhoods will provide the open cover needed for termination. For each, choose its shell sufficiently small by Theorem D.3, then hold that scale fixed and extend its caps, potential, and core by Theorems D.4 and D.5. Supports can be as small as needed to stay within the prescribed strict anchor region. Opposite updates for one label therefore never conflict. Profiles retain Equation (D.14) by their active-branch inequalities, and cores enlarge monotonically.

To prove finite termination, after N additions set

$$O_N = \{x \in E : \#\{j \neq j_0 : x \in G_j^{(N)}\} < m - 2\}, \quad K_E^{(N)} = K_E^{(0)} \setminus O_N.$$

Each O_N is open: with finitely many labels its complement is a finite union of intersections of closed $G_j^{(N)}$. Core enlargement implies $O_N \subset O_{N+1}$. Fix $x \in K_E^{(0)}$. For each of its strict anchors, some enumerated compact piece contains x , and the addition on that piece puts x in an open enlarged core, irrespective of how small its positive shell scale was chosen. Only finitely many anchors are involved. Once those additions have occurred, only threshold-equality labels can remain undecided at x . By Theorem C.1 there are at most $m - 8 < m - 2$ of those. Thus $x \in O_N$ for some finite N . The increasing open sets O_N cover $K_E^{(0)}$, so compactness supplies a finite subcover; taking its largest index gives $K_E^{(N)} = \emptyset$. Stop there. This argument proves an actual finite stopping time before forming any infinite sum of profiles. All subsequent updates only shrink the guard.

There may be countably many packets at one depth. An originating packet has finitely many directed coface occurrences, so each individual extension has already been chosen in finitely many places. Conversely, a fixed target packet has finitely many source faces, including through chains, so only finitely many packets at that depth affect it. Each of those sources required finitely many additions. This uses both directions of finite incidence from Theorem 5.1. Its updated functions are finite sums or finite minima and maxima; there is no infinite limiting profile on that packet. Its product thresholds are bounded by finitely many chosen thresholds, and all its potential derivatives remain bounded. The common form bound after a whole depth follows from the same summable allowance. This justifies proceeding to the next depth, and proves the asserted smoothness, compatibility, and form bounds throughout the system.

The new guard is closed because the number of labels is finite and each G_j' is closed. On a far collar, exact product cores give the same count upstairs and downstairs. Lower face dimension weakens the required count, proving the projection assertion in (ii). These are precisely the closed-set and product-limit properties needed for the guarded regularization in Appendix E; a quantitative smooth tube is obtained there from regular graphical convergence.

For the initial radial cut, process its locally finite compact transition shells in order, choosing the supports of all additions inside buffered neighborhoods of their respective shells. By Theorems B.4 and 5.4, the barrier subbands separate these neighborhoods, and the true boundary lies in their interiors. The true singular set in a shell is closed inside its compact buffered annulus, and the guard is closed. Their intersection is therefore compact. Apply the same increasing-open-cover

argument there, keeping the closure of every forcing region inside this one shell buffer. There is no coface extension at this step. Before any addition only finitely many previous additions have been made, so their total is a bounded compactly supported potential and Theorem D.3 applies. The base localization is obtained from a proper distance cutoff; all estimates after localization use only fixed compact data. Use summable allowances as before. Local finiteness leaves smooth potentials and well-defined profiles, and a compact test sees only finitely many shell additions. The form bound therefore persists after all shells are treated. This proves the radial assertion and completes the construction. $\square$

E Regularization and restoration of the curvature invariant

The minimizing partition is smooth away from a singular set, whereas the next cutting step needs smooth complete packets. Shielding has made the closed guard disjoint from the singular set; the first lemma below obtains the regular tubes needed for comparison. We construct globally smooth partitions that accurately approximate the true partitions near these guards. We then transfer the scalar form to the new walls and joined sides, smooth the cylinder seams, and convert the form bound into a pointwise bound by adding circle factors. Throughout this section a stage r and its current label j_0 are fixed. The profiles and cores of the remaining labels have already been updated by Theorem D.6. Write

$$\begin{aligned}\tilde{\Pi}_j &= \Pi_j + D_j, & G'_j &= E \setminus (U_j^{+, \text{updated}} \cup U_j^{-, \text{updated}}), \\ H_E^* &= \left\{ x \in E : \#\{j \in J_r \setminus \{j_0\} : x \in G'_j\} \geq m - 2 \right\}.\end{aligned}$$

Here $m = \dim E$ is the parent base dimension. The set H_E^* is closed and misses the singular set of the true boundary. A nonpositive counting threshold means the whole packet. Scalar curvatures, mean curvatures, Laplacians, and measures include the old torus factors; functions and tests are invariant in those factors. Constants in this section may depend on a packet and its finite incidence diagram. At the first, radial step the corresponding constants are assigned on enlarged compact radial blocks. In particular no uniform geometry bound for the original manifold is used.

Use adjacent parameters $0 \leq t_r < t_{r+1} \leq 1$, and put

$$\begin{aligned}\Delta_r &= t_{r+1} - t_r, & a_r &= .60 + .03t_r, & b_r &= 1.12 - .05t_r, & \lambda_r &= .25 - t_r/8, \\ \eta_r &= \Delta_r/200, & e_0 &= (\lambda_r - \lambda_{r+1})/100 = \Delta_r/800, & C &= 3.99.\end{aligned}$$

Only positivity of the particular increment is needed; none of the geometric choices is required to be uniform in r . The proofs below are stated for finite label data. Their constant-preserving form, spelled out in Theorem E.5, will also be used for a finite exceptional list together with a summable constant tail.

E.1 Guarded partitions and transfer of the raw form

Lemma E.1 (Uniform control on the guarded region). *Fix a finite integer $J \geq 4$. On each parent packet E , let ς_E be signed distance to the true boundary support, truncated at ± 1 , positive in the plus phase. If the support is empty on a component, use the corresponding constant sign. After the radial split, the following choices can be made on each whole parent packet.*

(i) *There is a positive $\rho_E < 1/10$ such that near*

$$\{x : \text{dist}(x, H_E^*) < 4\rho_E, \ |\varsigma_E(x)| < 4\rho_E\}$$

the true boundary is regular, has an embedded normal tube, and signed distance has bounded derivatives through order $J + 2$ on a slightly smaller tube. The current forced core closures have positive uniform distance from the true boundary.

- (ii) *On an elementary collar $B \times (T, \infty)$, signed distances converge uniformly to the pullback of ς_B as $T \rightarrow \infty$. They converge in C^J on the guarded tubes, with the coface tube chosen inside the face's tube. These statements are uniform over the whole face, including its noncompact ends.*

For the initial radial parent, assertion (i) holds separately on each enlarged compact radial block, with its own radius and derivative bounds; assertion (ii) is not required. The radii depend on the true partitions and the updated guards, and can be chosen before any smooth-approximation tolerances.

Proof. On a compact set, closedness of H_E^* and its disjointness from the singular set give a regular neighborhood of every point where the guard meets the boundary. A finite subcover and the normal inverse function theorem give a positive tube radius and all the asserted finite-order bounds there. The same compactness argument applies to the separation of the current core closures, which lie inside the barrier domains of Theorems B.1 and B.6 and whose neighborhoods miss the support.

We explain why these statements remain uniform at infinity. If a sequence escapes in E , the finite face description in Theorem A.3 gives, after passage to a subsequence, a product window over a further face F in which the F coordinates converge in a compact set and all the remaining normal coordinates tend to infinity. Bounded metric balls lie in these product windows eventually. The true phases converge there to the product of the true phase of F , by Theorem B.7. The updated cores and their complements are exact products. Consequently a limit of guarded points lies in H_F^* : a count of at least $m - 2$ is at least $\dim F - 2$. The limiting boundary point is therefore regular. Smooth, single-sheet convergence near this point supplies an embedded normal tube of a fixed positive radius, and excludes additional support in a smaller neighborhood. In particular no sequence at distance tending to zero from both the guard and the support can have degenerating tube radius or unbounded prescribed finite derivatives. This proves the first assertion by contradiction, with buffers reduced if necessary. The barrier statement at infinity follows in the same way from its face statement and phase/support convergence.

For uniform convergence in an entire elementary collar, suppose a discrepancy persists at (x_i, t_i) with $t_i \rightarrow \infty$. If x_i stays in a compact part of B , local product convergence gives a contradiction. Otherwise descend through the finite face diagram of B . Commutation of its collars with the t collar puts both the partition of E and that of B over the same further face F , with compact tangential coordinates and all escaping coordinates tending to infinity. Both limiting partitions are the same unique product partition. Local support convergence implies convergence of truncated distances: a nearest point at distance less than 1 is detected in a slightly larger fixed ball, and if there is no such point both truncated distances approach the saturated value. Away from the limiting support, phase convergence fixes the sign. This proves uniform C^0 convergence. On a guarded tube the limiting boundary is regular; smooth graph convergence and the normal inverse function theorem give C^J convergence of signed distance on smaller tubes.

Choose radii from the deepest faces upward. Each coface radius is smaller than a fixed fraction of every relevant further-face radius. For example, a point within ρ_E of H_E^* projects, far out, to within $2\rho_E$ of H_B^* , and $|\varsigma_E| < \rho_E$ implies $|\varsigma_B| < 2\rho_E$ there. These are statements about the fixed true geometry and exact cores, without approximation tolerances. This order also handles a simultaneous escape in several coordinates and proves the last assertion. At the radial step the same proof on compact blocks requires no assertion about an initial spatial end. $\square$

Lemma E.2 (Compatible smooth partitions). *There are compatible smooth defining functions g_E with 0 a regular value, positive and negative on the respective current forced core closures, and with*

$L_E = g_E^{-1}(0) \subset \text{int } G_{j_0}$. After the radial split they can be chosen uniformly as close as prescribed to ς_E in value and in C^J on smaller guarded tubes. In a buffered neighborhood of H_E^* , a smooth diffeomorphism arbitrarily close to the identity in C^{J-1} identifies the smooth partition with the true partition. It uses the same map on the wall and on the boundaries of its two sides. At the initial radial step, the approximation tolerances and comparison buffers are chosen separately on enlarged compact radial blocks. The defining functions, but not this comparison map, are exact products in sufficiently far old collars.

Proof. First choose all the geometric radii in Theorem E.1. Prescribe positive value and C^J error targets on the guarded regions, small enough also to preserve the core signs. Allocate smaller targets to faces to meet the requests of all their cofaces, including through chains, as in Theorem A.4. There are only finitely many such requests per face and finite depth. This allocation does not change any tube radius.

Work now from deeper faces upward. Smooth ς_E by local convolution and a smooth partition of unity, with arbitrarily small value error everywhere and derivative error on the regular buffers. These errors can be prescribed separately: use the smooth function itself on an inner regular buffer and blend to a value approximation outside a larger one. On a far collar blend to the previously chosen face function. Theorem E.1 makes the discrepancy from the face's true signed distance arbitrarily small by moving the collar start. The discrepancy of the already chosen face function from that distance is instead paid from its preallocated error target. These two errors are kept separate.

More explicitly, take cutoffs f_i from 0 to 1 in the elementary normal coordinates, in valid far product ranges. Use weights

$$w_0 = \prod_i (1 - f_i), \quad w_i = f_i \prod_{a < i} (1 - f_a).$$

They sum to 1. All derivatives through order J are bounded by constants fixed before the targets are allocated (translated cutoffs have the same bounds). Thus blending costs a fixed multiple of the participating errors. Face copies agree on all relevant overlaps once the starts exceed their product thresholds. When any $f_i = 1$, every participating copy equals that face copy on the overlap; hence the result is exactly compatible. This proves the claimed approximations.

Near a possible zero in a guarded tube, the defining function has nonzero normal derivative and its zero is a small normal graph over the true wall. Off the tube, uniform value closeness preserves the phase where its sign is required. The copied end functions already have regular zeros. On the remaining compact region, a relative regular-value perturbation makes all zeros regular without changing the copied ends. For completeness, if g is the preliminary function, choose a smooth $\chi \geq 0$, positive on its nonregular zeros and zero near the regions to be kept fixed. For an arbitrarily small regular value c of $-g/\chi$ on $\{\chi > 0\}$, the function $g + c\chi$ has regular zeros there. Where $\chi = 0$, its differential is that of g , since $d\chi = 0$ there. Bounded derivatives of χ and small c preserve all strict error and sign requirements. On the initial space use a positive smooth χ with locally prescribed small derivatives and the same regular-value argument; the radial separators are preserved.

If the perturbed wall is $t = \alpha(x)$ in a true normal tube, the map

$$(x, t) \mapsto (x, t - \chi_1(x, t)\alpha(x))$$

with $\chi_1 = 1$ on the needed graph region and supported inside a larger regular buffer is a diffeomorphism for sufficiently small graph norm. It carries the graph to $t = 0$, preserves its two local sides, and tends to the identity in C^{J-1} . Choose the buffer cutoff on the regular wall slightly beyond the projection of the smaller guard neighborhood and extend by the identity. Uniform regular charts

give bounded cutoff derivatives. The same formula restricts to the wall used as a cylinder base. No product property of this auxiliary comparison is asserted or needed. $\square$

Perform the split of Theorem A.1 using these defining functions. For the moment use the raw metric: the old metric on the bulk and the frozen induced metric on each fresh infinite cylinder. Let p_A denote restriction/inclusion on a wall or bulk and cylinder projection on a fresh cylinder. Pull back the remaining profiles, cores, and $\tilde{\Pi}_j$ by p_A . These data are continuous at a seam and compatible in old ends. Define the new required set D_A from the remaining cores and the new base dimension. The raw forms of Theorem B.10 were defined on the true minimizing wall and its joined sides. We first estimate those forms, then use the comparison map to transfer the estimate to the raw metrics of the smooth split just constructed.

Lemma E.3 (Counts and transfer of the raw form). *The defining-function accuracies can be chosen so that on a uniform neighborhood of D_A in each new packet the raw scalar form controls*

$$\lambda_{r+1} + (1 + 3\eta_r)b_{r+1} \sum_{j \neq j_0} \tilde{\Pi}_j, \quad (\text{E.1})$$

with a strictly positive mass and gradient reserve. The form has gradient coefficient $C = 3.99$ and includes the seam term $2 \int H_{\text{out}} u(0)^2$ on a joined side. Tests are compactly supported, with matching seam traces.

Proof. For a side packet $\dim A = m$, a point in D_A projects to at least $m - 1$ remaining undecided labels, hence into H_E^* and the parent's old required set. For a hit packet $\dim A = m - 1$, it projects to at least $m - 2$ remaining undecided labels. The wall lies in G_{j_0} , which supplies the additional old undecided label. The projection is again in both required sets. These inclusions are unchanged on the fresh cylinder, where all relevant data are pulled back from its base. All these maps decrease lengths in the raw metrics; therefore a small neighborhood of D_A projects into small neighborhoods of the two parent sets. The closed-set count, rather than a continuity claim for the counting function, is what is used here.

On compared true pieces in a sufficiently small neighborhood of the old required set, continuity and the bounded derivatives of the fixed parent data give

$$R \geq \lambda_r + b_r \sum_j \Pi_j - e_0, \quad e_0 = \frac{\lambda_r - \lambda_{r+1}}{100}.$$

By Theorems B.10 and B.11, the raw form on these pieces dominates $R - 1.02\Pi_{j_0}$, with spare gradient coefficients .94 on bulk and 1.99 on a wall or cylinder. The bound in Theorem D.6 pays for the additional potential $1.16 \sum_{j \neq j_0} D_j$ from these spare gradient terms and a mass loss of at most $.9e_0$. Since $b_r > 1.02$, dropping the remaining nonnegative current-label term leaves

$$Q_{\text{true}}(u) \geq \int \left(\lambda_r - 2e_0 + b_r \sum_{j \neq j_0} \Pi_j + 1.16 \sum_{j \neq j_0} D_j \right) u^2 + .2 \int |\nabla u|^2. \quad (\text{E.2})$$

Indeed the unrounded gradient remainders are $.94 - .9 \cdot .8 = .22$ and $1.99 - .9 \cdot 1.7 = .46$, and the mass loss is at most $1.9e_0$. The exact numerical margins are

$$\begin{aligned} a_{r+1} - (1 + \eta_r)a_r &= \frac{3\Delta_r(180 - t_r)}{20000} \geq .02685\Delta_r > 0, \\ b_r - (1 + 3\eta_r)b_{r+1} &= \frac{\Delta_r[664 + 15(t_r + \Delta_r)]}{20000} \geq .0332\Delta_r > 0, \\ 1.16 - (1 + 3\eta_r)b_{r+1} &= \frac{800 + 1000t_r + 664\Delta_r + 15\Delta_r(t_r + \Delta_r)}{20000} > 0. \end{aligned} \quad (\text{E.3})$$

These identities use $0 \leq t_r < t_r + \Delta_r \leq 1$. Thus Equation (E.2) exceeds Equation (E.1) by at least $98e_0$ in mass and .2 in gradient, in addition to strictly positive potential margins.

Use the comparison diffeomorphism of Theorem E.2 on bulk and on the common wall/cylinder base, leaving the cylinder coordinate and torus variables unchanged. A test supported in a smaller buffered neighborhood of D_A becomes a compact matching test on the true regular pieces, extended by zero outside the comparison region. For a test in one component set it to zero on other components. In the bulk away from the wall the map is the identity and the phase signs already agree. All supports stay away from singular points and their cylinders.

The pulled-back metric, volume weight, potentials, intrinsic stabilized wall curvature, and outward mean curvature converge uniformly with the chosen finite derivatives. For example the induced scalar curvature can be computed from Gauss' equation, so the second fundamental form and the ambient curvature give the requisite bounds. Consequently, for arbitrarily small $\delta > 0$ the discrepancy of the compared forms is bounded by

$$\delta \left[\int (u^2 + |\nabla u|^2) + \int_{\text{seam}} u(0)^2 \right].$$

The last term is controlled using the first unit interval of the raw product cylinder:

$$\int_{\text{seam}} u(0)^2 \leq 2 \int_{S \times [0,1]} (u^2 + |\partial_t u|^2). \quad (\text{E.4})$$

This follows by writing $u(0) = u(t) - \int_0^t \partial_t u$ and integrating in t . It needs no zero boundary condition. Choosing all approximation errors sufficiently small leaves, for example, a reserve of $90e_0$ in mass and .15 in gradient. These choices can be prescribed on parent buffers before the individual new components are known: the projection inclusions place every relevant point there. At the radial step the enlarged compact blocks supply the bounds; preserved separators trap each new component in one such block before cylinder attachment. Finally impose all coface requests through the finite tolerance allocation already used in Theorem E.2. $\square$

E.2 Seam smoothing and restriction of profiles

The proof below uses scalar-curvature concentration at a mean-curvature jump, as in the corner smoothing of Miao [30, Sections 2–3]. Here the smoothing width varies tangentially, and the form estimate must preserve compatible product ends.

Lemma E.4 (Smoothing a seam with variable width). *The raw joined metrics admit smooth compatible metrics with arbitrarily small C^0 error, and their scalar forms converge to the raw forms in the following quantitative sense. On each joined packet and for every $\delta > 0$, the smoothing can be chosen so that the absolute form error is at most*

$$\delta \int (u^2 + |\nabla u|^2)$$

for every compact matching test, using the raw metric to measure the right side. The old warped-product structure and every sufficiently far old product end are preserved. The choices on a wall can satisfy the requests of both incident sides at once.

Proof. In a normal collar of a smooth wall write the old stabilized metric as $ds^2 + \beta_s$ for $s \leq 0$, where s increases outward. Here β_s includes the old torus factors. The raw joined metric is $ds^2 + \beta_{\min(s,0)}$. Choose a smooth concave function Φ equal to t for $t \leq -1$ and to 0 for $t \geq 1$, with $0 \leq \Phi' \leq 1$.

Such a function is obtained by integrating a smooth decreasing step function symmetric about 0. It satisfies $\Phi \leq 0$ and $\int \Phi'' = -1 = -\int |\Phi''|$.

For a smooth positive function $\epsilon(x)$ on the wall put

$$\phi(x, s) = \epsilon(x)\Phi(s/\epsilon(x)), \quad G_\epsilon = ds^2 + \beta_{\phi(x,s)}.$$

It equals the raw metric outside $|s| < \epsilon(x)$ and differs from it in C^0 by $O(\sup \epsilon)$. In a fixed tangential chart the extra derivatives of ϕ satisfy

$$\begin{aligned} |\partial_x \phi| &\leq c|d\epsilon|, \\ |\partial_x^2 \phi| &\leq c(|d^2 \epsilon| + |d\epsilon|^2/\epsilon). \end{aligned}$$

Choose $\epsilon = \zeta^2$ with $\zeta > 0$ as small in C^2 as prescribed. Then $|d\epsilon|^2/\epsilon = 4|d\zeta|^2$ and $d^2 \epsilon = 2\zeta d^2 \zeta + 2d\zeta \otimes d\zeta$. Thus these tangential errors tend to zero. On a packet, positive compatible choices have a positive infimum and bounded derivatives by the compact-part/product-end description. Their size, but not the resulting higher metric bounds, can be made arbitrarily small. Normal collar widths are also positive packetwise by the same description.

Let $B_\tau^{\text{old}}, H_\tau^{\text{old}}$ be the slice second fundamental form and mean curvature of $ds^2 + \beta_s$. Exactly,

$$B_s = \phi_s B_\phi^{\text{old}}, \quad H_s = \phi_s H_\phi^{\text{old}}.$$

For a metric $ds^2 + \beta(x, s)$ the Gauss and normal variation identities give

$$R(G_\epsilon) = R(\beta(x, s)) - |B_s|^2 - H_s^2 - 2\partial_s H_s.$$

The tangential derivative bounds just proved show that all terms on the strip are uniformly bounded except

$$-2\phi_{ss} H_\phi^{\text{old}}. \tag{E.5}$$

In particular the formula differentiates H_s only in s ; potentially large mixed xs derivatives do not give another scalar spike. The remaining contribution from this derivative is $-2\phi_s^2 \partial_\tau H_\phi^{\text{old}}$, which is bounded.

Let $J_\epsilon(x, s)$ be the density of β_ϕ relative to the zero slice. Uniformly on the strip, $J_\epsilon = 1 + O(\epsilon)$ and $H_\phi^{\text{old}} = H_0^{\text{old}} + O(\epsilon)$. For each fixed x ,

$$\int -\phi_{ss} ds = 1, \quad \int |\phi_{ss}| ds = 1.$$

Writing $E_x = \int_{-\epsilon(x)}^{\epsilon(x)} |\partial_s u|^2 ds$, the fundamental theorem of calculus gives

$$|u(s) - u(0)|^2 \leq \epsilon(x)E_x, \quad |u(s)^2 - u(0)^2| \leq 2\sqrt{\epsilon(x)}|u(0)|E_x^{1/2} + \epsilon(x)E_x.$$

After integrating against $|\phi_{ss}|ds$, the error in replacing $u(s)^2$ by $u(0)^2$ is at most $c\sqrt{\sup \epsilon}(u(0)^2 + E_x)$, for $\sup \epsilon \leq 1$. The density and mean-curvature errors cost at most $c\sup \epsilon(u(0)^2 + E_x)$. Thus the form contribution of Equation (E.5) converges to $2 \int H_0^{\text{old}} u(0)^2$, with the outward sign required in Theorem B.9.

The bounded strip terms have vanishing form cost because

$$\int_{-\epsilon(x)}^{\epsilon(x)} u(s)^2 ds \leq c\epsilon(x)u(0)^2 + c\epsilon(x)^2 E_x.$$

The metric and density errors in the gradient and fixed bounded potential terms tend to zero in H^1 form norm. Integrate these estimates in x and use Equation (E.4); the stated error bound follows. One first proves it for smooth matching tests and then for the matching H^1 completion by density.

Choose ζ on new hit packets, from deeper faces upward, after propagating the two incident-side requests through their finite diagrams. The function is an exact pullback in old product ends, so the displayed smoothing is also an exact product there. It is unchanged far out in the fresh cylinder coordinate. Since the normal coordinate and old metric already split off every old normal factor, the operation commutes with all the inherited collar identifications. $\square$

Lemma E.5 (Approximation that reproduces constants). *A continuous scalar function on a packet can be approximated by smooth functions, with any prescribed positive value error, by operators that reproduce constants. For compatible input on a packet diagram, the approximations can be chosen compatible in all sufficiently far product ends of that diagram. If the input and all face copies used at a packet equal the same constant c , the result equals c there exactly. The same assertion holds componentwise for a finite list of functions and for a list having only finitely many nonconstant entries on each prescribed finite directed face/coface diagram.*

Proof. Choose a locally finite coordinate cover, relatively compact smaller charts, and a smooth partition of unity $\{\psi_\alpha\}$ subordinate to the smaller charts. Extend each chart slightly and average the function against a smooth nonnegative kernel of integral one, at a sufficiently small chart-dependent scale. Call the resulting local function $T_\alpha f$. Uniform continuity on the chart closure makes $|T_\alpha f - f|$ smaller than the prescribed error on $\text{supp } \psi_\alpha$. Thus

$$Tf = \sum_{\alpha} \psi_{\alpha} T_{\alpha} f$$

is a smooth approximation with that error. Every local average sends c to c , and $\sum \psi_{\alpha} = 1$, so $Tc = c$ exactly. Locally prescribed derivative accuracy is obtained in the same way wherever f is already smooth, using a slightly larger regular buffer.

Now use the finite tolerance allocation of Theorem A.4, and blend from deeper faces upward with the weights w_i in the proof of Theorem E.2. Their sum is one. Consequently blending preserves a common constant as well as exact face identities. The finite exceptional union over the prescribed directed dependencies is chosen before applying this construction. For every other entry the entire local input and all participating face copies coincide with the same constant, so all its averaging and blending errors are zero. No minimum of tolerances over these constant entries is needed. On the initial radial space the construction is local on buffered compact blocks and uses a locally finite chart cover. It has the same constant-reproduction property there. $\square$

On a new packet A , write R_{old} and Δ_{old} for the scalar curvature and invariant Laplacian of the smoothed split metric with its retained torus factors, before any new circle factors are added.

Lemma E.6 (Restricted profiles and smooth potentials). *After the seam smoothing the pulled-back profiles and cores satisfy the stage $r + 1$ profile conditions with smooth compatible potentials Π_j^{new} chosen so that*

$$(1 + \eta_r) \tilde{\Pi}_j \leq \Pi_j^{\text{new}} \leq (1 + 2\eta_r) \tilde{\Pi}_j.$$

If

$$V_A^* = \lambda_{r+1} + b_{r+1} \sum_{j \neq j_0} \Pi_j^{\text{new}}, \quad L_A^0 = -C \Delta_{\text{old}} + R_{\text{old}} - V_A^*, \quad (\text{E.6})$$

then on some positive uniform neighborhood of D_A its form is bounded below by a positive multiple of $\|u\|_2^2$.

Proof. We justify restriction to a hypersurface, which does not follow merely by restricting an ambient almost-everywhere inequality. If f is locally Lipschitz on an open Riemannian set and $|df| \leq F$ almost everywhere for continuous F , then at every point its local Lipschitz slope is at most F . In a small coordinate ball, convolve f on a slightly larger ball. The derivative bound of the convolutions is at most the supremum of F there times a metric distortion factor tending to 1 as the ball shrinks. Integrating this bound along straight segments, passing to the uniform limit, and shrinking the ball proves the assertion. It follows also that every rectifiable curve satisfies the corresponding integrated bound, by subdividing into such balls.

Apply this on each open finite band of a profile, with $F = (a_r \mu_j^2 + \tilde{\Pi}_j)/2$. Restriction to the wall, inclusion of bulk, and projection on a cylinder therefore preserve the bound everywhere in local slope, and hence almost everywhere for the restricted locally Lipschitz function. Finite truncations remain locally Lipschitz, including where a pole is present. Across a seam the same conclusion follows by splitting a path into its bulk and cylinder portions; traces agree. A sufficiently small C^0 metric change increases all these slopes by at most $1 + \eta_r$.

For each label j , the raw potential has its own positive lower bound $p_j^* > 0$. Apply Theorem E.5 to $(1 + 3\eta_r/2)\tilde{\Pi}_j$, with strict value error smaller than $\eta_r p_j^*/2$. This proves the displayed relative bounds. Different labels may require different accuracies; a common positive lower bound for an infinite label list is unnecessary. When the raw potential is a spatial constant c_j on the whole relevant incident diagram, choose its new value to be exactly $(1 + 3\eta_r/2)c_j$. By Equation (E.3),

$$2|\nabla \mu_j|_{\text{new}} \leq (1 + \eta_r)(a_r \mu_j^2 + \tilde{\Pi}_j) \leq a_{r+1} \mu_j^2 + \Pi_j^{\text{new}}.$$

Continuous projections preserve the inclusion of every core closure inside its pole interior and preserve all prescribed q constraints.

Choose the seam form error small relative to the reserves of Theorem E.3. The potential in Equation (E.6) is at most that in Equation (E.1); the extra η_r supplies additional margin for the potential smoothing. Metric equivalence allows the required-set neighborhood to be slightly reduced after smoothing. Thus one may retain, for example, an $e_0 \|u\|_2^2$ lower bound for the form of L_A^0 on tests in that neighborhood. No compatibility of the neighborhoods themselves is asserted. $\square$

We have obtained a smooth packet metric and the next profile inequalities. The scalar lower bound is still only a quadratic-form bound near the new required set. We next extend its positive gap to the whole packet by a correction that vanishes near that set, solve a bounded elliptic equation, and approximate its solutions compatibly. The final group of 399 circle factors converts this form information into the pointwise scalar inequality.

E.3 Compatible corrections and positive supersolutions

At this point the smooth split packets have compact truncations and exact product ends by Theorem A.1. At the radial step the preserved separating bands give the compact initial pieces. Hence Theorem A.2 already applies to these metrics and retained warps, before the following correction is constructed. In particular all bounds used below hold separately on each whole packet; they are not assumptions on the original manifold.

Lemma E.7 (A compatible correction with a global gap). *There are smooth nonnegative bounded compatible potentials K_A , with all derivatives bounded packetwise, such that $K_A = 0$ in a positive uniform neighborhood of D_A and*

$$Q_A(u) := \int (C|\nabla u|^2 + (R_{\text{old}} - V_A^* + K_A)u^2) \geq g_A \|u\|_2^2 \quad (g_A > 0) \quad (\text{E.7})$$

for every compact smooth test. Let ω_A^{old} be the product of the retained circle length functions for the smoothed metric on A , before the new circle factors at this split are appended. With K_A acting by multiplication, define

$$L_A = L_A^0 + K_A = -C \Delta_{\text{old}} + R_{\text{old}} - V_A^* + K_A.$$

Its quadratic form in the measure $\omega_A^{\text{old}} d \text{vol}_A$ is Q_A . In an exact end over B , the base metric is a product and every retained warp pulls back from B . Consequently

$$\Delta_{\text{old},A} = \Delta_{\text{old},B} + \Delta_{\text{normal}}, \quad L_A = L_B - C \Delta_{\text{normal}}.$$

Proof. First disregard compatibility on a fixed packet. Put $W_A = R_{\text{old}} - V_A^*$. Choose a smooth angle θ equal to 0 near D_A and $\pi/2$ outside the neighborhood of local positivity, with bounded derivatives and with the support of $\cos \theta$ inside that neighborhood. Uniform charts and the positive neighborhood width give such a function by smoothing a distance cutoff with buffers. Set

$$K_{\text{ind}} = J \sin^2 \theta + C |\nabla \theta|^2, \quad J > \max\{0, -\inf W_A\}.$$

The identity for the squared partition $\cos \theta, \sin \theta$ shows that $Q_A^0 + K_{\text{ind}}$ has gap $\gamma_A = \min\{e_0, J + \inf W_A\} > 0$. Indeed the gradient partition error is exactly $C |\nabla \theta|^2 u^2$; local positivity controls the cosine part and $W_A + J > 0$ controls the sine part. The correction is zero on a smaller uniform neighborhood of D_A .

Proceed from deeper faces upward. In sufficiently far elementary collars the already constructed K_B copies agree on intersections: on a common quadrant each is the pullback of the same further-face potential. They consequently define a *single* nonnegative function K_{end} on the collar union. Copies are not added together on an intersection. The metric, measure, and V_A^* are likewise exact products there. In particular V^* has the same formula on a face: C , λ_{r+1} , b_{r+1} , the label set, and the old torus list do not depend on base dimension, after padding unused flat factors on each finite compared diagram.

For the finitely many elementary collars choose angles θ_i which rise from 0 to $\pi/2$ over a normal interval of length L , inside the valid common product ranges. Use the squared partition

$$\eta_0 = \prod_i \cos \theta_i, \quad \eta_i = \sin \theta_i \prod_{a < i} \cos \theta_a.$$

It has $\sum_i \eta_i^2 = 1$ and $\sum_i |d\eta_i|^2 \leq \sum_i |d\theta_i|^2 \leq c_A L^{-2}$; the inequality follows successively from the two-term sine/cosine identity, so no number of intersections is being summed separately. The support of η_0 lies in a compact collar truncation.

Choose smooth multipliers $\alpha, \beta \in [0, 1]$ with $\alpha = 1$ on $\text{supp } \eta_0$ and $\beta = 1$ on the union of the face-cutoff supports. Let α vanish only farther out than those transition intervals. Let β vanish on the inner side of a buffered collar union, so βK_{end} extends smoothly by zero. Both can be formed from products of one-coordinate cutoffs. Define

$$K_A = \alpha K_{\text{ind}} + \beta K_{\text{end}}.$$

On the central support this dominates K_{ind} ; on each collar support it dominates the single face copy. Eventually $\alpha = 0, \beta = 1$, giving exact compatibility. Nonnegativity permits the harmless overlap of these two terms in the finite transition region.

This correction vanishes in a uniform neighborhood of D_A . The individual correction already does. For the end term, a point of D_A in a product collar projects into D_B , because a face has a smaller required counting threshold. Start the support of β with a buffer from the collar entry. A

point sufficiently close to D_A in this support then has a nearby required point in the same collar, and their face projections are just as close. It lies in the vanishing neighborhood of K_B . Take the minimum of the finitely many face widths and entry buffers; at corners the copies agree, so the same conclusion applies.

Apply the individual gap to $\eta_0 u$. Apply each face gap by slicing to $\eta_i u$; its support is away from the collar entry, and the normal gradient term is nonnegative. If $g_* = \min(\gamma_A, \{g_B\}) > 0$, the finite IMS identity gives

$$Q_A(u) \geq (g_* - C c_A L^{-2}) \|u\|_2^2.$$

Choose L so that the error is at most $g_*/2$. All quantities used for this choice are fixed before solving any elliptic equation. Boundedness and the derivative assertions follow from compact parts and finitely many product copies. Empty required sets or absent faces cause no change: one may take $\theta = \pi/2$ or omit the corresponding cutoff. This completes the compatible induction and proves the operator identity. $\square$

Lemma E.8 (The bounded gapped resolvent). *Let A be one of the resulting complete packets, with its weighted base measure. Suppose W is smooth and bounded with bounded derivatives, and $L = -C \Delta_\omega + W$ has form gap $g > 0$. Then there is a unique bounded smooth solution s of $Ls = 1$. It satisfies $0 < \inf s \leq \sup s < \infty$ and has bounded derivatives of every fixed order. Bounded smooth supersolutions of $Lw \geq 0$ are nonnegative. The same conclusions hold on Euclidean products of A .*

Proof. Use the bounded geometry, positive bounded weight, and center-uniform polynomial volume estimate of Theorem A.2. The form gap and boundedness of W make the closed form coercive on H^1 : combine $Q \geq g \|u\|_2^2$ with $Q \geq C \|\nabla u\|_2^2 - \|W_-\|_\infty \|u\|_2^2$. It therefore has a self-adjoint inverse on L^2 , of norm at most g^{-1} . If $f \geq 0$, testing $Lu = f$ with the negative part gives $Q(u_-) \leq 0$, so the inverse is positive.

Here is the decay estimate needed for bounded sources. We use an Agmon-type exponential-weight argument [1], proved here for the weighted packet operator. If f is supported in a metric unit ball I and $u = L^{-1}f$, put $\chi = \exp(\delta \min\{d(\cdot, I), D\})$, where $C\delta^2 < g$. The form identity for a bounded Lipschitz multiplier gives

$$Q(\chi u) = \langle Lu, \chi^2 u \rangle + C \int |\nabla \chi|^2 u^2.$$

It holds by compact approximation in H^1 ; on the support of f , $\chi = 1$. Hence

$$(g - C\delta^2) \|\chi u\|_2^2 \leq \langle f, u \rangle \leq g^{-1} \|f\|_2^2. \quad (\text{E.8})$$

Let $D \rightarrow \infty$. The polynomial volume bound, uniform in the center of I , implies

$$\int_A e^{-2\delta d(x, I)} d\mu(x) \leq C_{A, \delta} < \infty.$$

Indeed sum the exponential bound on successive unit annuli against their polynomial volume upper bound.

Choose smooth compactly supported $0 \leq f_R \leq 1$ equal to 1 on an increasing compact exhaustion, and put $u_R = L^{-1}f_R \geq 0$. These are initially H^1 weak solutions. For each fixed R , the coefficients and f_R are smooth, and weak interior regularity makes u_R smooth [37, Lecture 6, Theorem 1 and Corollary 1]. This assertion requires no derivative bounds for f_R uniform in R .

For any $h \in L^2(I)$, self-adjointness, Equation (E.8), and Cauchy–Schwarz with opposite exponential weights give

$$|\langle u_R, h \rangle| = |\langle f_R, L^{-1}h \rangle| \leq \frac{C_{A, \delta}^{1/2}}{\sqrt{g(g - C\delta^2)}} \|h\|_2.$$

Thus u_R has a uniform local L^2 bound on every unit ball, with constant independent of its center and of R . Choose the uniform chart radius small enough that each larger chart ball lies in a metric unit ball. Writing B_1 and $B_{1/2}$ for the fixed rescaling of two such nested chart balls, interior local boundedness gives

$$\|u_R\|_{L^\infty(B_{1/2})} \leq c_A(\|u_R\|_{L^2(B_1)} + \|f_R\|_{L^\infty(B_1)}).$$

Here we use Simon [37, Lecture 18, Theorem 2, with $p = 2$]; the fixed radius and coordinate changes are absorbed in c_A . The principal coefficients are uniformly elliptic and bounded, the lower-order coefficients are bounded, and the weight is bounded above and below. Since $0 \leq f_R \leq 1$, this gives a bound independent of the center and R .

Fix two nested relatively compact coordinate domains. For all sufficiently large R , the source is exactly 1 on the larger domain. The interior Schauder estimate [37, Lecture 12, Theorem 2], followed by differentiation, therefore bounds every derivative of u_R on the smaller domain, uniformly for these R . A diagonal subsequence converges smoothly on compact subsets to a bounded nonnegative solution s of $LS = 1$. Applying the same estimates to this limiting equation in the uniform packet charts gives bounded derivatives of every fixed order on A .

The maximum argument must allow W to have either sign. Let w be a bounded smooth supersolution, put $w_- = \max\{-w, 0\}$, and fix $0 < \delta < \sqrt{g/C}$. Write $\psi = e^{-\delta d(x, x_0)}$ and take compact Lipschitz cutoffs ζ_R equal to 1 on $B(x_0, R)$, supported in $B(x_0, 2R)$, with $|d\zeta_R| \leq 2/R$. Testing the supersolution with $(\psi\zeta_R)^2 w_-$, and expanding the gradient, gives directly

$$g\|\psi\zeta_R w_-\|_2^2 \leq Q(\psi\zeta_R w_-) \leq C \int |d(\psi\zeta_R)|^2 w_-^2.$$

All tests are compact, so no prior global gradient bound on w is needed. Boundedness of w and polynomial volume growth make ψw_- square integrable. The terms containing $d\zeta_R$ tend to zero, including the cross term by Cauchy–Schwarz. Dominated convergence therefore yields

$$g\|\psi w_-\|_2^2 \leq C\delta^2 \|\psi w_-\|_2^2,$$

so $w_- = 0$. On an open subdomain the same compact-test argument applies when the negative part has zero boundary trace, using the Dirichlet form gap. In particular it applies to nonnegative boundary values. Apply comparison to both signs of the difference of two bounded solutions to obtain uniqueness. If $M = 1 + \|W\|_\infty$, then $L(s - M^{-1}) = 1 - W/M \geq 0$, and comparison gives $s \geq M^{-1}$.

Euclidean products retain bounded geometry and polynomial volume, and their form gap follows by slicing. The preceding proof therefore applies verbatim; uniqueness identifies their solution with the pullback of s . $\square$

Lemma E.9 (Exact solutions and compatible approximations). *Fix all the corrections of Theorem E.7 and let s_A denote the exact bounded solution $L_A s_A = 1$. There exist positive smooth compatible functions v_A with*

$$L_A v_A \geq \frac{1}{2} \tag{E.9}$$

on every packet. They have positive infimum and bounded derivatives packetwise. These v_A are approximations; they are not relabeled as the original exact solutions s_A .

Proof. We first establish a uniform end estimate over a whole face, which also avoids any compactness ambiguity at its deeper ends. On an exact elementary collar $B \times [T, \infty)$, the difference $w = s_A - s_B$ is bounded and satisfies $L_A w = 0$. Set $m_B = \inf s_B > 0$ and choose $a > 0$ with $Ca^2 \|s_B\|_\infty \leq 1/2$. The function

$$b(x, t) = e^{-a(t-T)} s_B(x)$$

satisfies $L_A b = e^{-a(t-T)}(1 - Ca^2 s_B) \geq \frac{1}{2}e^{-a(t-T)}$. Comparison on the half collar, using its sliced Dirichlet gap and the Dirichlet comparison argument in the proof of Theorem E.8, yields

$$|s_A - s_B|(x, t) \leq \frac{\|s_A - s_B\|_\infty}{m_B} e^{-a(t-T)} s_B(x). \quad (\text{E.10})$$

The boundary inequalities hold at $t = T$ for both signs. Uniform interior elliptic estimates for the homogeneous difference give the same exponential decay in C^2 on $t \geq T + 1$. All constants are uniform in $x \in B$, including its deeper ends. Thus the estimate holds on each elementary collar, also where collars overlap. Exact agreement of the approximations on those overlaps will come from the compatible face copies used in the following construction.

All operator coefficients and every exact s_A are now fixed. Let M_A be a finite bound for $L_A : C^2 \rightarrow C^0$ in uniform charts, and choose a target

$$\tau_A < \min\{\frac{1}{4} \inf s_A, (4M_A)^{-1}\}.$$

The finite family of collar blending cutoffs has a C^2 operator bound $H_A \geq 1$, independent of translation of their starts. Include in H_A the fixed finite factor for the number of copies. For each packet B , let ε_B be the minimum of

$$\frac{\tau_A}{\prod_{i=0}^{\ell-1} (4H_{A_i})} \quad (A = A_0 > A_1 > \dots > A_\ell = B),$$

over all directed coface-to-face chains ending at B , including the length-zero chain. There are finitely many such chains, so this minimum is positive. Appending the edge $A > B$ to a minimizing chain for A gives $\varepsilon_B \leq \varepsilon_A/(4H_A)$. All these numbers are fixed after the corrections and operator norms, and before constructing any v_A .

Construct v_A from deeper faces upward. Keep s_A in the central part and use the ordinary weights in the proof of Theorem E.2 to blend to the already constructed copies of v_B . Move the starts until $\|s_A - s_B\|_{C^2} < \varepsilon_A/(4H_A)$ on all participating collars, using Equation (E.10). The persistent difference $v_B - s_B$ is paid by its previously allocated tolerance, not by moving the start. Start also beyond the finitely many compatibility thresholds for the v_B 's, so their copies agree on overlaps. The derivative bounds for the weights then give

$$\|v_A - s_A\|_{C^2} \leq \varepsilon_A \leq \tau_A.$$

The result is an exact face copy sufficiently far out, is smooth, and has bounded higher derivatives by compact parts and finitely many smooth products. Finally $v_A \geq \frac{3}{4} \inf s_A > 0$ and $L_A v_A \geq 1 - M_A \tau_A \geq 3/4$, which implies Equation (E.9). No geometry radius, correction, or operator is changed during this approximation. $\square$

E.4 Restoring the invariant

The passage from a positive solution to scalar-curvature control by a circle warp appears in Fischer-Colbrie and Schoen [16, Theorem 4(ii)]. Here bounded positive supersolutions must also be exactly compatible with every packet end; Theorem E.9 supplies these functions.

Proposition E.10 (The full invariant step). *Assume the stage r invariant and the conclusion of Theorem D.6. For every nonempty parent packet of base dimension at least two, processing j_0 produces the compatible smooth split packets of Theorem A.1, with the required forced signs and continuous projection maps. There are smooth compatible old metrics and 399 new torus factors*

on every packet such that the profiles, cores, potentials, and scalar curvature satisfy the stage $r + 1$ invariant. This includes a possible one-dimensional output, on which the scalar lower bound is global. After the radial split the packets have the complete compact-part/product-end geometry of Theorem A.2.

Proof. Use Theorems E.2, E.4 and E.6 for the smooth base metrics and the profile data, and Theorems E.7 and E.9 for v_A . Append $k = 399$ circle variables, each with length function $v_A^{2/(k+1)}$. The warped-product identity is

$$R_{\text{new}} = R_{\text{old}} - \frac{4k}{k+1} \frac{\Delta_{\text{old}} v_A}{v_A}.$$

Indeed a common scale f contributes $-2kf^{-1} \Delta_{\text{old}} f - k(k-1)f^{-2}|df|^2$; substitution of $f = v_A^{2/(k+1)}$ cancels the separate $|dv_A|^2$ term. The old Laplacian on invariant functions is the weighted base Laplacian, including the product of all old warp lengths. Here k counts only the newly appended circle factors. Any unit factors used earlier to compare old lists on a finite diagram have zero logarithmic derivative and leave both Δ_{old} and this coefficient unchanged. Since $4k/(k+1) = C$, Equations (E.6) and (E.9) give

$$R_{\text{new}} = V_A^* - K_A + \frac{L_A v_A}{v_A} \geq V_A^* \quad \text{on } D_A.$$

Here $K_A = 0$ on a neighborhood of D_A . Adding these factors does not change base distances or profile gradients. The new factors are positive, bounded above and away from zero, and exactly compatible. The same number is appended in every status at the stage, so all end identities compare the same list of factors. This proves the analytic invariant, including when the new required threshold is zero. The componentwise metric operations preserve every unused old circle length identically equal to 1: restriction, product extension, and $\beta_s \mapsto \beta_{\phi(x,s)}$ all fix that constant entry. The 399 appended lengths are the only common new entries. Likewise restriction and cylinder pullback preserve a constant pole profile and its full core, and Theorem E.5 preserves the specified multiplied constant potentials.

The zero wall lies in $\text{int } G_{j_0}$ and all current forced signs hold by Theorem E.2; earlier forced cores are carried by the continuous projection and retain their pole neighborhoods. Thus this is exactly the regular compatible splitting operation required by the packet construction. At the radial split, the preserved alternating compact barrier bands separate all wall components into bounded transition shells and trap every closed sign-side component between two separating bands (with a compact initial block). These components are compact before prolongation. The normal collars and smooth metrics used above are products in all old ends. Theorems A.1 and A.2 therefore supply the claimed geometry after prolongation. The main argument applies the one-dimensional obstruction and reconstructs the compatible partitions in the original manifold. $\square$

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